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MEDICAL DEVICE FOR REMOVING AN IMPLANTED OBJECT USING LASER CUT HYPOTUBES

Abstract

Methods and devices for separating an implanted object, such as a pacemaker lead, from tissue surrounding such object in a patient's vasculature system. A surgical device includes a handle, an elongate inner sheath and a circular cutting blade that extends from the distal end of the sheath upon actuating the handle. The circular cutting blade is configured to engage the tissue surrounding an implanted lead and cut such tissue in a coring fashion as the surgical device translates along the length of the lead, thereby allowing the lead to enter the device's elongate shaft. The surgical device has a barrel cam cylinder in the handle assembly that imparts rotation of the blade. The inner sheath and outer sheath are constructed of laser-cut hypotubes, thereby allowing the surgical device, particularly the sheath assembly, to have a smaller profile for navigating smaller sized vasculature.

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Background/Summary

CROSS-REFERENCE TO PRIOR APPLICATIONS [0001] This application is a Continuation of U.S. application Ser. No. 17/000,606, filed Aug. 24, 2020, now U.S. Pat. No. 12,303,165, which is a Continuation of U.S. application Ser. No. 15/755,931, filed Feb. 27, 2018, now U.S. Pat. No. 10,751,529, issued Aug. 5, 2020, which is the U.S. National Phase application under 35 U.S.C. § 371 of International Application No. PCT/US2016/049108, filed Aug. 26, 2016, which claims the benefit of U.S. Provisional Application No. 62/211,151, filed Aug. 28, 2015, each of which is hereby incorporated by reference. CROSS-REFERENCE TO RELATED APPLICATIONS [0002] U.S. application Ser. No. 15/755,931, filed Feb. 27, 2018 is a Continuation-In-Part of U.S. application Ser. No. 14/627,851, filed Feb. 20, 2015 and entitled MEDICAL DEVICE FOR REMOVING AN IMPLANTED OBJECT, which claims the benefit of and priority to 35 U.S.C. § 119 and/or 35 U.S.C. § 120, which is a Continuation-In-Part of International Application No. PCT/US2014/026496, filed Mar. 13, 2014 and entitled SURGICAL INSTRUMENT FOR REMOVING AN IMPLANTED OBJECT, which claims the benefit of and priority to, under 35 U.S.C. § 119(e), U.S. Provisional Application No. 61/793,597, filed Mar. 15, 2013, entitled SURGICAL INSTRUMENT FOR REMOVING AN IMPLANTED OBJECT. U.S. application Ser. No. 14/627,851, filed Feb. 20, 2015 and entitled MEDICAL DEVICE FOR REMOVING AN IMPLANTED OBJECT, also claims the benefit of and priority to, under 35 U.S.C. § 119(e), U.S. Provisional Application No. 61/947,377, filed Mar. 3, 2014, entitled MEDICAL DEVICE FOR REMOVING AN IMPLANTED OBJECT, U.S. Provisional Application No. 62/058,790, filed Oct. 2, 2014, entitled MEDICAL DEVICE FOR REMOVING AN IMPLANTED OBJECT, and U.S. Provisional Application No. 62/113,865, filed Feb. 9, 2015, entitled MEDICAL DEVICE FOR REMOVING AN IMPLANTED OBJECT. The above applications are hereby incorporated by reference in their entireties for all that they teach and for all purposes.

FIELD OF THE DISCLOSURE

[0003] The present disclosure relates generally to devices, methods and systems for separating tissue in a patient's vascular system, and more specifically, to devices for separating tissue attached to implanted objects, such as leads, in a patient's vascular system and removing such objects.

BACKGROUND

[0004] Surgically implanted cardiac pacing systems, such as pacemakers and defibrillators, play an important role in the treatment of heart disease. In the 50 years since the first pacemaker was implanted, technology has improved dramatically, and these systems have saved or improved the quality of countless lives. Pacemakers treat slow heart rhythms by increasing the heart rate or by coordinating the heart's contraction for some heart failure patients. Implantable cardioverter-defibrillators stop dangerous rapid heart rhythms by delivering an electric shock.

[0005] Cardiac pacing systems typically include a timing device and a lead, which are placed inside

the body of a patient. One part of the system is the pulse generator containing electric circuits and a battery, usually placed under the skin on the chest wall beneath the collarbone. To replace the battery, the pulse generator must be changed by a simple surgical procedure every 5 to 10 years. Another part of the system includes the wires, or leads, which run between the pulse generator and the heart. In a pacemaker, these leads allow the device to increase the heart rate by delivering small timed bursts of electric energy to make the heart beat faster. In a defibrillator, the lead has special coils to allow the device to deliver a high-energy shock and convert potentially dangerous rapid rhythms (ventricular tachycardia or fibrillation) back to a normal rhythm. Additionally, the leads may transmit information about the heart's electrical activity to the pacemaker.

[0006] For both of these functions, leads must be in contact with heart tissue. Most leads pass through a vein under the collarbone that connects to the right side of the heart (right atrium and right ventricle). In some cases, a lead is inserted through a vein and guided into a heart chamber where it is attached with the heart. In other instances, a lead is attached to the outside of the heart. To remain attached to the heart muscle, most leads have a fixation mechanism, such as a small screw and/or hooks at the end.

[0007] Within a relatively short time after a lead is implanted into the body, the body's natural healing process forms scar tissue along the lead and possibly at its tip, thereby fastening it even more securely in the patient's body. Leads usually last longer than device batteries, so leads are simply reconnected to each new pulse generator (battery) at the time of replacement. Although leads are designed to be implanted permanently in the body, occasionally these leads must be removed, or extracted. Leads may be removed from patients for numerous reasons, including but not limited to, infections, lead age, and lead malfunction.

[0008] Removal or extraction of the lead may be difficult. As mentioned above, the body's natural healing process forms scar tissue over and along the lead, and possibly at its tip, thereby encasing at least a portion of the lead and fastening it even more securely in the patient's body. In addition, the lead and/or tissue may become attached to the vasculature wall. Both results may, therefore, increase the difficulty of removing the leads from the patient's vasculature.

[0009] A variety of tools have been developed to make lead extraction safer and more successful. Current lead extraction techniques include mechanical traction, mechanical devices, and laser devices. Mechanical traction may be accomplished by inserting a locking stylet into the hollow portion of the lead and then pulling the lead to remove it. An example of such a lead locking device is described and illustrated in U.S. Pat. No. 6,167,315 to Coe et al., which is hereby incorporated herein by reference in its entirety for all that it teaches and for all purposes.

[0010] A mechanical device to extract leads may include one or more a flexible tubes called a sheath that passes over the lead and/or the surrounding tissue. One of the sheaths may include a tip having a dilator, a separator and/or a cutting blade, such that upon advancement, the tip (and possibly the sheath cooperate to) dilates, separates and/or cuts to separate the scar tissue from other scar tissue including the scar tissue surrounding the lead. In some cases, the tip (and sheath) may also separate the tissue itself from the lead. Once the lead is separated from the surrounding tissue and/or the surrounding tissue is separated from the remaining scar tissue, the lead may be inserted into a hollow lumen of the sheath for removal and/or be removed from the patient's vasculature using some other mechanical devices, such as the mechanical traction device previously described in United States Patent Publication No. 2008/0154293 to Taylor, which is hereby incorporated herein by reference in its entirety for all that it teaches and for all purposes.

[0011] Some lead extraction devices include mechanical sheaths that have trigger mechanisms for extending the blade from the distal end of the sheath. An example of such devices and method used to extract leads is described and illustrated in U.S. Pat. No. 5,651,781 to Grace, which is hereby incorporated herein by reference in its entirety for all that it teaches and for all purposes. Another example of these device that has a trigger mechanism for extending the blade from the distal end of the sheath is described and illustrated in United States Patent Publication No. 2014/0277037 having

application Ser. No. 13/834,405 filed Mar. 14, 2013, which is hereby incorporated herein by reference in its entirety for all that it teaches and for all purposes.

[0012] Controlling the amount of extension and retraction of the blade within a patient's vasculature may be critical, particularly when the sheath and blade negotiate tortuous paths that exist in certain vascular or physiological environments and/or when the blade is attempting to cut and/or separate tough surrounding tissue. Furthermore, in certain cases, using such mechanical devices for lead removal may require more meticulous control, such as when the leads are located in, and/or attached to a structurally-weak portion of the vasculature. For instance, typical leads in a human may pass through the innominate vein, past the superior vena cava ("SVC"), and into the right atrium of the heart. Tissue growth occurring along the SVC and other locations along the innominate vein may increase the risk and difficulty in extracting the leads from such locations, particularly when the vein(s)' walls are thin and the surrounding tissue is notably fibrous.

SUMMARY

[0013] Accordingly, there is a need for a device, method and/or system such as a surgical device that has the capability to precisely control the extension, retraction and rotation of a blade from within an outer sheath. For example, it may be desirable for the blade to rotate in one direction as the blade initially extends from and retracts within the outer sheath, then rotate in an opposite direction upon subsequent extension and retraction during the same actuation of the surgical device. The present disclosure discusses a surgical device that has a barrel cam cylinder in the handle assembly that imparts rotation of the blade and a separate cam mechanism in the tip of outer sheath assembly that imparts and controls the extension and retraction of the blade. The barrel cam cylinder and cam mechanism cooperate to cause the blade to rotate in one direction as it initially extends from and retracts in the outer sheath and to rotate in a second direction as it extends and retracts a second time. For each actuation of the handle, the blade rotates in one direction as it initially extends from and retracts into outer sheath and subsequently rotates in a second direction as it extends and retracts a second time. Alternating the direction of rotation in conjunction with the extension and retraction of the blade during rotation creates a slicing action in one direction for each extension and retraction of the blade, thereby minimizing the potential for the blade to become jammed in the surrounding tissue.

[0014] A device in accordance with this disclosure for removing an implanted object from a body vessel, may comprise a sheath assembly comprising an outer sheath assembly and an inner sheath assembly, and a pin, the outer sheath assembly comprising an outer sheath and an outer band, the outer band coupled to the pin, the inner sheath assembly comprising an inner sheath and a tip, wherein the tip has a cutting surface, the inner sheath comprising a proximal end and a distal end, wherein the distal end of the inner sheath is coupled to the tip, the tip comprising a cam slot for receipt of and cooperation with the pin, and a handle assembly comprising a trigger and a barrel cam cylinder, the trigger comprising a trigger pin, the barrel cam cylinder comprising a barrel cam cylinder slot for receipt and cooperation with the trigger pin, wherein the proximal end of the inner sheath is coupled to the barrel cam cylinder such that upon the trigger pin moving proximally in a longitudinal direction, the barrel cam cylinder rotates in both a clockwise direction and a counter clockwise direction, thereby causing the tip to rotate in both the clockwise direction and the counter clockwise direction while the tip moves longitudinally.

[0015] A device in accordance with this disclosure for removing an implanted object from a body vessel, may alternatively comprise a sheath assembly comprising an outer sheath assembly and an inner sheath assembly, and a pin, wherein the outer sheath assembly and the inner sheath assembly each comprise a proximal end and a distal end, wherein the distal end of the outer sheath assembly is coupled to the distal end of the inner sheath assembly by the pin, the inner sheath assembly comprising an inner sheath and a tip at its distal end, wherein the tip has a cutting surface, the tip comprising a slot for receipt of and cooperation with the pin, and a handle assembly comprising a trigger and a barrel cam cylinder, the trigger comprising a trigger pin, the barrel cam cylinder

comprising a barrel cam cylinder slot for receipt and cooperation with the trigger pin, wherein the proximal end of the inner sheath is coupled to the barrel cam cylinder by the trigger pin such that upon the trigger pin moving proximally in a longitudinal direction, the barrel cam cylinder rotates in a first direction and a second direction, wherein the first direction is different than the second direction, wherein the tip moves longitudinally while the barrel cam cylinder rotates in the first direction, and wherein the tip moves longitudinally while the barrel cam cylinder rotates in the second direction.

[0016] There is a need for a device, method and/or system such as a surgical device that has the capability to precisely control the extension, retraction and rotation of a blade from within an outer sheath. For example, it may be desirable for the blade to rotate in one direction as the blade initially extends from and retracts within the outer sheath, then rotate in an opposite direction upon subsequent extension and retraction during the same actuation of the surgical device. The present disclosure discusses a surgical device that has a barrel cam cylinder in the handle assembly that imparts rotation of the blade and a separate cam mechanism in the tip of outer sheath assembly that imparts and controls the extension and retraction of the blade. The barrel cam cylinder and cam mechanism cooperate such that (1) upon a first actuation of the device, the barrel cam cylinder rotates in a first direction, thereby causing the blade to rotate in the first direction while the blade extends and retracts, and (2) upon a second actuation of the device, the barrel cam cylinder rotates in a second direction, thereby causing the blade to rotate in the second direction while the blade extends and retracts. Alternating the direction of rotation in conjunction with the extension and retraction of the blade during rotation creates a slicing action that minimizes the potential for the blade to become jammed in the surrounding tissue.

[0017] A device in accordance with this disclosure for removing an implanted object from a body vessel, may comprise a sheath assembly comprising an outer sheath assembly and an inner sheath assembly, and a pin, the outer sheath assembly comprising an outer sheath and an outer band, the outer band coupled to the pin, the inner sheath assembly comprising an inner sheath and a tip, wherein the tip has a cutting surface, the inner sheath comprising a proximal end and a distal end, wherein the distal end of the inner sheath is coupled to the tip, the tip comprising a cam slot for receipt of and cooperation with the pin, and a handle assembly comprising a trigger and a barrel cam cylinder, the trigger comprising a trigger pin, the barrel cam cylinder comprising a barrel cam cylinder slot for receipt and cooperation with the trigger pin, wherein the proximal end of the inner sheath is coupled to the barrel cam cylinder such that (1) upon a first actuation of the trigger to proximally move the trigger pin in a longitudinal direction, the barrel cam cylinder rotates in a first direction, thereby causing the tip to rotate in the first direction while the tip moves longitudinally; and (2) upon a second actuation of the trigger to proximally move the trigger pin in the longitudinal direction, the barrel cam cylinder rotates in a second direction, thereby causing the tip to rotate in the second direction while the tip moves longitudinally.

[0018] A device in accordance with this disclosure for removing an implanted object from a body vessel, may comprise a sheath comprising a proximal end and a distal end, a tip coupled to the distal end of the sheath, wherein the tip has a cutting surface, a handle assembly rotatably carrying the sheath, the handle assembly comprising a trigger comprising a trigger pin; and a barrel cam assembly comprising a barrel cam cylinder comprising a barrel cam cylinder slot for receipt and cooperation with the trigger pin, the barrel cam cylinder slot comprising a first slot portion and a second slot portion; a follower guide rotatably carried by the barrel cam cylinder; wherein upon a first actuation of the trigger to proximally move the trigger pin in a longitudinal direction, the follower guide urges the trigger pin to traverse the first slot portion, thereby causing the barrel cam cylinder and the tip to rotate in the first direction; and wherein upon a second actuation of the trigger to proximally move the trigger pin in a longitudinal direction, the follower guide urges the trigger pin to traverse the second slot portion, thereby causing the barrel cam cylinder and the tip to rotate in the second direction.

[0019] A device in accordance with this disclosure for removing an implanted object from a body vessel, may comprise a sheath assembly comprising an outer sheath assembly and an inner sheath assembly, and a pin, the outer sheath assembly comprising an outer sheath and an outer band, the outer band coupled to the pin, wherein at least a portion of the outer sheath comprises an outer hypotube, the inner sheath assembly comprising an inner sheath and a tip, wherein the tip has a cutting surface, wherein at least a portion of the outer sheath comprises an inner hypotube, the inner sheath comprising a proximal end and a distal end, wherein the distal end of the inner sheath is coupled to the tip, the tip comprising a cam slot for receipt of and cooperation with the pin, and a handle assembly comprising a trigger and a barrel cam cylinder, the trigger comprising a trigger pin, the barrel cam cylinder comprising a barrel cam cylinder slot for receipt and cooperation with the trigger pin, wherein the proximal end of the inner sheath is coupled to the barrel cam cylinder such that (1) upon a first actuation of the trigger to proximally move the trigger pin in a longitudinal direction, the barrel cam cylinder rotates in a first direction, thereby causing the tip to rotate in the first direction while the tip moves longitudinally and (2) upon a second actuation of the trigger to proximally move the trigger pin in the longitudinal direction, the barrel cam cylinder rotates in a second direction, thereby causing the tip to rotate in the second direction while the tip moves longitudinally. Because the inner sheath and outer sheath are constructed of laser-cut hypotubes, the surgical device, particularly the sheath assembly has a smaller overall profile, which improves the surgical devices ability to navigate smaller sized vasculature.

[0020] A device wherein the outer sheath assembly is stationary and the inner sheath assembly is capable of rotating.

[0021] A device wherein the pin couples the tip of the inner sheath assembly to the outer band of the outer sheath assembly.

[0022] A device wherein the handle assembly further comprises a spring assembly coupled to the trigger.

[0023] A device wherein the spring is a constant force spring.

[0024] A device wherein the outer hypotube is laser cut and wherein the inner hypotube is laser cut.

[0025] A device wherein the outer hypotube comprises a first outer segment and a second outer segment and wherein the inner hypotube comprises a first inner segment and a second inner segment.

[0026] A device wherein the first outer segment is distal the second outer segment and wherein the first inner segment is distal the second inner segment.

[0027] A device wherein the first outer segment has a first outer flexibility and a first outer length and the second outer segment has a second outer flexibility and a second outer length and wherein the first inner segment has a first inner flexibility and a first inner length and the second inner segment has a second inner flexibility and a second inner length, wherein the first outer flexibility is greater than the second outer flexibility, and wherein the first inner flexibility is greater than the second inner flexibility.

[0028] A device wherein the first outer flexibility is consistent along the first outer length and the second outer flexibility is variable along the second outer length, and wherein the first inner flexibility is consistent along the first inner length and the second inner flexibility is variable along the second inner length.

[0029] A device wherein the first outer length is less than the first inner length.

[0030] A device wherein the first outer flexibility is less than the first inner flexibility.

[0031] A device wherein the second outer flexibility is less than the second inner flexibility.

[0032] A device wherein an inner distal end of the first inner length axially aligns with an outer distal end of the first outer length, wherein the first inner length is greater than the first outer length such that the first inner length axially overlaps the first outer length.

[0033] A device wherein the second outer length has an outer distal end and an outer proximal end, wherein the second inner length has an inner distal end and an inner proximal end, and wherein the

outer distal end of the second inner length axially overlaps with the second outer distal end of the second outer length and a proximal end of a third outer length.

[0034] A device wherein the first inner segment has a constant pitch.

[0035] A device wherein the second inner segment has a variable pitch that increases from its distal end to its proximal end.

[0036] A device wherein the first outer segment has a constant pitch.

[0037] A device wherein the second outer segment has a variable pitch that increases from its distal end to its proximal end.

[0038] A device wherein the second outer segment has a variable angle that increases from its distal end to its proximal end.

[0039] The phrases “at least one”, “one or more”, and “and/or” are open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C”, “at least one of A, B, or C”, “one or more of A, B, and C”, “one or more of A, B, or C” and “A, B, and/or C” means A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B and C together. When each one of A, B, and C in the above expressions refers to an element, such as X, Y, and Z, or class of elements, such as X.sub.1-X.sub.n, Y.sub.1-Y.sub.m, and Z.sub.1-Z.sub.o, the phrase is intended to refer to a single element selected from X, Y, and Z, a combination of elements selected from the same class (e.g., X.sub.1 and X.sub.2) as well as a combination of elements selected from two or more classes (e.g., Y.sub.1 and Z.sub.o).

[0040] The term “a” or “an” entity refers to one or more of that entity. As such, the terms “a” (or “an”), “one or more” and “at least one” may be used interchangeably herein. It is also to be noted that the terms “comprising”, “including”, and “having” may be used interchangeably.

[0041] A “barrel cam cylinder”, which is sometimes referred to as a “cylindrical cam”, typically includes a groove (slot or channel) cut into the surface of a cylinder, and a follower, such as a pin, which rides in the groove. A barrel cam cylinder is generally used to convert rotational motion to linear motion parallel to the rotational axis of the cylinder or to convert linear motion, parallel to the axis of the cylinder, to rotational motion. For the purposes of this disclosure, unless otherwise specified, the barrel cam cylinder may refer to the cylinder and the follower.

[0042] A “kerf” is a slit. For example, in this disclosure a slit may be made in the inner and outer sheaths, which can be constructed of hypotubes. The kerf may be made by using a laser, which cuts a slit in the hypotube.

[0043] A “lead” is a conductive structure, typically an electrically insulated coiled wire. The electrically conductive material may be any conductive material, with metals and intermetallic alloys common. The outer sheath of insulated material is biocompatible and bio stable (e.g., non-dissolving in the body) and generally includes organic materials such as polyurethane and polyimide. Lead types include, by way of non-limiting example, epicardial and endocardial leads. Leads are commonly implanted into a body percutaneously or surgically.

[0044] The term “means” as used herein shall be given its broadest possible interpretation in accordance with 35 U.S.C. Section 112(f). Accordingly, a claim incorporating the term “means” shall cover all structures, materials, or acts set forth herein, and all of the equivalents thereof. Further, the structures, materials or acts and the equivalents thereof shall include all those described in the summary of the invention, brief description of the drawings, detailed description, abstract, and claims themselves.

[0045] A “serration” or “serrated edge” or “serrated blade” or other variations, as used herein, shall mean the configuration of a cutting surface having a notched edge or saw-like teeth. The notched edges create a plurality of smaller points that contact (and therefore less contact area with) the material being cut in comparison to an un-notched blade. Additionally, the pressure applied by each serrated point of contact is relatively greater and the points of contact are at a sharper angle to the material being cut. One example of a serrated blade may include one notch adjacent to and abutting another notch such that there is very little, if any, blade between such notches, thereby creating

points of contact. There are multiple variations and/or features of serrations. For example, one type of serrated feature is referred to as a “crown.” As used herein, a serrated blade, or other variation, in the shape of a “crown,” shall mean a blade comprising a plurality of notches and adjacent un-notched areas such that the combination of notched and un-notched areas resembles a crown for a royal member (e.g., king, queen, etc.), particularly when the blade is circular. A further type of “crown” includes a “hook crown.” As used herein, a serrated blade, or other variation, in the shape of a “hook crown,” shall mean a blade comprising of a plurality of notches and adjacent un-notched areas, wherein the length of the un-notched areas ascend to the next adjacent point at an angle to increase the slicing action in one rotary direction and the notches are created at an angle to create a hook feature at the points to promote engagement with the tissue at the hook-shaped point.

[0046] A “surgical implant” or “implanted object” is a medical device manufactured to replace a missing biological structure, support, stimulate, or treat a damaged biological structure, or enhance, stimulate, or treat an existing biological structure. Medical implants are man-made devices, in contrast to a transplant, which is a transplanted biomedical tissue. In some cases implants contain electronics, including, without limitation, artificial pacemaker, defibrillator, electrodes, and cochlear implants. Some implants are bioactive, including, without limitation, subcutaneous drug delivery devices in the form of implantable pills or drug-eluting stents.

[0047] “Vasculature” or “vascular system” is any part of the circulatory system, which includes the heart, blood, and blood vessels such as arteries, veins and capillaries.

[0048] It should be understood that every maximum numerical limitation given throughout this disclosure is deemed to include each and every lower numerical limitation as an alternative, as if such lower numerical limitations were expressly written herein. Every minimum numerical limitation given throughout this disclosure is deemed to include each and every higher numerical limitation as an alternative, as if such higher numerical limitations were expressly written herein. Every numerical range given throughout this disclosure is deemed to include each and every narrower numerical range that falls within such broader numerical range, as if such narrower numerical ranges were all expressly written herein.

[0049] The preceding is a simplified summary of the disclosure to provide an understanding of some aspects of the disclosure. This summary is neither an extensive nor exhaustive overview of the disclosure and its various aspects, embodiments, and configurations. It is intended neither to identify key or critical elements of the disclosure nor to delineate the scope of the disclosure but to present selected concepts of the disclosure in a simplified form as an introduction to the more detailed description presented below. As will be appreciated, other aspects, embodiments, and configurations of the disclosure are possible utilizing, alone or in combination, one or more of the features set forth above or described in detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0050] The accompanying drawings are incorporated into and form a part of the specification to illustrate several examples of the present disclosure. These drawings, together with the description, explain the principles of the disclosure. The drawings simply illustrate preferred and alternative examples of how the disclosure may be made and used and are not to be construed as limiting the disclosure to only the illustrated and described examples. Further features and advantages will become apparent from the following, more detailed, description of the various aspects, embodiments, and configurations of the disclosure, as illustrated by the drawings referenced below.

[0051] FIG. 1 is a perspective view of a human having a pacemaker lead located in the venous system and a terminating electrode anchored to the ventricular heart chamber, with an embodiment of a surgical device being shown inserted into the body and partly advanced over the lead;

[0052] FIG. 2 is an elevation view of an embodiment of a surgical device;

[0053] FIG. 3 is a cross-sectional view of a sheath assembly within a blood vessel with an extendable and rotatable blade for removing a lead according to an embodiment of the disclosure;

[0054] FIG. 4A is an internal view of an embodiment of a handle assembly of the surgical device illustrated in FIG. 2;

[0055] FIG. 4B is a perspective view of an embodiment of a trigger for the handle assembly illustrated in FIG. 4A;

[0056] FIG. 4C is an elevation view of an embodiment of the barrel cam cylinder for the handle assembly illustrated in FIG. 4A;

[0057] FIG. 4D is a cross-sectional view of an embodiment of the barrel cam cylinder illustrated in FIG. 4C;

[0058] FIG. 4E is an end view of an embodiment of the barrel cam cylinder illustrated in FIG. 4C;

[0059] FIG. 4F is an enlarged perspective view of an embodiment of a spool for the handle assembly illustrated in FIG. 4A;

[0060] FIG. 5A is an elevation view of the barrel cam cylinder illustrated in FIG. 4C in its home position;

[0061] FIG. 5B is an elevation view of the barrel cam cylinder illustrated in FIG. 4C upon being rotated to about 136.5 degrees in clockwise direction and/or counter-clockwise direction;

[0062] FIG. 5C is an elevation view of the barrel cam cylinder illustrated in FIG. 4C upon being rotated to about 273.1 degrees in clockwise direction;

[0063] FIG. 5D is an elevation view of the barrel cam cylinder illustrated in FIG. 4C upon being rotated to 307.6 degrees in a counter-clockwise direction and toward its home position;

[0064] FIG. 5E is an end view of the barrel cam cylinder illustrated in FIG. 4E including an indication of the amount of angular rotation for the cam cutter at each of the barrel cam cylinder positions illustrated in FIGS. 5A, 5B, 5C and 5D;

[0065] FIG. 6 is an elevation view of an embodiment of the sheath assembly;

[0066] FIG. 6A is a break-away, elevation view of an embodiment of the distal end of the sheath assembly illustrated in FIG. 6;

[0067] FIG. 6B is a break-away, elevation view of an embodiment of the proximal end of the sheath assembly illustrated in FIG. 6;

[0068] FIG. 7A is an elevation view of an embodiment of the outer sheath assembly;

[0069] FIG. 7B is an elevation view of an embodiment of the inner sheath assembly;

[0070] FIG. 8 is a cross-sectional view of an embodiment of the sheath assembly illustrated in FIG. 6;

[0071] FIG. 8A is an enlarged cross-sectional view of the distal end of the inner sheath assembly located within the outer sheath assembly illustrated in FIG. 8, wherein the blade is retracted and located within the outer sheath assembly;

[0072] FIG. 8A' is an enlarged cross-sectional view of the distal end of the inner sheath assembly located within the outer sheath assembly illustrated in FIG. 8, wherein the blade is extended and located outside the outer sheath assembly;

[0073] FIG. 8B is a distal end view of the inner sheath assembly located within the outer sheath assembly illustrated in FIG. 8;

[0074] FIG. 8C is an enlarged cross-sectional view of the inner key of the inner sheath assembly located within the outer key of the outer sheath assembly illustrated in FIG. 8;

[0075] FIG. 9A is a perspective view of an outer band member according to an embodiment of the disclosure;

[0076] FIG. 9B is an end view of the outer band member illustrated in FIG. 9A;

[0077] FIG. 9C is cross-sectional view of the outer band member illustrated in FIG. 9A taken along line 9C-9C of FIG. 9B;

[0078] FIG. 10A is a perspective view of a cutting tip according to an embodiment of the

disclosure;

[0079] FIG. **10B** is side view of the cutting tip illustrated in FIG. **10A**;

[0080] FIG. **10C** is end view of the cutting tip member illustrated in FIG. **10A**;

[0081] FIG. **10D** is cross-sectional view of the cutting tip illustrated in FIG. **10A** taken along line **10D-10D** in FIG. **10C**;

[0082] FIG. **11** is an illustration of the cam slot profile of the cutting tip and the cam slot of the barrel cam cylinder depicting the longitudinal position of the cutting tip in combination with the longitudinal position of the trigger for a particular amount of angular rotation by both the cutting tip and the barrel cam cylinder;

[0083] FIG. **12** is a perspective view of a human having a pacemaker lead located in the venous system and a terminating electrode anchored to the ventricular heart chamber, with an embodiment of a surgical device being shown inserted into the body and partly advanced over the lead;

[0084] FIG. **13** is an elevation view of an embodiment of a surgical device;

[0085] FIG. **14** is a cross-sectional view of a sheath assembly within a blood vessel with an extendable and rotatable blade for removing a lead according to an embodiment of the disclosure;

[0086] FIG. **15A** is an internal view of an embodiment of a handle assembly of the surgical device illustrated in FIG. **13**;

[0087] FIG. **15B** is a perspective view of an embodiment of a trigger for the handle assembly illustrated in FIG. **15A**;

[0088] FIG. **15C** is an elevation view of an embodiment of the barrel cam assembly for the handle assembly illustrated in FIG. **15A**;

[0089] FIG. **15D** is an elevation view of a barrel cam cylinder of the barrel cam assembly illustrated in FIG. **15C**;

[0090] FIG. **15E** is an illustration of the cam slot profile of the cam slot of the barrel cam cylinder illustrated in FIG. **15D**;

[0091] FIG. **15F** is a longitudinal-sectional view of the barrel cam cylinder illustrated in FIG. **15D**;

[0092] FIG. **15G** is a cross-sectional view of the barrel cam cylinder illustrated in FIG. **15D**;

[0093] FIG. **15H** is an elevation view of a follower guide of the barrel cam assembly illustrated in FIG. **15C**;

[0094] FIG. **15I** is an illustration of the aperture profile of the follower guide illustrated in FIG. **15H**;

[0095] FIG. **15J** is an end view of the barrel cam assembly of FIG. **15C** illustrating a relative rotation-inhibiting mechanism in a first relative rotation-inhibiting position;

[0096] FIG. **15K** is an end view of the barrel cam assembly of FIG. **15C** illustrating the relative rotation-inhibiting mechanism in a second relative rotation-inhibiting position;

[0097] FIG. **15L** is an enlarged perspective view of an embodiment of a spring assembly for the handle assembly illustrated in FIG. **15A**;

[0098] FIG. **16** is an elevation view of an embodiment of the sheath assembly;

[0099] FIG. **16A** is a break-away, elevation view of an embodiment of the distal end of the sheath assembly illustrated in FIG. **16**;

[0100] FIG. **16B** is a break-away, elevation view of an embodiment of the proximal end of the sheath assembly illustrated in FIG. **16**;

[0101] FIG. **17A** is an elevation view of an embodiment of the outer sheath assembly;

[0102] FIG. **17B** is an elevation view of an embodiment of the inner sheath assembly;

[0103] FIG. **18** is a cross-sectional view of an embodiment of the sheath assembly illustrated in FIG. **16**;

[0104] FIG. **18A** is an enlarged cross-sectional view of the distal end of the inner sheath assembly located within the outer sheath assembly illustrated in FIG. **18**, wherein the blade is retracted and located within the outer sheath assembly;

[0105] FIG. **18A'** is an enlarged cross-sectional view of the distal end of the inner sheath assembly

located within the outer sheath assembly illustrated in FIG. 18, wherein the blade is extended and located outside the outer sheath assembly;

[0106] FIG. 18B is a distal end view of the inner sheath assembly located within the outer sheath assembly illustrated in FIG. 18;

[0107] FIG. 18C is an enlarged cross-sectional view of the inner key of the inner sheath assembly located within the outer key of the outer sheath assembly illustrated in FIG. 18;

[0108] FIG. 19A is a perspective view of an outer band member according to an embodiment of the disclosure;

[0109] FIG. 19B is an end view of the outer band member illustrated in FIG. 19A;

[0110] FIG. 19C is cross-sectional view of the outer band member illustrated in FIG. 19A taken along line 19C-19C of FIG. 19B;

[0111] FIG. 20A is a perspective view of a cutting tip according to an embodiment of the disclosure;

[0112] FIG. 20B is side view of the cutting tip illustrated in FIG. 20A;

[0113] FIG. 20C is end view of the cutting tip member illustrated in FIG. 20A;

[0114] FIG. 20D is cross-sectional view of the cutting tip illustrated in FIG. 20A taken along line 20D-20D in FIG. 20C;

[0115] FIG. 21 depicts two-dimensional illustrations of a profile of a cam slot of an embodiment of a cutting tip, a profile of a cam slot of an embodiment of a barrel cam cylinder, and a profile of an aperture of an embodiment of a follower guide;

[0116] FIG. 22A is an illustration of the cam slot profile of the cutting tip, the cam slot of the barrel cam cylinder, and the profile of the aperture of the follower guide depicting the longitudinal position of the cutting tip in combination with the longitudinal position of the trigger for a particular amount of angular rotation by both the cutting tip and the barrel cam cylinder; the follower guide is illustrated in a first relative rotation-inhibiting position compared to the barrel cam cylinder;

[0117] FIG. 22B is another illustration of the cam slot profile of the cutting tip, the cam slot of the barrel cam cylinder, and the profile of the aperture of the follower guide depicting the longitudinal position of the cutting tip in combination with the longitudinal position of the trigger for a particular amount of angular rotation by both the cutting tip and the barrel cam cylinder; the follower guide is illustrated in a second relative rotation-inhibiting position compared to the barrel cam cylinder;

[0118] FIG. 22C is another illustration of the cam slot profile of the cutting tip, the cam slot of the barrel cam cylinder, and the profile of the aperture of the follower guide depicting the longitudinal position of the cutting tip in combination with the longitudinal position of the trigger for a particular amount of angular rotation by both the cutting tip and the barrel cam cylinder; the follower guide is again illustrated in the first relative rotation-inhibiting position compared to the barrel cam cylinder;

[0119] FIG. 23A is an elevation view of the barrel cam assembly illustrated in FIG. 15C in its first home position;

[0120] FIG. 23B is an elevation view of the barrel cam assembly illustrated in FIG. 15C upon being rotated away from its first home position;

[0121] FIG. 23C is an elevation view of the barrel cam assembly illustrated in FIG. 15C upon being rotated further away from its first home position;

[0122] FIG. 23D is an elevation view of the barrel cam assembly illustrated in FIG. 15C in its second home position;

[0123] FIG. 23E is an elevation view of the barrel cam assembly illustrated in FIG. 15C upon being rotated away from its second home position;

[0124] FIG. 23F is an elevation view of the barrel cam assembly illustrated in FIG. 15C upon being rotated further away from its second home position;

[0125] FIG. **24** is a perspective view of an embodiment of the barrel cam assembly for a surgical device;

[0126] FIG. **25** is an exploded perspective view of the barrel cam assembly of FIG. **24**;

[0127] FIG. **26** is a side view of a barrel cam cylinder of the barrel cam assembly of FIG. **24**;

[0128] FIG. **27** is a side view of a follower guide of the barrel cam assembly of FIG. **24**;

[0129] FIG. **28** is another side view of the follower guide of FIG. **24**;

[0130] FIG. **29** is a partial perspective view of an embodiment of the barrel cam assembly for a surgical device;

[0131] FIG. **30** is another partial perspective view of the barrel cam assembly of FIG. **29**;

[0132] FIG. **31** is a perspective view of an embodiment of the barrel cam assembly for a surgical device; a follower guide of the barrel cam assembly is translucent for illustrative purposes;

[0133] FIG. **32** is a perspective view of an embodiment of the barrel cam assembly for a surgical device; a follower guide of the barrel cam assembly is translucent for illustrative purposes;

[0134] FIG. **33** is a perspective view of an embodiment of the barrel cam assembly for a surgical device; hidden features are shown in light gray lines;

[0135] FIG. **34** is another perspective view of the barrel cam assembly of FIG. **33**;

[0136] FIG. **35** depicts two-dimensional illustrations of a profile of a cam slot of an embodiment of a cutting tip, a profile of a cam slot of an embodiment of a barrel cam cylinder, and a profile of an aperture of an embodiment of a follower guide;

[0137] FIG. **36A** is an illustration of the cam slot profile of the cutting tip, the cam slot of the barrel cam cylinder, and the profile of the aperture of the follower guide depicting the longitudinal position of the cutting tip in combination with the longitudinal position of the trigger for a particular amount of angular rotation by both the cutting tip and the barrel cam cylinder; the follower guide is illustrated in a first relative rotation-inhibiting position compared to the barrel cam cylinder;

[0138] FIG. **36B** is another illustration of the cam slot profile of the cutting tip, the cam slot of the barrel cam cylinder, and the profile of the aperture of the follower guide depicting the longitudinal position of the cutting tip in combination with the longitudinal position of the trigger for a particular amount of angular rotation by both the cutting tip and the barrel cam cylinder; the follower guide is illustrated in a second relative rotation-inhibiting position compared to the barrel cam cylinder;

[0139] FIG. **36C** is another illustration of the cam slot profile of the cutting tip, the cam slot of the barrel cam cylinder, and the profile of the aperture of the follower guide depicting the longitudinal position of the cutting tip in combination with the longitudinal position of the trigger for a particular amount of angular rotation by both the cutting tip and the barrel cam cylinder; the follower guide is again illustrated in the first relative rotation-inhibiting position compared to the barrel cam cylinder;

[0140] FIG. **37** depicts a two-dimensional illustration of a profile of a cam slot of an embodiment of a barrel cam cylinder;

[0141] FIG. **38** is an elevation view of an embodiment of a distal end of the sheath assembly;

[0142] FIG. **38A** is an enlarged cross-sectional view of the distal end of the sheath assembly in FIG. **38**, including the inner sheath assembly located within the outer sheath assembly illustrated, wherein the blade is retracted and located within the outer sheath assembly;

[0143] FIG. **39** is a break-away, elevation view of an embodiment of the distal end of the sheath assembly illustrated in FIG. **38** and FIG. **38A**;

[0144] FIG. **40** is an elevation view of an embodiment of a distal end of the outer sheath assembly;

[0145] FIG. **40A** is an enlarged cross-sectional view of the distal end of the outer sheath assembly in FIG. **40**;

[0146] FIG. **41** is an elevation view of an embodiment of a distal end of the inner sheath assembly;

[0147] FIG. **41A** is an enlarged cross-sectional view of the distal end of the inner sheath assembly

in FIG. 41;

[0148] FIG. 42 is an elevation view of a hypotube for use as an outer sheath;

[0149] FIG. 42A is a cross-sectional view of the hypotube taken along line A-A in FIG. 42;

[0150] FIG. 42B is an enlarged view of a segment of the hypotube taken about circle B in FIG. 42;

[0151] FIG. 43 is an elevation view of a hypotube for use as an inner sheath;

[0152] FIG. 43A is a cross-sectional view of the hypotube taken along line A-A in FIG. 43;

[0153] FIG. 43B is an enlarged view of a segment of the hypotube taken about circle B in FIG. 43;

[0154] FIG. 44 is an enlarged view of the hypotube of FIG. 42 and/or the hypotube of FIG. 43

depicting the angle (θ) of a kerf imparted within the respective hypotubes;

[0155] FIG. 45A is a block diagram depicting the axial alignment of segments of the outer sheath (outer hypotube) with respect to the inner sheath (inner hypotube);

[0156] FIG. 45B is an alternative block diagram depicting the axial alignment of segments of the outer sheath (outer hypotube) with respect to the inner sheath (inner hypotube);

[0157] FIG. 45C is another alternative block diagram depicting the axial alignment of segments of the outer sheath (outer hypotube) with respect to the inner sheath (inner hypotube);

[0158] FIG. 45D is a further alternative block diagram depicting the axial alignment of segments of the outer sheath (outer hypotube) with respect to the inner sheath (inner hypotube).

[0159] It should be understood that the drawings are not necessarily to scale. In certain instances, details that are not necessary for an understanding of the disclosure or that render other details difficult to perceive may have been omitted. It should be understood, of course, that the disclosure is not necessarily limited to the particular embodiments illustrated herein.

DETAILED DESCRIPTION

[0160] Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,”

“comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items.

[0161] Embodiments according to this disclosure provide a surgical device that includes a sheath assembly, which can be deployed safely within a vascular system of a patient and separate implanted objects, such as leads, from a patient's vasculature system. FIG. 1 depicts a surgical device 106 having a sheath assembly 112 inserted within an exemplary patient 104. The sheath assembly 112 surrounds an implanted lead (not shown) running along the left innominate vein past the SVC and connected into, or about, the right ventricle of the heart. Upon surrounding the lead with the sheath assembly 112, the user of the surgical device 106 may actuate the handle assembly 108, thereby rotating and extending a cutting blade (not shown) beyond the distal end of the sheath assembly 112 to dilate, separate and/or cut the tissue surrounding the lead within the patient's SVC.

[0162] The cutting blade may extend from and retract into the sheath multiple times upon actuation of the handle assembly according to the profile of the cam slot in the cutting tip disclosed below. The cutting blade may also rotate in both a clockwise and counter-clockwise direction per the profile of the cam slot in the barrel cam cylinder discussed below. When the clinician releases the handle assembly, the cutting blade is ensured to remain or return within the sheath assembly 112, thereby allowing the clinician to force and advance the distal portion of the sheath assembly against additional uncut tissue. The clinician repeats the actuation step, thereby causing the cutting blade to re-appear and extend beyond the distal end of the sheath assembly 112 to cut the adjacent tissue. Each time actuation occurs, the proximal portion of the implanted lead and/or surrounding tissue enters further into a hollow passageway within the sheath assembly 112. This process is again repeated until the implanted lead and/or surrounding tissue is completely or substantially dilated,

separated, and/or cut from the tissue attached to the SVC. At that time, the implanted lead may safely be removed from the patient's SVC.

[0163] With reference to FIG. 2, an exemplary surgical device **106** is depicted. The surgical device **106** includes a handle assembly **108** and a flexible sheath assembly **112**. The flexible sheath assembly **112**, which is discussed in more detail below, generally includes a flexible inner sheath assembly (not shown) located within a flexible outer sheath assembly. It may be preferable for the outer sheath to remain stationary while the inner sheath is capable of moving (e.g., rotating and extending) with respect to the outer sheath. The inner sheath and outer sheath can both be flexible, rigid or a combination thereof.

[0164] With reference to FIG. 4A, an exemplary handle assembly is depicted. The handle assembly **108** may include some or all of the following components: a handle frame **404**, a trigger **408**, a spring assembly **412**, a strain relief component **416**, a barrel cam cylinder **420**, a bushing **424** and an end cap **427**. The handle frame **404** may be constructed of a singular component or multiple component, such as two halves as illustrated in FIG. 4A.

[0165] Referring to FIG. 4B, an exemplary trigger **408** is illustrated. The trigger **408** depicted in FIG. 4A includes one opening **430** into which a clinician can insert his/her fingers. A trigger, however, may have more than one opening. Additionally, a trigger may also be comprised of a straight or non-linear member without any openings. Furthermore, a trigger may be in the shape of a button capable of being depressed. As long as the trigger, either alone or in conjunction with the handle frame, is ergonomically correct and comfortable for the clinician, the trigger may have a variety of sizes and shapes.

[0166] The trigger **408** illustrated in FIG. 4B includes a trigger pin **428** that extends vertically from the top of the trigger **408**. The trigger pin **428** may be formed of a metal, such as a copper alloy (for example, brass or bronze, particularly C **630** nickel aluminum bronze), and may include a frusto-conically shaped end to facilitate insertion into the handle frame **404**. The trigger pin **428**, which cooperates with the groove in the barrel cam cylinder **420**, acts as a follower for the barrel cam. The trigger **408** also includes a pair of sliders **432** protruding laterally from the proximal end of the trigger **408** and a pair of sliders **436** protruding laterally from the distal end of the trigger **408**. When the trigger **408** is located within the handle assembly **108**, the sliders **432**, **436** sit and slide in corresponding grooves within the handle frame **404**. The trigger **408** also includes a post **440** extending vertically from the top of trigger **408**, and preferably from the distal end of the top of the trigger **408**. The post **440** connects to spring assembly **412**.

[0167] The handle assembly **108**, including the trigger **408** and barrel cam cylinder **420** discussed above is an example of a mechanical actuation means to rotate the inner sheath assembly. In an alternate embodiment, the actuation means may comprise electromechanical components. For example, the actuation means may comprise an electric motor (not shown) having a driven shaft that is directly or indirectly coupled to the inner sheath, the barrel cam cylinder, the trigger pin, and/or any combination thereof. The motor's shaft may be indirectly coupled to the inner sheath by one or more gears discussed hereinbefore. The motor may be controlled by a switch, thereby causing the inner sheath to rotate in a clockwise and/or a counter-clockwise direction upon actuating a switch that may also act as the trigger. The electric motor may be either a direct current (DC) motor or an alternating current (AC) motor. Accordingly, the motor may be powered by a DC source, such as a battery, or an AC source, such as a conventional power cord. Additionally, those skilled in the art will appreciate that there are numerous other ways in which a surgical device comprising a rotatable sheath may be actuated and driven.

[0168] As mentioned above, the handle assembly **108** may include a strain relief component **416**. The strain relief component **416**, as illustrated in FIG. 4A, is attached to the distal end of the handle frame **404** and tapers from its proximal end toward its distal end. The strain relief component **416** also has a lumen passing through it, thereby allowing the sheath assembly **112** to extend there through and into the handle assembly **108**. The strain relief component **416** may be constructed of a

flexible material such as, Santoprene™ thermoplastic vulcanizate produced by ExxonMobil. The material from which the strain relief component is made and the shape of the strain relief component provide a flexural modulus to protect the flexible shaft as it extends the rigid handle. The lumen of the strain relief may also contain a counter bore that enables ancillary outer sheaths to be docked during device preparations.

[0169] Referring to FIGS. 4C, 4D and 4E, there is depicted an elevation view, cross-sectional view and end view of the barrel cam cylinder **420**, respectively. As illustrated in FIGS. 4C and 4D, the barrel cam cylinder **420** has an exterior surface comprising a cam slot (or channel) **444** that cooperates with the trigger pin **428** to create the barrel cam. The cam slot **444** may create a two dimensional linear and/or non-linear cam profile, which is discussed in further detail below. The barrel cam cylinder **420** has a proximal end **448** and a distal end **452** through which a lumen **456** extends.

[0170] FIG. 4E illustrates the end view of the distal end **452** of the barrel cam cylinder **420**. The distal end **452** of the lumen **456** of the barrel cam cylinder **420** is designed to mate with exterior of the proximal end of the inner key **612**, which is discussed in further detail below. The cross section of the distal end **452** of the lumen **456** of the barrel cam cylinder **420** is preferably non-circular. For example, one embodiment of a non-circular lumen includes two chamfered sides **464**, wherein one chamfered side **464** is not offset, and the other chamfered side **464** is offset (e.g., about 8 degrees). Because the distal end of the barrel cam cylinder **420** is designed to mate with exterior of the proximal end of the inner key **612** and transfer torque from the barrel cam cylinder **420** to the inner sheath assembly via the inner key **612**, the cross section of the exterior of proximal end of the inner key **612** will have a complimentary profile of the lumen **456**. Although the cross sectional shape of the non-circular lumen is described as having two chamfered sides **464**, the disclosure shall not be limited to such shape and may include alternative non-circular shapes, such as a square, rectangle, D-shape, triangle, rhombus, trapezoid, pentagon, hexagon, octagon, parallelogram, ellipse, etc. Alternatively, the inner key could couple to the outside of the barrel cam cylinder.

[0171] The proximal end of the barrel cam cylinder **420** mates with the bushing **424**. Specifically, the exterior, distal end of the bushing **424** is located within the proximal end of the lumen **456**. Both the exterior, distal end of the bushing **424** and the proximal end of the lumen **456** are circularly shaped, thereby allowing the bushing **424** and the barrel cam cylinder **420** to rotate with respect to one another. The proximal end of the exterior of the bushing **424**, however, is located within a groove within the handle frame **404**, thereby preventing the bushing **424** and the barrel cam cylinder **420** from moving longitudinally within the handle assembly **108**.

[0172] Referring to FIG. 4F, an exemplary spring assembly **412** is depicted. The spring assembly **412** includes a constant force spring **472** and a spool **474**. One end of the constant force spring **472** is connected to the spool **474**, and the other end of the constant force spring **472** is connected to the post **440** extending from the trigger **408**. As a clinician pulls the trigger **408** proximally, the sliders **432**, **436** travel and slide in the grooves within the handle frame **404**, thereby preventing the trigger **408** from moving vertically within the handle assembly **108** and only allowing the trigger **408** to move along the longitudinal axis of the surgical device **106** from its distal end toward its proximal end and/or vice versa. As the trigger **408** moves proximally, the constant force spring **472** uncoils, thereby creating tension and a distally directed force. Accordingly, when the trigger **408** is released by the clinician, the constant force spring **472** recoils and pulls the trigger **408** back towards its original and most distal position.

[0173] Referring to FIG. 6, there is depicted an elevation view of an embodiment of an assembled sheath assembly **112** of the present disclosure. The sheath assembly **112** includes an inner sheath assembly and an outer sheath assembly. Referring to FIG. 6A, which illustrates an exploded view of the distal end of the sheath assembly **112**, and referring to FIG. 6B, which is an exploded illustration of the proximal end and central portion of the sheath assembly **112**, the sheath assembly **112** may include some or all of the following components: an outer band **636**; a guide

pin **640**; a cutting tip **632**; a flexible inner sheath **620**; a flexible outer sheath **624**; an outer jacket **628**; an inner key **612**; an outer key **608**; and a rigid inner tube **616**.

[0174] Referring to FIG. 7A, there is depicted an embodiment of the outer sheath assembly **602** of the present disclosure. The outer sheath assembly **602** includes an outer band **636** located at and attached to the distal end of an elongated flexible outer sheath **624**, and an outer key **608** located at and attached to the proximal end of the flexible outer sheath **624**. The outer band **636** may be attached to the distal end of a flexible outer sheath **624** via a weld, an adhesive, a press-fitting technique, an interlock such as a barbed joint or other known means of attachment. All such attachment techniques within the knowledge of one skilled in the art are considered within the scope of this disclosure. Similarly, the outer key **608** may be attached to the proximal end of the flexible outer sheath **624** via a weld, an adhesive, a press-fitting technique, interlock such as a barbed joint, or other known means of attachment. Although it is not shown on FIG. 7A, the outer sheath assembly may also include a flexible outer jacket **628** that covers the outer sheath **624** and abuts the outer band **636**, thereby providing the outer sheath assembly with a relatively smooth, continuous and uninterrupted exterior profile. The flexible jacket also contains the egress of blood from the system.

[0175] Referring to FIG. 7B, there is depicted an embodiment of the inner sheath assembly **604** of the present disclosure. The inner sheath assembly **604** includes a cutting tip **632**, a flexible inner sheath **620**, an inner key **612**, and a rigid inner tube **616**. The proximal end of the cutting tip **632** is attached to the distal end of a flexible inner sheath **620**; the distal end of an inner tube **616** is attached to the proximal end of the flexible inner sheath **620**; and an inner key **612** is attached to the proximal end of the inner tube **616**. The means of attaching these components may include a weld, an adhesive, a press-fitting technique, or other known means of attachment. As will be discussed below, the guide pin **640** couples the outer band **636** with the cutting tip **632**, and the guide pin **640** may be included with either the inner sheath assembly **604** or the outer sheath assembly **602**.

[0176] It may be preferable for a portion of either the inner sheath **620** and/or the outer sheath **624** to be rigid and a portion of the outer sheath to be flexible. Both the rigid portion and the flexible portion may be constructed of materials suitable for insertion into the human body. For example, the rigid portion may be constructed of stainless steel, and the flexible portion may be constructed of a flexible polymer such as polytetrafluoroethylene or thermoplastic elastomers. Assuming that both a rigid portion and a flexible portion are used, they will form a unitary inner sheath and/or outer sheath. As depicted in FIG. 7B, the rigid inner tube **616** is not only attached to the inner key **612**, the rigid tube **616** also is inserted through the inner key **612** and extends from both the proximal end and distal end of the inner key **612**. The attachment and extension of the rigid tube **616** to the inner key **612** allows for an increased amount of torque that can be transferred from the barrel cam to the rigid tube **616** via the inner key **612** and eventually to the cutting tip **632** via the inner sheath assembly **604**. The extension of the rigid tube through the handle provides an access point for introduction of other medical devices. The extension also provides a means of controlling blood egress after the lead has been extracted.

[0177] It may be preferable that at least a portion of the outer sheath **624** and the inner sheath **620** be generally flexible in order to accept, accommodate and navigate the patient's vasculature system. In addition to being flexible, the inner sheath **620** may also have a high degree of stiffness in order to receive the torque transferred from the barrel cam cylinder/inner key and transfer sufficient torque to the cutting tip **632** discussed in more detail below. The inner sheath **620** (and/or the outer sheath **624**) may be formed of a polymer extrusion, braided reinforced polymer extrusion, coils, bi-coils, tri-coils, laser cut metal tubing and any combination of the above. The inner sheath (and/or the outer sheath **624**) may be a unitary structure comprised of multiple portions.

[0178] Referring to FIG. 8, there is depicted a cross-sectional view of an embodiment of the sheath assembly **112** comprising the inner sheath assembly **604** located within the outer sheath assembly

602. Referring to FIG. **8C**, there is depicted an enlarged view of the inner key **612** of the inner sheath assembly **604** located within the outer key **608** of the outer sheath assembly **602**. As discussed above, the exterior of the inner key **612** is designed to mate with lumen **456** of the distal end of the barrel cam cylinder **420**. Accordingly, the cross section of the exterior of proximal end of the inner key **612** will have a profile complimentary to the distal end of the lumen **456** within the barrel cam cylinder **420**. For example, assuming the cross section of the distal end **452** of the lumen **456** of the barrel cam cylinder **420** is non-circular and has two chamfered sides, wherein one chamfered side is not offset, and the other chamfered side is offset (e.g., about 8 degrees), then the exterior of the proximal end of the inner key **612** will also have a non-circular profile with two chamfered sides, wherein one chamfered side is not offset, and the other chamfered side is offset (e.g., about 8 degrees). The inner key **612** and outer key **608** provide means for rotationally coupling. The inner key **612** is a means for rotationally coupling the inner sheath assembly **604** to the barrel cam, and the outer key is a means for rotationally coupling the outer shaft assembly to the handle. The inner key **612** and outer key **608** provide journal bearing for the other key.

[0179] As further illustrated in FIG. **8C**, the inner key **612** is able to rotate freely within the outer key **608** due, at least in part, to the distal end of the exterior of the inner key **612** having a circular cross section that mates with a circular cross section of the proximal end of a lumen within the outer key **608**. Additionally, because the inner key **612** and outer key **608** are loosely coupled, the inner key **612** and outer key **608** are able to move longitudinally with respect to one another. For instance, supposing the outer key **608** is fixed such that it neither rotates nor moves longitudinally, the inner key **612** is able to both rotate and travel longitudinally within the outer key **608**.

Accordingly, as the barrel cam cylinder **420** rotates, the inner key **612** will rotate within the outer key **608**, and the inner sheath assembly **604** will rotate within the outer sheath assembly **602**, including the rotation of the cutting tip **632** within the outer band **636**. And the cam slot profile in the cutting tip **632** controls the longitudinal movement of the inner sheath assembly **604** within the outer sheath assembly **602**, including the longitudinal movement of the inner key **612** relative to the outer key **608** and the longitudinal movement of the cutting tip **632** relative to the outer band **636**.

[0180] Continuing to refer to FIG. **8C**, the lumen within the outer key **608** is larger toward its proximal end and smaller toward its distal end because there is a step down or an abutment in the lumen as it progresses from the proximal end to the distal end. Due to the transition from a larger lumen to a smaller lumen within the outer key **608**, there is depicted an adjustable gap **610** between the distal end of the inner key **612** and the abutment within the distal end of the larger lumen in the outer key **608**. This gap increases, decreases and/or remains the same according to the cam slot profile of the cutting tip **632**. The abutment in the outer key **608** insures that the inner key **612** will only travel a limited longitudinal distance within the outer key **608**, thereby limiting the inner sheath assembly **604** potential longitudinal movement within the outer sheath assembly **602**, including limiting the longitudinal movement of the cutting tip **632** relative to the outer band **636** in the distal direction.

[0181] Referring to FIG. **8A**, there is depicted an enlarged cross-sectional view of the distal end of the sheath assembly **112** with the inner sheath assembly **604** coupled with the outer sheath assembly **602** via guide pin **640**, wherein the blade **822** of the cutting tip **632** is in a retracted position and located within the outer sheath assembly **602**. As discussed above, the distal end of the outer sheath assembly **602** includes an outer band **636**, which may be constructed of a biocompatible metal, such as stainless steel, and polished so that it is generally smooth and evenly rounded at its most distal point, thereby allowing it to act as a dilator when pressed and forced against tissue. The distal end **822** of cutting tip **632** includes a cutting surface capable of cutting tissue. The inner sheath assembly **604** is coupled to the outer sheath assembly **602** through the cutting tip **632** and the outer band **636**, respectively, via guide pin **640**. One end of the guide pin **640** is fixed within the outer band **636**, and the other end of the guide pin **640** is located within the

cam slot **814** of the cutting tip **632**. As the inner sheath **620** rotates, upon actuation of the trigger assembly discussed above, the cutting tip **632** also rotates because the inner sheath **620** is fixedly attached to the cutting tip **632**. As the cutting tip **632** rotates, the cutting tip **632** may also extend distally in the direction of the arrow (.fwdarw.) according to the profile of the cam slot **814** as depicted in FIG. **8A'**. As the cutting tip **632** extends distally and rotates, the guide pin **640** and the outer sheath assembly **602**, particularly the outer band **636**, remain stationary. Thus, as the cutting tip **632** extends distally (and potentially retracts proximally according to the cam slot profile) and rotates, the cutting surface at the distal end **822** of the cutting tip **632** is able to perform a slicing action against the tissue and cut it.

[0182] Again, FIG. **8A** depicts the cutting tip **632** within a retracted (and potentially un-actuated) position because the cutting tip **632** is in a proximal position. Stated differently, the distal end **822** of the cutting tip **632** of FIG. **8A** is located within the interior of the outer sheath assembly **602**, particularly the outer band **636**, and does not extend beyond the distal end of the outer band **636**. With reference to FIG. **8A'**, the cutting tip **632** is depicted in an extended (and actuated) position because the cutting tip **632** is extending beyond the distal end of the outer sheath assembly **602** and the outer band **636**.

[0183] FIG. **3** depicts the distal portion of the flexible outer sheath and flexible inner sheath of FIG. **8A** surrounding a lead **330** within a patient's vein **334** with the cutting tip **632** in its extended position. The circumferential nature of the cutting surface (e.g., notched blade) at the distal end of the cutting tip **632** causes the surgical device to act as a coring device, thereby cutting tissue **338** either partially (i.e., less than 360 degrees) or completely (i.e., 360 degrees) around the lead or implanted object being extracted. The amount of tissue that the cutting surface cuts depends upon the size, shape and configuration of the lead, as well as the diameter and thickness of the circular cutting blade. For example, if the diameter of the circular cutting surface is substantially greater than the diameter of the lead, then the cutting surface will cut and core more tissue in comparison to a cutting surface having a smaller diameter. Once the desired cut has been made, the operator releases trigger and the cutting tip **632** (including the cutting surface) returns to a retracted position. Upon the cutting surface returning to a retracted position, the distal tip of the outer band **636** (and/or other portions of the outer sheath assembly) safely acts as a dilating device, thereby stretching tissue as the outer sheath assembly move over the lead or implanted object to be extracted.

[0184] With each full squeeze of the trigger, the cutting tip (and inner sheath) will rotate clockwise and counterclockwise while extending and retracting. The cutting tip (and inner sheath) retracts into the tip of the outer sheath when the trigger is fully compressed and/or remains retracted at the release of the trigger after a full squeeze of the trigger. If the trigger is partially squeezed, it will not reset and the cutting tip (and inner sheath) will reverse its motion upon release of the trigger, returning the blade to the retracted position. The return to the fully returned trigger position results in a rotation of about 35 degrees that due to the profile at the distal cam, which retains the cutting in the sheathed position.

[0185] Although the inner sheath and outer sheath are coupled to one another via the cutting tip, the outer band, and the guide pin, the inner sheath assembly and outer sheath assembly may be coupled to one another in other ways. Stated differently, those of skill in the art will understand how to make and use the disclosed aspects, embodiments, and/or configurations after understanding the present disclosure to couple the sheaths in a manner to allow a cutting surface to extend and rotate beyond the distal end of the outer sheath. All such configurations within the knowledge of one skilled in the art are considered within the scope of this disclosure.

[0186] With reference to FIGS. **9A**, **9B** and **9C**, an exemplary outer band **636** is depicted. The outer band **636** may be a sleeve in the general shape of a hollow cylinder. Although the exterior of the outer band **636** is non-uniform, it may be uniform. The interior of the outer band **636** is non-uniform. For example, the interior of the outer band **636** includes an abutment **916** to prevent the

cutting tip (not shown in FIGS. **9A**, **9B** and **9C**) from traveling further from the proximal end **912** to the distal end **908** within the outer band **636**. The outer band **636** also includes a hole **904** for receipt and possible attachment of a guide pin (not shown in FIGS. **9A**, **9B** and **9C**) which protrudes radially inward. As discussed in more detail above, the guide pin engages the cam slot of the cutting tip. The size, shape and configuration of the outer band **636** may differ depending upon how it is attached to the flexible outer sheath. As discussed above, the outer sheath may be stationary. If so, the outer band **636** and the guide pin remain stationary as the cutting tip moves (e.g., rotates and travel longitudinally) relative thereto. The outer band may also contain a journal bearing surface to align the cutting blade during actuation and provide a surface to disengage the tissue at the retraction of the cutting blade within the device.

[0187] With reference to FIGS. **10A**, **10B**, **10C** and **10D**, an exemplary cutting tip **632** is depicted. The cutting tip **632** has a generally hollow cylindrical shape. The cutting tip **632** comprises a proximal portion **1024**, an intermediate portion **1028**, and a distal portion **1032**. The outside diameter of the proximal portion **1024** is sized to allow it to be inserted to and/or engage (or otherwise attached to) the interior diameter of the inner flexible sheath (not shown in FIGS. **10A**, **10B**, **10C** and **10D**). The distal end of cutting tip **632** comprises a cutting surface **1012** having a serrated, sharp blade profile. The intermediate portion **1028** comprises a channel (or cam slot) **1016** cut within its exterior surface. As the inner flexible sheath rotates and moves within the outer sheath—from its proximal end to distal end—the outer sheath and pin may remain stationary. If so, the inner sheath (not shown), which is connected to cutting tip **632**, forces the cutting tip **632** to rotate. The cam slot **1016** engages the guide pin, and the shape and profile of the cam slot **1016** controls the rate and distance with which the cutting tip **632** travels longitudinally. That is, the configuration of the cam slot **1016** controls the cutting tip's direction and amount of longitudinal travel, such as moving distally toward an extended position and/or proximally toward a retracted position, while the cutting tip **632** rotates in either a clockwise or counter-clockwise direction.

[0188] Referring again to FIGS. **10A**, **10B**, **10C** and **10D**, the cutting tip **632** may also comprise a step up **1020** such that the diameter of the intermediate portion **1028** is greater than the distal portion **1032**. As the cutting tip **632** rotates, and the cutting surface **1012** extends beyond the distal end of the outer band into an extended position, the step up **1020** of the cutting tip **632** contacts the abutment of the outer band, thereby limiting the distance that the cutting tip **632** may travel and/or may prevent the cutting tip **632** from exiting or extending beyond the distal tip of the outer sheath assembly, particularly the outer band, in the event that the pin is sheared.

[0189] The profile of the cam slot in the cutting tip may have various configurations, such as those disclosed in U.S. patent application Ser. No. 13/834,405 filed Mar. 15, 2013 and entitled Retractable Blade For Lead Removal Device, which is hereby incorporated herein by reference in its entirety for all that it teaches and for all purposes. For example, the cam slot may have a substantially linear profile, a substantially sinusoidal profile, or a combination of individual and/or multiple linear and non-linear profiles. Additionally, the cam slot may have an open and continuous configuration, thereby allowing the cutting tip to continuously rotate, or the cam slot may have a closed and discontinuous configuration such that when the cutting tip reaches its fully extended position, the trigger assembly must be released or reversed so that the cutting tip returns to initially retracted position before being re-actuated. For instance, the cam slot **1016** in FIG. **10A** is discontinuous because the cam slot does not travel around the entire circumference of the exterior of the cutting tip **632**. Although certain figures in this disclosure only illustrate either the open or closed cam slot configuration, either configuration may be used with any of the inner cam embodiments disclosed and/or discussed herein and are considered within the scope of this disclosure. Furthermore, various types of cams, such as a partial lobe cam (which includes a cam slot surrounding less than 360 degrees of the circumference of the exterior surface of the cutting tip), a single lobe cam (which includes a cam slot surrounding 360 degrees of the circumference of the exterior surface of the cutting tip), double lobe cams (which includes a cam slot surrounding

720 degrees of the circumference of the exterior surface of the cutting tip) and/or other multiple lobe cams.

[0190] The distal end of cutting tip **632** may comprise a cutting surface **1012** having different blade profiles, such as those disclosed in U.S. patent application Ser. No. 13/834,405 filed Mar. 15, 2013 and entitled Retractable Blade For Lead Removal Device, which is hereby incorporated herein by reference in its entirety for all that it teaches and for all purposes. For example, the plane of the cutting surface **1012** of the distal end of the cutting tip depicted in the figures of this disclosure is parallel to the plane of the proximal end of the cutting tip. The plane of the cutting surface, however, may be offset (0 degrees to 90 degrees) from the plane of the proximal end of the cutting tip. Also, as discussed above, the profile of the cutting surface **1012** in FIGS. **10A-10D** has a plurality of serrations. The profile of the cutting surface **1012** need not be serrated and may comprise other configurations, such as a constant and/or smooth sharp profile. The profile of the cutting surface **1012** in FIGS. **10A-10D** has a plurality of six (6) serrations. However, it may be preferable to have less than or more than six (6) serrations. It may also be preferable to have between five (5) and seven (7) serrations, or between four (4) and eight (8) serrations, or between six (6) and ten (10) serrations.

[0191] Although the cutting surface **1012** illustrates a certain number of serrations, FIGS. **10A-10D** are not intended to represent the only number and type of serrations that may be included in a serrated cutting surface. Depending upon the size of the surgical device, including the sheaths, and cutting tip, those of skill in the art will understand how to make and use the disclosed aspects, embodiments, and/or configurations after understanding the present disclosure to adjust the number, size and configurations of the serrations. All such configurations within the knowledge of one skilled in the art are considered within the scope of this disclosure. Furthermore, the serrations may comprise a myriad of different shapes and configurations, including but not limited to any variation of a square, rectangle, rhombus, parallelogram, trapezoid, triangle, circle, ellipse, kite, etc.

[0192] As discussed above, FIGS. **10A**, **10B** and **10D** depict the intermediate portion **1028** of the cutting tip **632** having a cam slot (or channel) **1016** cut within its exterior surface, and FIGS. **4C** and **4D** depict the barrel cam cylinder **420** having a channel (or cam slot) **444** on its exterior surface that creates a non-linear cam profile. Referring to FIG. **11** there is depicted a two-dimensional illustration of the profile of the cam slot **1016** for the cutting tip **632** at the top of the figure and a two-dimensional illustration of the profile of the cam slot **444** for the barrel cam cylinder **420** at the bottom of the figure. The horizontal axis, which is the same for the top illustration and the bottom illustration, is the degree(s) of rotation of the cutting tip **632** and the barrel cam cylinder **420**. For example, assuming that the profile of the cam slot **1016** in the cutting tip **632** is discontinuous, as depicted in FIG. **10A**, the cutting tip **632** will rotate less than 360 degrees. It may be preferable for the cutting tip **632** to rotate between 5 and 355 degrees, 180 degrees and 355 degrees, 210 degrees and 325 degrees, 240 degrees and 295 degrees, or 270 degrees and 275 degrees. It may also be preferable for the cutting tip **632** rotate about 180, 185, 190, 200, 205, 210, 215, 220, 225, 230, 235, 240, 245, 250, 255, 260, 265, 270, 275, 280, 285, 290, 300, 305, 310, 315, 320, 325, 330, 335, 340, 345, 350 or 355 degrees. The first half of a trigger pull results in about a 2732 rotation in one direction-clockwise when looking from the handle to the tip-thereby returning the cam blade to the sheathed position. The second half of the trigger pull results in about a 2732 in the opposite direction-counter-clockwise when looking from the handle to the tip-thereby, returning the cutting blade to the sheathed position again. The blade remains in the sheathed position for the full return of the trigger to the forward position. The vertical axis for the top illustration is the amount of longitudinal movement, if any, of the cutting tip **632** including its cutting surface. The vertical axis for the bottom illustration is the amount of longitudinal displacement (in inches) of the trigger assembly (and trigger pin).

[0193] Referring to FIG. **11** in combination with FIGS. **5A-5E**, the following discussion explains the interaction between the rotation of the barrel cam cylinder **420**, the rotation of the cutting tip

632, the longitudinal movement of the handle (via the position of its trigger pin **428**), and the longitudinal movement of the cutting tip **632**. The following is a description of the positions CT1-CT6 of the guide pin **640** within the cam slot of the cutting tip **632**: [0194] CT1—the guide pin **640** is at its home position within the cam slot **1016** of the cutting tip **632**, and the cutting tip **632** is in a retracted position within the outer sheath assembly **602** (including the outer band **636**); [0195] CT2—the cutting tip **632** has rotated in a clockwise direction over the guide pin **640** within the cam slot **1016** for about half of its predetermined rotation, and the cutting tip **632** is in its most extended position outside the outer sheath assembly **602**; [0196] CT3—the cutting tip **632** has completed its rotation in a clockwise direction over the guide pin **640** within the cam slot **1016**, and the cutting tip **632** is in a retracted position within the outer sheath assembly **602**; [0197] CT4—the cutting tip **632** has rotated in a counter-clockwise direction over the guide pin **640** within the cam slot **1016** for about half of its predetermined rotation, and the cutting tip **632** is in its most extended position outside the outer sheath assembly **602**; [0198] CT5—(not shown) the cutting tip **632** has completed its rotation in a counter-clockwise direction over the guide pin **640** within the cam slot **1016**, and the cutting tip **632** is in a retracted position within the outer sheath assembly **602**; and [0199] CT6—the cutting tip **632** has completed its rotation in a counter-clockwise direction over the guide pin **640** within the cam slot **1016**, and the cutting tip **632** is in its most retracted position within the outer sheath assembly **602**.

[0200] The positions CT1-CT6 of the guide pin **640** within the cam slot of the cutting tip **632** correspond with positions BC1-BC6 of the trigger pin **428** within the cam slot of the barrel cam cylinder **420**. The following is a description of the positions BC1-BC6 of the trigger pin **428** within the cam slot of the barrel cam cylinder **420**: [0201] BC1—the trigger pin **428** (along with the trigger **408** of the trigger assembly **106**) is at its home position within the cam slot **444** of the barrel cam cylinder **420**; [0202] BC2—the trigger pin **428** has moved longitudinally in a proximal direction, thereby causing the barrel cam cylinder **420** to rotate in a clockwise direction; at this point, the barrel cam cylinder has rotated clockwise about half of its predetermined amount; [0203] BC3—the trigger pin **428** has moved about half of its longitudinal movement is continuing to move longitudinally in a proximal direction and the barrel cam cylinder **420** has completed its rotation in a clockwise direction; [0204] BC4—the trigger pin **428** is moving longitudinally in a proximal direction, thereby causing the barrel cam cylinder **420** to rotate in a counter-clockwise direction; at this point, the barrel cam cylinder has rotated counter-clockwise about half of its predetermined amount; [0205] BC5—the trigger pin **428** has moved about its entire longitudinal movement in a proximal direction and the barrel cam cylinder **420** has completed its rotation in a counter-clockwise direction; and [0206] BC6—the trigger pin **428** has moved longitudinally in a distal direction, thereby causing the barrel cam cylinder **420** to rotate any remaining amount in a counter-clockwise direction.

[0207] Continuing to refer to FIG. **11** and FIGS. **5A-5E**, when the trigger assembly **106**, particularly the trigger **408**, is at its initial, distal position, the trigger pin **428** is at its home position (BC1). Referring to the top illustration of FIG. **11**, at the time the trigger pin **428** is at its home position (BC1), the guide pin **640** in the sheath assembly **112** is at its initial position (CT1), and the cutting tip **632** is at a retracted (or recessed) position within the outer sheath assembly **602**. Upon a clinician pulling the trigger **408** and moving the trigger pin **428** proximally, both the barrel cam cylinder **420** and the cutting tip **632** rotate in a clockwise direction (from the proximal perspective of barrel cam cylinder). Upon the cutting tip **632** rotating adjacent the guide pin **640** from position CT1 to CT2, the profile of the cam slot in the cutting tip **632** causes the cutting tip **632** to move longitudinally in a distal direction from a retracted position to an extended position. When the trigger pin **428** is at position BC2, (i) the barrel cam cylinder **420** and the cutting tip **632** have rotated about half of its predetermined allowable rotation in the clockwise direction, (ii) the guide pin **640** is at position CT2, and (iii) the cutting tip **632** is at its most extended position.

[0208] As the clinician continues to pull the trigger, the trigger pin **428** continues to move

proximally, and the barrel cam cylinder **420** and the cutting tip **632** continue to rotate in a clockwise direction. Specifically, the cutting tip **632** rotates adjacent the guide pin **640**, and the profile of the cam slot in cutting tip **632** causes the cutting tip **632** to move longitudinally from position CT2, which is an extended position, to CT3, which is a recessed position. When the trigger pin **428** is at position BC3, (i) the barrel cam cylinder **420** and the cutting tip **632** have rotated about half of their predetermined allowable rotation in the clockwise direction, (ii) the guide pin **640** is at position CT3, and (iii) the cutting tip **632** is at a retracted position within the outer sheath assembly **602** (including the outer band **636**).

[0209] Referring to the top illustration of FIG. **11**, the cam slot in the cutting tip **632** extends beyond position CT3. As discussed above, the inner sheath assembly and outer sheath assembly may both be flexible. In order to accommodate for the potentially additional length created by the flexing of the sheath assemblies, as well as accommodating for manufacturing tolerances, the cam slot **1016** in the cutting tip **632** extends beyond position CT3. For example, if during use of the surgical device, the home position of the guide pin **640** is slightly to the right of position CT1, rather than exactly at position CT1, then the extended length of the cam slot allows the guide pin **640** to travel to the right of position CT3, thereby allowing the cutting tip to rotate its total amount of allowable rotation in the clockwise direction without obstruction.

[0210] As the trigger pin **428** moves from position BC1 to BC3 in the barrel cam cylinder **420**, and the barrel cam cylinder **420** rotates in a clockwise direction, the trigger pin **428** rides along the inside edge of the cam slot **444** in the barrel cam cylinder **420**. However, when the trigger pin **428** moves from position BC3 to BC5 in the barrel cam cylinder, the barrel cam cylinder **420** rotates in a counter-clockwise direction, and the trigger pin **428** rides along the outside edge of the cam slot **444** in the barrel cam cylinder **420**.

[0211] When the guide pin **640** reaches position CT3 in the cutting tip **632** and the trigger pin **428** reaches position BC3 in the barrel cam cylinder **420**, the trigger assembly **106**, particularly the trigger **408**, has only travelled about half of its predetermined allowable distance in the longitudinal direction. As the user continues to pull the trigger assembly **106**, the trigger **408** and trigger pin **428** continue to move proximally. As this occurs, the barrel cam cylinder **420** and the cutting tip **632** switch from rotating in a clockwise direction to rotating in a counter-clockwise direction. And because the cutting tip **632** switches from rotating in a clockwise direction to a counter-clockwise direction as the cutting tip **632** moves past position CT3 to CT4 adjacent the guide pin **640**, the cutting tip **632** moves from a retracted position within the outer sheath assembly **602** (including the outer band **636**) to an extended position or partially extended position outside the outer sheath assembly **602** (including the outer band **636**). When the trigger pin **428** is at position BC4, (i) the barrel cam cylinder **420** and the cutting tip **632** have rotated counter-clockwise for slightly less than half (or half) of its predetermined allowable distance, (ii) the guide pin **640** is at position CT4, and (iii) the cutting tip **632** is at its most extended position.

[0212] As the user pulls the trigger **408** further, the trigger pin **428** continues to move proximally from position BC4 to position BC5 in the barrel cam cylinder **420**, thereby causing the barrel cam cylinder **420** and cutting tip **632** to continue rotating in a counter-clockwise direction. Specifically, the cutting tip **632** rotates adjacent the guide pin **640** from position CC4 toward CC6, and the profile of the cam slot in the cutting tip **632** causes the cutting tip **632** to move from an extended position to a retracted position. When the trigger pin **428** is located at position BC5 in cam slot of the barrel cam cylinder **420**, the trigger **408** has reached the end of its longitudinal movement in the proximal direction. Upon the trigger pin **428** surpassing position BC5 in the cam slot of the barrel cam cylinder **420**, the constant force spring causes the trigger **408** and trigger pin **428** to reverse direction and travel toward its distal position.

[0213] As discussed above, for a discontinuous cam slot in the cutting tip **432**, the cutting tip **432** rotates less than 360 degrees in either the clockwise or counter-clockwise direction. Assuming that the predetermined amount of allowable rotation is about 275 degrees, the amount of angular

rotation by the barrel cam cylinder **420** in the clockwise direction from BC1 to BC3 and by the cutting tip **632** from CT1 to CT3 is about 275 degrees. The amount of angular rotation by the barrel cam cylinder **420** in the counter-clockwise direction from BC3 to BC5 and the cutting tip **632** from CT3 to CT3 is greater than 275 degrees by about nine degrees. This additional rotation (or over rotation) by the barrel cam cylinder **420** and the cutting tip **632** in the counter-clockwise direction ensures that the cutting tip **632**, including its cutting surface, is covered by the outer sheath assembly **602**, particularly the outer band **636**. Upon the trigger **408** reaching the end of its longitudinal movement in the proximal direction at position BC5, the barrel cam cylinder **420** and cutting tip **632** continue to move in a counter-clockwise direction to position BC6 and position CT6, respectively. Specifically, barrel cam cylinder **420** rotates counter-clockwise from position BC5 to BC6 about 17 degrees, thereby causing the cutting tip **632** to rotate counter-clockwise the same amount from position CT5 (not shown) to position CT6, which is the cutting tip's most recessed position.

[0214] When trigger pin **408** is at position BC6 in the barrel cam cylinder **420** and the guide pin **640** is at position CT6 within the cutting tip **632**, the barrel cam cylinder **420** and cutting tip **632** still need to return to their home positions BC1, CT1. In order for the barrel cam cylinder **420** and cutting tip **632** to return to their home positions BC1, CT1, the barrel cam cylinder **420** and cutting tip **632** rotate about 34.5 degrees in the clockwise direction from position BC6 to BC1 and from CT6 to CT1. When the trigger pin **428** is back to its home position (BC1), (i) the barrel cam cylinder **420** and cutting tip **632** have rotated counter-clockwise 307.6 degrees (and rotated clockwise 34.5 degrees), (ii) the guide pin **640** is at position CT1, and (iii) the cutting tip **632** is at a recessed position. That is, the cutting tip **632** (and barrel cam cylinder **420**) have rotated a net 273.1 degrees in the clockwise direction in a retracted(home)-extended-retracted sequence of positions and 273.1 degrees in the counter-clockwise direction a retracted-extended-retracted(home) sequence of positions, even though the cutting tip **632** (and barrel cam cylinder **420**) rotated in both a counter-clockwise direction and a clockwise direction in a retracted-extended-retracted (home) sequence of positions. The user may then repeat the process, if so desired.

[0215] Embodiments according to this disclosure provide a surgical device that includes a sheath assembly, which can be deployed safely within a vascular system of a patient and separate implanted objects, such as leads, from a patient's vasculature system. FIG. 12 depicts a surgical device **1206** having a sheath assembly **1212** inserted within an exemplary patient **1204**. The sheath assembly **1212** surrounds an implanted lead (not shown) running along the left innominate vein past the SVC and connected into, or about, the right ventricle of the heart. Upon surrounding the lead with the sheath assembly **1212**, the user of the surgical device **1206** may actuate the handle assembly **1208**, thereby rotating and extending a cutting blade (not shown) beyond the distal end of the sheath assembly **1212** to dilate, separate and/or cut the tissue surrounding the lead within the patient's SVC.

[0216] The cutting blade may extend from and retract into the sheath upon actuation of the handle assembly according to the profile of the cam slot in the cutting tip disclosed below. The cutting blade may rotate in a first, or clockwise, direction upon an initial, or first, actuation of the handle assembly per the profile of the cam slot in the barrel cam cylinder discussed below. When the clinician releases the handle assembly, the cutting blade is ensured to remain or return within the sheath assembly **1212**, thereby allowing the clinician to force and advance the distal portion of the sheath assembly against additional uncut tissue. The cutting blade may rotate in a second, or counter-clockwise, direction upon a subsequent, or second, actuation of the handle assembly per the profile of the cam slot in the barrel cam cylinder discussed below. Each time actuation occurs, the proximal portion of the implanted lead and/or surrounding tissue enters further into a hollow passageway within the sheath assembly **1212**. This process is again repeated until the implanted lead and/or surrounding tissue is completely or substantially dilated, separated, and/or cut from the tissue attached to the SVC. At that time, the implanted lead may safely be removed from the

patient's SVC.

[0217] With reference to FIG. 13, an exemplary surgical device **1206** is depicted. The surgical device **1206** includes a handle assembly **1208** and a flexible sheath assembly **1212**. The flexible sheath assembly **1212**, which is discussed in more detail below, generally includes a flexible inner sheath assembly (not shown) located within a flexible outer sheath assembly. It may be preferable for the outer sheath to remain stationary while the inner sheath is capable of moving (e.g., rotating and extending) with respect to the outer sheath. The inner sheath and outer sheath can both be flexible, rigid or a combination thereof.

[0218] With reference to FIG. 15A, an exemplary handle assembly **1208** is depicted. The handle assembly **1208** may include some or all of the following components: a handle frame **1504**, a trigger **1508**, a spring assembly **1512**, a strain relief component **1516**, a barrel cam assembly **1519**, a bushing **1524** and an end cap **1527**. The handle frame **1504** may be constructed of a singular component or multiple component, such as two halves as illustrated in FIG. 15A.

[0219] Referring to FIG. 15B, an exemplary trigger **1508** is illustrated. The trigger **1508** depicted in FIG. 15B includes one opening **1530** into which a clinician can insert his/her fingers. A trigger, however, may have more than one opening. Additionally, a trigger may also be comprised of a straight or non-linear member without any openings. Furthermore, a trigger may be in the shape of a button capable of being depressed. As long as the trigger, either alone or in conjunction with the handle frame, is ergonomically correct and comfortable for the clinician, the trigger may have a variety of sizes and shapes.

[0220] The trigger **1508** illustrated in FIG. 15B includes a trigger pin **1528** that extends vertically from the top of the trigger **1508**. The trigger pin **1528** may be formed of a metal, such as a copper alloy (for example, brass or bronze, particularly C **630** nickel aluminum bronze), and may include a frusto-conically shaped end to facilitate insertion into the handle frame **1504**. The trigger pin **1528**, which cooperates with a groove in a barrel cam cylinder of the barrel cam assembly **1519**, acts as a follower for the barrel cam cylinder. The trigger **1508** also includes a pair of sliders **1532** protruding laterally from the proximal end of the trigger **1508** and a pair of sliders **1536** protruding laterally from the distal end of the trigger **1508**. When the trigger **1508** is located within the handle assembly **1208**, the sliders **1532**, **1536** sit and slide in corresponding grooves within the handle frame **1504**. The trigger **1508** also includes a post **1540** extending vertically from the top of trigger **1508**, and preferably from the distal end of the top of the trigger **1508**. The post **1540** connects to the spring assembly **1512**.

[0221] As mentioned above, the handle assembly **1208** may include a strain relief component **1516**. The strain relief component **1516**, as illustrated in FIG. 15A, is attached to the distal end of the handle frame **1504** and tapers from its proximal end toward its distal end. The strain relief component **1516** also has a lumen passing through it, thereby allowing the sheath assembly **1212** to extend there through and into the handle assembly **1208**. The strain relief component **1516** may be constructed of a flexible material such as, Santoprene™ thermoplastic vulcanizate produced by ExxonMobil. The material from which the strain relief component is made and the shape of the strain relief component provide a flexural modulus to protect the flexible shaft as it extends the rigid handle. The lumen of the strain relief may also contain a counter bore that enables ancillary outer sheaths to be docked during device preparations.

[0222] Referring to FIG. 15C, an exemplary barrel cam assembly **1519** is illustrated. The barrel cam assembly **1519** includes a barrel cam cylinder **1520** that rotatably carries a follower guide **1521**. As described in further detail below, the barrel cam cylinder **1520** and the follower guide **1521** cooperate with the trigger pin **1528** to create the barrel cam.

[0223] The barrel cam cylinder **1520** may be formed from one or more biocompatible materials, such as polyethylene-filled Delrin®, stainless steel, anodized aluminum, brass, titanium, or the like. As illustrated in FIG. 15D, the barrel cam cylinder **1520** has an exterior surface comprising a cam groove (or slot or channel) **1544** that translatably receives the trigger pin **1528**. FIG. 15E depicts a

two-dimensional illustration of the profile of the cam slot **1544** for the barrel cam cylinder **1520**. The cam slot **1544** defines a generally “hourglass”-like or “figure eight”-like path for the follower (that is, the trigger pin **1528**). As described in further detail below, the trigger pin **1528** traverses about half of the cam slot **1544** when an initial, or first, actuation is applied to the trigger **1508**, and the trigger pin **1528** traverses the remainder of the cam slot **1544** (that is, about half of the cam slot **1544**) when a subsequent, or second, actuation is applied to the trigger **1508**. In each case, and as described in further detail below, the follower guide **1521** causes the trigger pin **1528** to travel straight through the intersection (or crossing portion) **1545** of the cam slot **1544** during each actuation of the trigger **1508**. Stated another way, the follower guide **1521** causes the trigger pin **1528** to travel from a first leg **1547** of the cam slot **1544** to a second parallel leg **1549** of the cam slot **1544**, and then from a third leg **1551** of the cam slot **1544** to a fourth parallel leg **1553** of the cam slot **1544**.

[0224] FIGS. **15F** and **15G** illustrate a longitudinal-sectional view and a cross-sectional view, respectively, of the barrel cam cylinder **1520**. The barrel cam cylinder **1520** has a proximal end **1548** and a distal end **1552** through which a lumen **1556** extends. The distal end **1552** of the lumen **1556** of the barrel cam cylinder **1520** is designed to mate with exterior of the proximal end of the inner key **1612**, which is discussed in further detail below. The cross section of the distal end **1552** of the lumen **1556** of the barrel cam cylinder **1520** is preferably non-circular. For example, one embodiment of a non-circular lumen includes two chamfered sides **1564**, wherein one chamfered side **1564** is not offset, and the other chamfered side **1564** is offset (e.g., about 8 degrees). Because the distal end of the barrel cam cylinder **1520** is designed to mate with exterior of the proximal end of the inner key **1612** and transfer torque from the barrel cam cylinder **1520** to the inner sheath assembly via the inner key **1612**, the cross section of the exterior of proximal end of the inner key **1612** will have a complimentary profile of the lumen **1556**. Although the cross sectional shape of the non-circular lumen is described as having two chamfered sides **1564**, the disclosure shall not be limited to such shape and may include alternative non-circular shapes, such as a square, rectangle, D-shape, triangle, rhombus, trapezoid, pentagon, hexagon, octagon, parallelogram, ellipse, etc. Alternatively, the inner key could couple to the outside of the barrel cam cylinder.

[0225] The proximal end of the barrel cam cylinder **1520** mates with the bushing **1524**. Specifically, the exterior, distal end of the bushing **1524** is located within the proximal end of the lumen **1556**. Both the exterior, distal end of the bushing **1524** and the proximal end of the lumen **1556** are circularly shaped, thereby allowing the bushing **1524** and the barrel cam cylinder **1520** to rotate with respect to one another. The proximal end of the exterior of the bushing **1524**, however, is located within a groove within the handle frame **1504**, thereby preventing the bushing **1524** and the barrel cam cylinder **1520** from moving longitudinally within the handle assembly **1208**.

[0226] The follower guide **1521** may be formed from one or more biocompatible materials, such as stainless steel, anodized aluminum, titanium, or the like. In some embodiments, the follower guide **1521** and the barrel cam cylinder **1520** have a relatively high coefficient of friction therebetween. The follower guide **1521** is rotatably carried by the barrel cam cylinder **1520**; as such, it may be preferable for the follower guide **1521** to be a dissimilar material from the barrel cam cylinder **1520** to inhibit galling. In some embodiments, the inner surface of the follower guide **1521** may have a slightly different cross-sectional shape than that of the outer surface of the barrel cam cylinder **1520** to inhibit unintentional rotation of the follower guide **1521** relative to the barrel cam cylinder **1520**. For example, the outer surface of the barrel cam cylinder **1520** may have a circular cross-sectional shape, and the inner surface of the follower guide **1520** may have a slightly non-circular cross-sectional shape.

[0227] As illustrated in FIG. **15H**, the follower guide **1521** is a generally cylindrical component that includes an aperture **1565**. The trigger pin **1528** extends through the aperture **1565** to enter the cam slot **1544** of the barrel cam cylinder **1520**. As explained in further detail below, first and second diagonally extending walls **1567** and **1569** of the aperture **1565** (that is, diagonally

extending relative to the longitudinal axis of the barrel cam assembly **1519**) engage the trigger pin **1528** to cause the trigger pin **1528** to travel straight through the intersection **1545** of the cam slot **1544**. In addition and as explained in further detail below, the follower guide **1521** rotates relative to the barrel cam cylinder **1520** to appropriately position the first and second diagonally extending walls **1567** and **1569** during first and second actuations of the trigger **1508**. The aperture **1565** includes first and second longitudinally extending walls **1571** and **1573** (that is, walls extending parallel to the longitudinal axis of the barrel cam assembly **1519**) that engage the trigger pin **1528** to facilitate rotation of the follower guide **1521** relative to the barrel cam cylinder **1520**.

[0228] The first and second diagonally extending walls **1567** and **1569**, the first and second longitudinally extending walls **1571** and **1573**, and the other walls that define the aperture **1565** extend from an inner surface to an outer surface of the follower guide **1521**. In some embodiments, these walls extend in a radial direction between the inner surface and the outer surface. In some embodiments, these walls extend diagonally between the inner surface and the outer surface (that is, these walls form a chamfer between the inner surface and the outer surface).

[0229] Referring now to FIGS. **15J** and **15K**, the barrel cam assembly **1519** further includes a relative rotation-inhibiting mechanism **1575** that, as the name implies, inhibits some rotation of the follower guide **1521** relative to the barrel cam cylinder **1520**. Generally, the relative rotation-inhibiting mechanism **1575** permits the follower guide **1521** to occupy a first relative rotation-inhibiting position and a second relative rotation-inhibiting position. In the first relative rotation-inhibiting position, the mechanism **1575** permits the follower guide **1521** to rotate in a first direction relative to the barrel cam cylinder **1520** (that is, toward the second relative rotation-inhibiting position) and inhibits rotation of the follower guide **1521** relative to the barrel cam cylinder **1520** in a second direction. In the second relative rotation-inhibiting position, the mechanism **1575** permits the follower guide **1521** to rotate in the second direction relative to the barrel cam cylinder **1520** (that is, toward the first relative rotation-inhibiting position) and inhibits rotation of the follower guide **1521** relative to the barrel cam cylinder **1520** in the first direction.

[0230] In some embodiments and as shown in FIGS. **15J** and **15K**, the relative rotation-inhibiting mechanism **1575** may include a longitudinally extending tab (or arm) **1577** (see also FIGS. **15C** and **15H**) carried by the follower guide **1521** and a semi-annular flange **1579** (see also FIGS. **15C** and **D**) carried by the barrel cam cylinder **1520**. FIG. **15J** illustrates such an embodiment of the relative rotation-inhibiting mechanism **1575** in the first relative rotation-inhibiting position (the arm **1577** engages a first side of the semi-annular flange **1579**), and FIG. **15K** illustrates such an embodiment of the relative rotation-inhibiting mechanism **1575** in the second relative rotation-inhibiting position (the arm **1577** engages a second side of the semi-annular flange **1579**).

[0231] In some embodiments, the relative rotation-inhibiting mechanism **1575** may take other forms. For example, the mechanism **1575** may include one or more magnets that hold the follower guide **1521** in the first and second relative rotation-inhibiting positions.

[0232] Referring to FIG. **15L**, an exemplary spring assembly **1512** is depicted. The spring assembly **1512** includes a constant force spring **1572** and a spool **1574**. One end of the constant force spring **1572** is connected to the spool **1574**, and the other end of the constant force spring **1572** is connected to the post **1540** extending from the trigger **1508**. As a clinician pulls the trigger **1508** proximally, the sliders **1532**, **1536** travel and slide in the grooves within the handle frame **1504**, thereby preventing the trigger **1508** from moving vertically within the handle assembly **1208** and only allowing the trigger **1508** to move along the longitudinal axis of the surgical device **1206** from its distal end toward its proximal end and/or vice versa. As the trigger **1508** moves proximally, the constant force spring **1572** uncoils, thereby creating tension and a distally directed force. Accordingly, when the trigger **1508** is released by the clinician, the constant force spring **1572** recoils and pulls the trigger **1508** back towards its original and most distal position.

[0233] Referring to FIG. **16**, there is depicted an elevation view of an embodiment of an assembled sheath assembly **1212** of the present disclosure. The sheath assembly **1212** includes an inner sheath

assembly and an outer sheath assembly. Referring to FIG. 16A, which illustrates an exploded view of the distal end of the sheath assembly **1212**, and referring to FIG. 16B, which is an exploded illustration of the proximal end and central portion of the sheath assembly **1212**, the sheath assembly **1212** may include some or all of the following components: an outer band **1636**; a guide pin **1640**; a cutting tip **1632**; a flexible inner sheath **1620**; a flexible outer sheath **1624**; an outer jacket **1628**; an inner key **1612**; an outer key **1608**; and a rigid inner tube **1616**. [0234] Referring to FIG. 17A, there is depicted an embodiment of the outer sheath assembly **1602** of the present disclosure. The outer sheath assembly **1602** includes an outer band **1636** located at and attached to the distal end of an elongated flexible outer sheath **1624**, and an outer key **1608** located at and attached to the proximal end of the flexible outer sheath **1624**. The outer band **1636** may be attached to the distal end of a flexible outer sheath **1624** via a weld, an adhesive, a press-fitting technique, an interlock such as a barbed joint or other known means of attachment. All such attachment techniques within the knowledge of one skilled in the art are considered within the scope of this disclosure. Similarly, the outer key **1608** may be attached to the proximal end of the flexible outer sheath **1624** via a weld, an adhesive, a press-fitting technique, interlock such as a barbed joint, or other known means of attachment. Although it is not shown on FIG. 17A, the outer sheath assembly **1602** may also include a flexible outer jacket **1628** (see FIG. 16A) that covers the outer sheath **1624** and abuts the outer band **1636**, thereby providing the outer sheath assembly **1602** with a relatively smooth, continuous and uninterrupted exterior profile. The flexible jacket also contains the egress of blood from the system.

[0235] Referring to FIG. 17B, there is depicted an embodiment of the inner sheath assembly **1604** of the present disclosure. The inner sheath assembly **1604** includes a cutting tip **1632**, a flexible inner sheath **1620**, an inner key **1612**, and a rigid inner tube **1616**. The proximal end of the cutting tip **1632** is attached to the distal end of a flexible inner sheath **1620**; the distal end of an inner tube **1616** is attached to the proximal end of the flexible inner sheath **1620**; and an inner key **1612** is attached to the proximal end of the inner tube **1616**. The means of attaching these components may include a weld, an adhesive, a press-fitting technique, or other known means of attachment. As will be discussed below, the guide pin **1640** couples the outer band **1636** with the cutting tip **1632**, and the guide pin **1640** may be included with either the inner sheath assembly **1604** or the outer sheath assembly **1602**.

[0236] It may be preferable for a portion of either the inner sheath **1620** and/or the outer sheath **1624** to be rigid and a portion of the outer sheath to be flexible. Both the rigid portion and the flexible portion may be constructed of materials suitable for insertion into the human body. For example, the rigid portion may be constructed of stainless steel, and the flexible portion may be constructed of a flexible polymer such as polytetrafluoroethylene or thermoplastic elastomers. Assuming that both a rigid portion and a flexible portion are used, they will form a unitary inner sheath and/or outer sheath. As depicted in FIG. 17B, the rigid inner tube **1616** is not only attached to the inner key **1612**, the rigid tube **1616** also is inserted through the inner key **1612** and extends from both the proximal end and distal end of the inner key **1612**. The attachment and extension of the rigid tube **1616** to the inner key **1612** allows for an increased amount of torque that can be transferred from the barrel cam to the rigid tube **1616** via the inner key **1612** and eventually to the cutting tip **1632** via the inner sheath assembly **1604**. The extension of the rigid tube through the handle provides an access point for introduction of other medical devices. The extension also provides a means of controlling blood egress after the lead has been extracted.

[0237] It may be preferable that at least a portion of the outer sheath **1624** and the inner sheath **1620** be generally flexible in order to accept, accommodate and navigate the patient's vasculature system. In addition to being flexible, the inner sheath **1620** may also have a high degree of stiffness in order to receive the torque transferred from the barrel cam cylinder/inner key and transfer sufficient torque to the cutting tip **1632** discussed in more detail below. The inner sheath **1620** (and/or the outer sheath **1624**) may be formed of a polymer extrusion, braided reinforced polymer

extrusion, coils, bi-coils, tri-coils, laser cut metal tubing and any combination of the above. The inner sheath (and/or the outer sheath **1624**) may be a unitary structure comprised of multiple portions.

[0238] Referring to FIG. **18**, there is depicted a cross-sectional view of an embodiment of the sheath assembly **1212** comprising the inner sheath assembly **1604** located within the outer sheath assembly **1602**. Referring to FIG. **18C**, there is depicted an enlarged view of the inner key **1612** of the inner sheath assembly **1604** located within the outer key **1608** of the outer sheath assembly **1602**. As discussed above, the exterior of the inner key **1612** is designed to mate with lumen **1556** of the distal end of the barrel cam cylinder **1520**. Accordingly, the cross section of the exterior of proximal end of the inner key **1612** will have a profile complimentary to the distal end of the lumen **1556** within the barrel cam cylinder **1520**. For example, assuming the cross section of the distal end **1552** of the lumen **1556** of the barrel cam cylinder **1520** is non-circular and has two chamfered sides, wherein one chamfered side is not offset, and the other chamfered side is offset (e.g., about 8 degrees), then the exterior of the proximal end of the inner key **1612** will also have a non-circular profile with two chamfered sides, wherein one chamfered side is not offset, and the other chamfered side is offset (e.g., about 8 degrees). The inner key **1612** and outer key **1608** provide means for rotationally coupling. The inner key **1612** is a means for rotationally coupling the inner sheath assembly **1604** to the barrel cam, and the outer key is a means for rotationally coupling the outer shaft assembly to the handle. The inner key **1612** and outer key **1608** provide journal bearing for the other key.

[0239] As further illustrated in FIG. **18C**, the inner key **1612** is able to rotate freely within the outer key **1608** due, at least in part, to the distal end of the exterior of the inner key **1612** having a circular cross section that mates with a circular cross section of the proximal end of a lumen within the outer key **1608**. Additionally, because the inner key **1612** and outer key **1608** are loosely coupled, the inner key **1612** and outer key **1608** are able to move longitudinally with respect to one another. For instance, supposing the outer key **1608** is fixed such that it neither rotates nor moves longitudinally, the inner key **1612** is able to both rotate and travel longitudinally within the outer key **1608**. Accordingly, as the barrel cam cylinder **1520** rotates, the inner key **1612** will rotate within the outer key **1608**, and the inner sheath assembly **1604** will rotate within the outer sheath assembly **1602**, including the rotation of the cutting tip **1632** within the outer band **1636**. And the cam slot profile in the cutting tip **1632** controls the longitudinal movement of the inner sheath assembly **1604** within the outer sheath assembly **1602**, including the longitudinal movement of the inner key **1612** relative to the outer key **1608** and the longitudinal movement of the cutting tip **1632** relative to the outer band **1636**.

[0240] Continuing to refer to FIG. **18C**, the lumen within the outer key **1608** is larger toward its proximal end and smaller toward its distal end because there is a step down or an abutment in the lumen as it progresses from the proximal end to the distal end. Due to the transition from a larger lumen to a smaller lumen within the outer key **1608**, there is depicted an adjustable gap **1610** between the distal end of the inner key **1612** and the abutment within the distal end of the larger lumen in the outer key **1608**. This gap increases, decreases and/or remains the same according to the cam slot profile of the cutting tip **1632**. The abutment in the outer key **1608** insures that the inner key **1612** will only travel a limited longitudinal distance within the outer key **1608**, thereby limiting the inner sheath assembly **1604** potential longitudinal movement within the outer sheath assembly **1602**, including limiting the longitudinal movement of the cutting tip **1632** relative to the outer band **1636** in the distal direction.

[0241] Referring to FIG. **18A**, there is depicted an enlarged cross-sectional view of the distal end of the sheath assembly **1212** with the inner sheath assembly **1604** coupled with the outer sheath assembly **1602** via guide pin **1640**, wherein the blade **1822** of the cutting tip **1632** is in a retracted position and located within the outer sheath assembly **1602**. As discussed above, the distal end of the outer sheath assembly **1602** includes an outer band **1636**, which may be constructed of a

biocompatible metal, such as stainless steel, and polished so that it is generally smooth and evenly rounded at its most distal point, thereby allowing it to act as a dilator when pressed and forced against tissue. The distal end **1822** of cutting tip **1632** includes a cutting surface capable of cutting tissue. The inner sheath assembly **1604** is coupled to the outer sheath assembly **1602** through the cutting tip **1632** and the outer band **1636**, respectively, via guide pin **1640**. One end of the guide pin **1640** is fixed within the outer band **1636**, and the other end of the guide pin **1640** is located within the cam slot **1814** of the cutting tip **1632**. As the inner sheath **1620** rotates, upon actuation of the trigger assembly discussed above, the cutting tip **1632** also rotates because the inner sheath **1620** is fixedly attached to the cutting tip **1632**. As the cutting tip **1632** rotates, the cutting tip **1632** may also extend distally in the direction of the arrow (.fwdarw.) according to the profile of the cam slot **1814** as depicted in FIG. **18A'**. As the cutting tip **1632** extends distally and rotates, the guide pin **1640** and the outer sheath assembly **1602**, particularly the outer band **1636**, remain stationary. Thus, as the cutting tip **1632** extends distally (and potentially retracts proximally according to the cam slot profile) and rotates, the cutting surface at the distal end **1822** of the cutting tip **1632** is able to perform a slicing action against the tissue and cut it.

[0242] Again, FIG. **18A** depicts the cutting tip **1632** within a retracted (and potentially un-actuated) position because the cutting tip **1632** is in a proximal position. Stated differently, the distal end **1822** of the cutting tip **1632** of FIG. **18A** is located within the interior of the outer sheath assembly **1602**, particularly the outer band **1636**, and does not extend beyond the distal end of the outer band **1636**. With reference to FIG. **18A'**, the cutting tip **1632** is depicted in an extended (and actuated) position because the cutting tip **1632** is extending beyond the distal end of the outer sheath assembly **1602** and the outer band **1636**.

[0243] FIG. **14** depicts the distal portion of the flexible outer sheath and flexible inner sheath surrounding a lead **330** within a patient's vein **334** with the cutting tip **1632** in its extended position. The circumferential nature of the cutting surface (e.g., notched blade) at the distal end of the cutting tip **1632** causes the surgical device to act as a coring device, thereby cutting tissue **338** either partially (i.e., less than 360 degrees) or completely (i.e., 360 degrees) around the lead or implanted object being extracted. The amount of tissue that the cutting surface cuts depends upon the size, shape and configuration of the lead, as well as the diameter and thickness of the circular cutting blade. For example, if the diameter of the circular cutting surface is substantially greater than the diameter of the lead, then the cutting surface will cut and core more tissue in comparison to a cutting surface having a smaller diameter.

[0244] Although the inner sheath and outer sheath are coupled to one another via the cutting tip, the outer band, and the guide pin, the inner sheath assembly and outer sheath assembly may be coupled to one another in other ways. Stated differently, those of skill in the art will understand how to make and use the disclosed aspects, embodiments, and/or configurations after understanding the present disclosure to couple the sheaths in a manner to allow a cutting surface to extend and rotate beyond the distal end of the outer sheath. All such configurations within the knowledge of one skilled in the art are considered within the scope of this disclosure.

[0245] With reference to FIGS. **19A**, **19B** and **19C**, an exemplary outer band **1636** is depicted. The outer band **1636** may be a sleeve in the general shape of a hollow cylinder. Although the exterior of the outer band **1636** is non-uniform, it may be uniform. The interior of the outer band **1636** is non-uniform. For example, the interior of the outer band **1636** includes an abutment **1916** to prevent the cutting tip (not shown in FIGS. **19A**, **19B** and **19C**) from traveling further from the proximal end **1912** to the distal end **1908** within the outer band **1636**. The outer band **1636** also includes a hole **1904** for receipt and possible attachment of a guide pin (not shown in FIGS. **19A**, **19B** and **19C**) which protrudes radially inward. As discussed in more detail above, the guide pin engages the cam slot of the cutting tip. The size, shape and configuration of the outer band **1636** may differ depending upon how it is attached to the flexible outer sheath. As discussed above, the outer sheath may be stationary. If so, the outer band **1636** and the guide pin remain stationary as the cutting tip

moves (e.g., rotates and travel longitudinally) relative thereto. The outer band may also contain a journal bearing surface to align the cutting blade during actuation and provide a surface to disengage the tissue at the retraction of the cutting blade within the device.

[0246] With reference to FIGS. **20A**, **20B**, **20C** and **20D**, an exemplary cutting tip **1632** is depicted. The cutting tip **1632** has a generally hollow cylindrical shape. The cutting tip **1632** comprises a proximal portion **2024**, an intermediate portion **2028**, and a distal portion **2032**. The outside diameter of the proximal portion **2024** is sized to allow it to be inserted to and/or engage (or otherwise attached to) the interior diameter of the inner flexible sheath (not shown). The distal end of cutting tip **1632** comprises a cutting surface **2012** having a serrated, sharp blade profile. The intermediate portion **2028** comprises a channel (or cam slot) **2016** cut within its exterior surface. As the inner flexible sheath rotates and moves within the outer sheath—from its proximal end to distal end—the outer sheath and pin may remain stationary. If so, the inner sheath (not shown), which is connected to cutting tip **1632**, forces the cutting tip **1632** to rotate. The cam slot **2016** engages the guide pin, and the shape and profile of the cam slot **2016** controls the rate and distance with which the cutting tip **1632** travels longitudinally. That is, the configuration of the cam slot **2016** controls the cutting tip's direction and amount of longitudinal travel, such as moving distally toward an extended position and/or proximally toward a retracted position, while the cutting tip rotates in either a clockwise or counter-clockwise direction.

[0247] Referring again to FIGS. **20A**, **20B**, **20C** and **20D**, the cutting tip **1632** may also comprise a step up **2020** such that the diameter of the intermediate portion **2028** is greater than the distal portion **2032**. As the cutting tip **1632** rotates, and the cutting surface **2012** extends beyond the distal end of the outer band into an extended position, the step up **2020** of the cutting tip **1632** contacts the abutment of the outer band, thereby limiting the distance that the cutting tip **1632** may travel and/or may prevent the cutting tip **1632** from exiting or extending beyond the distal tip of the outer sheath assembly, particularly the outer band, in the event that the pin is sheared.

[0248] The profile of the cam slot in the cutting tip may have various configurations, such as those disclosed in U.S. patent application Ser. No. 13/834,405 filed Mar. 15, 2013 and entitled Retractable Blade For Lead Removal Device, which is hereby incorporated herein by reference in its entirety for all that it teaches and for all purposes. For example, the cam slot may have a substantially linear profile, a substantially sinusoidal profile, or a combination of individual and/or multiple linear and non-linear profiles. Additionally, the cam slot may have an open and continuous configuration, thereby allowing the cutting tip to continuously rotate, or the cam slot may have a closed and discontinuous configuration such that when the cutting tip reaches its fully extended position, the trigger assembly must be released or reversed so that the cutting tip returns to initially retracted position before being re-actuated. For instance, the cam slot **2016** in FIG. **20A** is discontinuous because the cam slot does not travel around the entire circumference of the exterior of the cutting tip **1632**. In some embodiments and as shown in FIGS. **21** and **22A-22C**, the cam slot **2016** may be symmetric over a longitudinally-extending plane. Although certain figures in this disclosure only illustrate either the open or closed cam slot configuration, either configuration may be used with any of the inner cam embodiments disclosed and/or discussed herein and are considered within the scope of this disclosure. Furthermore, various types of cams, such as a partial lobe cam (which includes a cam slot surrounding less than 360 degrees of the circumference of the exterior surface of the cutting tip), a single lobe cam (which includes a cam slot surrounding 360 degrees of the circumference of the exterior surface of the cutting tip), double lobe cams (which includes a cam slot surrounding 720 degrees of the circumference of the exterior surface of the cutting tip) and/or other multiple lobe cams.

[0249] The distal end of cutting tip **1632** may comprise a cutting surface **2012** having different blade profiles, such as those disclosed in U.S. patent application Ser. No. 13/834,405 filed Mar. 15, 2013 and entitled Retractable Blade For Lead Removal Device, which is hereby incorporated herein by reference in its entirety for all that it teaches and for all purposes. For example, the plane

of the cutting surface **2012** of the distal end of the cutting tip depicted in the figures of this disclosure is parallel to the plane of the proximal end of the cutting tip. The plane of the cutting surface, however, may be offset (0 degrees to 90 degrees) from the plane of the proximal end of the cutting tip. Also, as discussed above, the profile of the cutting surface **2012** in FIGS. **10A-10D** has a plurality of serrations. The profile of the cutting surface **2012** need not be serrated and may comprise other configurations, such as a constant and/or smooth sharp profile. The profile of the cutting surface **2012** in FIGS. **20A-20D** has a plurality of six (6) serrations. However, it may be preferable to have less than or more than six (6) serrations. It may also be preferable to have between five (5) and seven (7) serrations, or between four (4) and eight (8) serrations, or between six (6) and ten (10) serrations.

[0250] Although the cutting surface **2012** illustrates a certain number of serrations, FIGS. **20A-20D** are not intended to represent the only number and type of serrations that may be included in a serrated cutting surface. Depending upon the size of the surgical device, including the sheaths, and cutting tip, those of skill in the art will understand how to make and use the disclosed aspects, embodiments, and/or configurations after understanding the present disclosure to adjust the number, size and configurations of the serrations. All such configurations within the knowledge of one skilled in the art are considered within the scope of this disclosure. Furthermore, the serrations may comprise a myriad of different shapes and configurations, including but not limited to any variation of a square, rectangle, rhombus, parallelogram, trapezoid, triangle, circle, ellipse, kite, etc.

[0251] FIG. **21** depicts two-dimensional illustrations of the profile of the cam slot **2016** for the cutting tip **1632**, the profile of the cam slot **1544** for the barrel cam cylinder **1520**, and the profile of the aperture **1565** of the follower guide **1521**. FIGS. **22A-22C** depict the how actuation of the trigger **1508**, and the resulting movement of the trigger pin **1528**, results in rotational movement of the barrel cam cylinder **1520**, the follower guide **1521**, and the cutting tip **1632**, and translation movement of the cutting tip **1632**. In these figures, a horizontal axis for the profiles of the slots **2016** and **1544** is the degree(s) of rotation of the cutting tip **1632** and the barrel cam cylinder **1520**. For example, assuming that the profile of the cam slot in the cutting tip **1632** is discontinuous, as depicted in FIG. **20A**, the cutting tip **1632** will rotate less than 360 degrees. It may be preferable for the cutting tip **1632** to rotate between 5 and 355 degrees, 180 degrees and 355 degrees, 210 degrees and 325 degrees, 240 degrees and 295 degrees, or 270 degrees and 275 degrees. It may also be preferable for the cutting tip **1632** rotate about 180, 185, 190, 200, 205, 210, 215, 220, 225, 230, 235, 240, 245, 250, 255, 260, 265, 270, 275, 280, 285, 290, 300, 305, 310, 315, 320, 325, 330, 335, 340, 345, 350 or 355 degrees. A vertical axis for the profile of the cam slot **2016** for the cutting tip **1632** is the amount of longitudinal displacement, if any, of the cutting tip **1632**. The vertical axis for the profile of the cam slot **1544** for the barrel cam cylinder **1520** is the amount of longitudinal displacement of the trigger assembly (and trigger pin **1528**).

[0252] In FIGS. **22A-22C**, the aperture **1565** of the follower guide **1521** is shown as a dashed line and is overlaid on the profile of the cam slot **1544** for the barrel cam cylinder **1520** to illustrate the rotational position of the aperture **1565** relative to the cam slot **1544**. As shown in FIGS. **22A-22C** and explained in further detail below, the rotational position of the aperture **1565** changes relative to the cam slot **1544** during actuation of the trigger **1508**.

[0253] Generally, an initial, or first, actuation of the trigger **1508** (that is, pulling the trigger **1508** as far as permitted by the handle assembly and then releasing the trigger **1508** so that it returns to its home position) results in about a 254 degree rotational displacement of the cutting tip **1632** and the barrel cam cylinder **1520** in one direction-clockwise when looking from the handle to the tip. The first actuation also extends the cutting tip **1632** from the outer band **1636** and then returns the cutting tip **1632** to the sheathed position as the cutting tip **1632** rotates. A subsequent, or second, actuation of the trigger **1508** results in about a 254 degree rotational displacement of the cutting tip **1632** and the barrel cam cylinder **1520** in the opposite direction-counter-clockwise when looking from the handle to the tip. The second actuation also extends the cutting tip **1632** from the outer

band **1636** and then returns the cutting tip **1632** to the sheathed position as the cutting tip **1632** rotates. Additional “odd” actuations (that is, a third actuation, a fifth actuation, and so on) cause the same device motions as the first actuation of the trigger **1508**, and additional “even” actuations (that is, a fourth actuation, a sixth actuation, and so on) cause the same device motions as the second actuation of the trigger **1508**.

[0254] The following discussion more specifically explains the interaction between the rotation of the barrel cam cylinder **1520**, the rotation of the follower guide **1521**, the rotation of the cutting tip **1632**, the longitudinal movement of the handle (via the position of the trigger pin **1528**), and the longitudinal movement of the cutting tip **1632**.

[0255] First and referring specifically to FIGS. **22A**, **23A**, **23B**, and **23C**, the following is a description of the positions **BC1'-BC3'** of the trigger pin **1528** within the cam slot **1544** of the barrel cam cylinder **1520**. In these positions, the follower guide **1521** is in the first relative rotation-inhibiting position, and the follower guide **1521** rotates together with the barrel cam cylinder **1520**.

[0256] **BC1'**—The trigger pin **1528** is at a first home position within the cam slot of the barrel cam cylinder **1520**. The trigger **1508** of the trigger assembly is also at its home position. [0257] **BC2'**—The trigger pin **1528** has moved longitudinally in a proximal direction, thereby causing the barrel cam cylinder **1520** to rotate in a clockwise direction. The trigger pin **1528** engages the first diagonally-extending wall **1567** of the aperture **1565** of the follower guide **1521**. [0258] **BC3'**—The trigger pin **1528** has moved longitudinally further in the proximal direction, thereby causing the barrel cam cylinder **1520** to rotate further in the clockwise direction. The trigger pin **1528** engages the first longitudinally-extending wall **1571** of the aperture **1565** of the follower guide **1521**.

[0259] The positions **BC1'-BC3'** of the trigger pin **1528** within the cam slot **1544** of the barrel cam cylinder **1520** correspond to positions **CT1'-CT3'** of the guide pin **1640** within the cam slot of the cutting tip **1632**. The following is a description of the positions **CT1'-CT3'** of the guide pin **1640** within the cam slot of the cutting tip **1632**. [0260] **CT1'**—The guide pin **1640** is at a first home position within the cam slot **2016** of the cutting tip **1632**, and the cutting tip **1632** is in a retracted position within the outer sheath assembly **1602** (including the outer band **1636**). [0261] **CT2'**—The cutting tip **1632** has rotated in a clockwise direction over the guide pin **1640** within the cam slot **2016** for about half of its predetermined rotation. The cutting tip **1632** may be in its most extended position outside the outer sheath assembly **1602**. [0262] **CT3'**—The cutting tip **1632** has rotated further in the clockwise direction over the guide pin **1640** within the cam slot **2016**. The cutting tip **1632** may be in an intermediate position between the extended position and the retracted position within the outer sheath assembly **1602**, or the cutting tip **1632** may be in the retracted position within the outer sheath assembly **1602**.

[0263] By beginning a first actuation of the trigger **1508** as described above, the trigger pin **1528** moves from its first home position **BC1'** and into the intersection **1545** of the cam slot **1544** of the barrel cam cylinder **1520** (position **BC2'**). This action causes the barrel cam cylinder **1520** and the cutting tip **1632** to rotate in a clockwise direction by about 127 degrees. In position **BC2'**, the trigger pin **1528** engages the first diagonally-extending wall **1567** of the aperture **1565** of the follower guide **1521** to ensure that the trigger pin **1528** travels straight through the intersection **1545** of the cam slot **1544** of the barrel cam cylinder **1520**. Stated another way, the trigger pin **1528** travels from the first leg **1547** of the cam slot **1544** to the second leg **1549** of the cam slot **1544**. Stated yet another way, the follower guide **1521** ensures that that trigger pin **1528** traverses a first slot portion defined by the first leg **1547** and the second leg **1549**. This ensures that the barrel cam cylinder **1520** continues to rotate in a clockwise direction. Engagement of the trigger pin **1528** against the wall of the aperture **1565** of the follower guide **1521** tends to rotate the follower guide **1521** in a clockwise direction relative to the barrel cam cylinder **1520**. However, such motion is inhibited by the relative rotation-inhibiting mechanism **1575** in the first relative rotation-inhibiting position.

[0264] By continuing the first actuation of the trigger **1508**, the trigger **1508** pin moves from position BC2' to position BC3'. This action causes the barrel cam cylinder **1520** and the cutting tip **1632** to rotate in the clockwise direction by about 50 degrees. In position BC3', the trigger pin **1528** engages the first longitudinally-extending wall **1571** of the aperture **1565** of the follower guide **1521**. As described in further detail below, movement of the trigger pin **1528** past position BC3' rotates the follower guide **1521** in a counter-clockwise direction relative to the barrel cam cylinder **1520**.

[0265] Referring now to FIGS. 22B and 23D, the following is a description of the positions BC4'-BC6' of the trigger pin **1528** within the cam slot **1544** of the barrel cam cylinder **1520**. In these positions, the follower guide **1521** is in the second relative rotation-inhibiting position, and the follower guide **1521** rotates together with the barrel cam cylinder **1520**. [0266] BC4'—The trigger pin **1528** has moved about its entire longitudinal movement in a proximal direction and the barrel cam cylinder **1520** has completed its rotation in a clockwise direction. The trigger pin **1528** has rotated the follower guide **1521** to the second relative rotation-inhibiting position. [0267] BC5'—The trigger pin **1528** has moved longitudinally in a distal direction. The barrel cam cylinder **1520** has remained stationary. [0268] BC6'—The trigger pin **1528** has moved longitudinally further in the distal direction. The trigger pin **1528** is at a second home position within the cam slot of the barrel cam cylinder **1520**. The trigger **1508** of the trigger assembly is also at its home position. [0269] The positions BC4'-BC6' of the trigger pin **1528** within the cam slot of the barrel cam cylinder **1520** correspond to positions CT4'-CT6' of the guide pin **1640** within the cam slot of the cutting tip **1632**. The following is a description of the positions CT4'-CT6' of the guide pin **1640** within the cam slot of the cutting tip **1632**. [0270] CT4'—The cutting tip **1632** has completed its rotation in a clockwise direction over the guide pin **1640** within the cam slot **2016**. The cutting tip **1632** is in its most retracted position within the outer sheath assembly **1602**. [0271] CT5'—(Not shown) the cutting tip **1632** has not moved relative to position CT4'. [0272] CT6'—The guide pin **1640** is at a second home position within the cam slot **2016** of the cutting tip **1632**, and the cutting tip **1632** is in a retracted position within the outer sheath assembly **1602** (including the outer band **1636**).

[0273] By continuing the first actuation of the trigger **1508**, the trigger pin **1528** moves from position BC3' (FIG. 22A) to position BC4'. This action causes the barrel cam cylinder **1520** and the cutting tip **1632** to rotate in the clockwise direction by about 110 degrees. This action also causes the trigger pin **1528** to rotate the follower guide **1521** in a counter-clockwise direction by about 110 degrees relative to the barrel cam cylinder **1520**. That is, the follower guide **1521** rotates from the first relative rotation-inhibiting position to the second relative rotation-inhibiting position. In the second relative rotation-inhibiting position, the follower guide **1521** is positioned to ensure that the trigger pin **1528** subsequently crosses its previous path and travels straight through the intersection **1545** of the cam slot **1544** of the barrel cam cylinder **1520**.

[0274] After the trigger pin **1528** reaches position BC4', the trigger **1508** is moved in a distal direction (for example, by releasing the trigger **1508**). This action causes the trigger pin **1528** to move from position BC4' to position BC5' and then position BC6'. When the trigger pin **1528** moves from position BC5' to position BC6', the barrel cam cylinder **1520** and the cutting tip **1632** rotate in the counter-clockwise direction by about 33 degrees. As such, the barrel cam cylinder **1520** and the cutting tip **1632** are rotationally displaced by about 254 degrees in the clockwise direction as the trigger pin **1528** moves from position BC1' to position BC6'. The trigger pin **1528** remains at position BC6', the second home position thereof, until the clinician begins a second actuation of the trigger **1508**.

[0275] Referring now to FIGS. 22B, 23E, and 23F, the following is a description of the positions BC7'-BC8' of the trigger pin **1528** within the cam slot **1544** of the barrel cam cylinder **1520**. In these positions, the follower guide **1521** is in the second relative rotation-inhibiting position, and the follower guide **1521** rotates together with the barrel cam cylinder **1520**. [0276] BC7'—The

trigger pin **1528** has moved longitudinally in a proximal direction, thereby causing the barrel cam cylinder **1520** to rotate in a counter-clockwise direction. The trigger pin **1528** engages the second diagonally-extending wall **1569** of the aperture **1565** of the follower guide **1521**. [0277] BC8'—The trigger pin **1528** has moved longitudinally further in the proximal direction, thereby causing the barrel cam cylinder **1520** to rotate further in the counter-clockwise direction. The trigger pin **1528** engages the second longitudinally-extending wall **1573** of the aperture **1565** of the follower guide **1521**.

[0278] The positions BC7'-BC8' of the trigger pin **1528** within the cam slot of the barrel cam cylinder **1520** correspond to positions CT7'-CT8' of the guide pin **1640** within the cam slot of the cutting tip **1632**. The following is a description of the positions CT7'-CT8' of the guide pin **1640** within the cam slot of the cutting tip **1632**. [0279] CT7'—The cutting tip **1632** has rotated in a counter-clockwise direction over the guide pin **1640** within the cam slot **2016** for about half of its predetermined rotation. The cutting tip **1632** may be in its most extended position outside the outer sheath assembly **1602**. [0280] CT8'—The cutting tip **1632** has rotated further in the counter-clockwise direction over the guide pin **1640** within the cam slot **2016**. The cutting tip **1632** may be in an intermediate position between the extended position and the retracted position within the outer sheath assembly **1602**, or the cutting tip **1632** may be in the retracted position within the outer sheath assembly **1602**.

[0281] By beginning a second actuation of the trigger **1508** as described above, the trigger pin **1528** moves from its second home position BC6' and into the intersection **1545** of the cam slot **1544** of the barrel cam cylinder **1520** (position BC7'). This action causes the barrel cam cylinder **1520** and the cutting tip **1632** to rotate in a counter-clockwise direction by 127 degrees. In position BC7', the trigger pin **1528** engages the second diagonally-extending wall **1569** of the aperture **1565** of the follower guide **1521** to ensure that the trigger pin **1528** travels straight through the intersection **1545** of the cam slot **1544** of the barrel cam cylinder **1520**. Stated another way, the trigger pin **1528** travels from the third leg **1551** of the cam slot **1544** to the fourth leg **1553** of the cam slot **1544**. Stated yet another way, the follower guide **1521** ensures that that trigger pin **1528** traverses a second slot portion defined by the third leg **1551** and the fourth leg **1553**. This ensures that the barrel cam cylinder **1520** continues to rotate in a counter-clockwise direction. Engagement of the trigger pin **1528** against the wall of the aperture **1565** of the follower guide **1521** tends to rotate the follower guide **1521** in a counter-clockwise direction relative to the barrel cam cylinder **1520**. However, such motion is inhibited by the relative rotation-inhibiting mechanism **1575** in the second relative rotation-inhibiting position.

[0282] By continuing the second actuation of the trigger **1508**, the trigger **1508** pin moves from position BC7' to position BC8'. This action causes the barrel cam cylinder **1520** and the cutting tip **1632** to further rotate in the counter-clockwise direction by 50 degrees. In position BC8', the trigger pin **1528** engages the second longitudinally-extending wall **1573** of the aperture **1565** of the follower guide **1521**. As described in further detail below, movement of the trigger pin **1528** past position BC8' rotates the follower guide **1521** in a clockwise direction relative to the barrel cam cylinder **1520**.

[0283] Referring now to FIGS. 22C and 23A, the following is a description of the positions BC9'-BC10' of the trigger pin **1528** within the cam slot of the barrel cam cylinder **1520**. In these positions, the follower guide **1521** is in the first relative rotation-inhibiting position, and the follower guide **1521** rotates together with the barrel cam cylinder **1520**. [0284] BC9'—The trigger pin **1528** has moved about its entire longitudinal movement in a proximal direction and the barrel cam cylinder **1520** has completed its rotation in a counter-clockwise direction. The trigger pin **1528** has rotated the follower guide **1521** back to the first relative rotation-inhibiting position. [0285] BC10'—The trigger pin **1528** has moved longitudinally in a distal direction. The barrel cam cylinder **1520** has remained stationary.

[0286] The positions BC9'-BC10' of the trigger pin **1528** within the cam slot of the barrel cam

cylinder **1520** correspond to positions CT9'-CT10' of the guide pin **1640** within the cam slot of the cutting tip **1632**. The following is a description of the positions CT9'-CT10' of the guide pin **1640** within the cam slot of the cutting tip **1632**. [0287] CT9'—The cutting tip **1632** has completed its rotation in a counter-clockwise direction over the guide pin **1640** within the cam slot **2016**. The cutting tip **1632** is in its most retracted position within the outer sheath assembly **1602**. [0288] CT10'—(Not shown) the cutting tip **1632** has not moved relative to position CT9'.

[0289] By continuing the second actuation of the trigger **1508**, the trigger pin **1528** moves from position BC8' (FIG. **22B**) to position BC9'. This action causes the barrel cam cylinder **1520** and the cutting tip **1632** to further rotate in the counter-clockwise direction by 110 degrees. This action also causes the trigger pin **1528** to rotate the follower guide **1521** in a clockwise direction relative to the barrel cam cylinder **1520** by 100 degrees. That is, the follower guide **1521** rotates from the second relative rotation-inhibiting position back to the first relative rotation-inhibiting position. In the first relative rotation-inhibiting position, the follower guide **1521** is again positioned to ensure that the trigger pin **1528** subsequently crosses its previous path and travels straight through the intersection **1545** of the cam slot **1544** of the barrel cam cylinder **1520**.

[0290] After the trigger pin **1528** reaches position BC9', the trigger **1508** is moved in a distal direction (for example, by releasing the trigger **1508**). This action causes the trigger pin **1528** to move from position BC9' to position BC10' and then position BC1'. When the trigger pin **1528** moves from position BC10' to position BC1', the barrel cam cylinder **1520** and the cutting tip **1632** rotate in the clockwise direction by about 33 degrees. As such, the barrel cam cylinder **1520** and the cutting tip **1632** are rotationally displaced by about 254 degrees in the counter-clockwise direction as the trigger pin **1528** moves from position BC6' to position BC1'.

[0291] The trigger pin **1528** remains at position BC1', the first home position thereof, until the clinician begins a third actuation of the trigger **1508**. As described above, additional “odd” actuations (that is, a third actuation, a fifth actuation, and so on) of the trigger **1508** cause the same device motions as the first actuation of the trigger **1508**, and additional “even” actuations (that is, a fourth actuation, a sixth actuation, and so on) of the trigger **1508** cause the same device motions as the second actuation of the trigger **1508**.

[0292] Referring now to FIGS. **24-28**, an exemplary barrel cam assembly **2419** is depicted. The barrel cam assembly **2419** may be used with a surgical device, such as the surgical device **1206** described above, in place of the barrel cam assembly **1519**. The barrel cam assembly **2419** includes a barrel cam cylinder **2420** that rotatably carries a follower guide **2421**. The barrel cam cylinder **2420** and the follower guide **2421** may have the same features as any of the barrel cam cylinders and the follower guides, respectively, described herein (for example, the cam groove **1544** and the follower aperture **1565**, respectively), with the exception of the relative rotation-inhibiting mechanism.

[0293] The barrel cam assembly **2419** includes a relative rotation-inhibiting mechanism **2475** that inhibits some rotation of the follower guide **2421** relative to the barrel cam cylinder **2420**. Generally, the relative rotation-inhibiting mechanism **2475** permits the follower guide **2421** to move from a first relative rotation-inhibiting position (referred to as the “first locked position” for simplicity) to a second relative rotation-inhibiting position (referred to as the “first locked position” for simplicity) and vice versa. In the first locked position, the mechanism **2475** initially inhibits the follower guide **2421** from rotating in a first direction relative to the barrel cam cylinder **2420** (that is, toward the second locked position) and inhibits rotation of the follower guide **2421** in a second direction relative to the barrel cam cylinder **2420**. In the second locked position, the mechanism **2475** initially inhibits the follower guide **2421** from rotating in the second direction relative to the barrel cam cylinder **2420** (that is, toward the first locked position) and inhibits rotation of the follower guide **2421** in the first direction relative to the barrel cam cylinder **2420**.

[0294] The relative rotation-inhibiting mechanism **2475** includes a radially-outwardly projecting protrusion **2402** formed near the proximal end of the barrel cam cylinder **2420**. The protrusion

2402 may be formed on the barrel cam cylinder **2420** in a machining process. The protrusion **2402** includes a proximally-facing curved recess **2404**. The protrusion **2402** also includes a first transversely-facing engagement surface **2406** and a second transversely-facing engagement surface **2408**.

[0295] The relative rotation-inhibiting mechanism **2475** also includes a first spring prong **2410**, a second spring prong **2412** (see FIG. 25), a first transversely-facing engagement surface **2413**, and a second transversely-facing engagement surface **2414** (see FIG. 25) formed near the proximal end of the follower guide **2421**. The spring prongs **2410**, **2412** and the engagement surfaces **2413**, **2414** may be formed on the follower guide **2421** in a laser cutting process.

[0296] The first spring prong **2410** is cantilevered from the remainder of the follower guide **2421** and extends partially about the circumference of the follower guide **2421**. The first spring prong **2410** includes a curved tip **2418** that selectively engages the curved recess **2404** of the protrusion **2402** to inhibit rotation of the follower guide **2421** relative to the barrel cam cylinder **2420**. This aspect is described in further detail below. In some embodiments, the curved tip **2418** has a radius of about 0.040 inches and 0.015 inches of interference with the curved recess **2404** of the protrusion **2402**. Such dimensions facilitate both securement and slidable detachment of the first spring prong **2410** relative to the protrusion **2402**.

[0297] The second spring prong **2412** is cantilevered from the remainder of the follower guide **2421**. The second spring prong **2412** extends partially about the circumference of the follower guide **2421** and faces in the opposite circumferential direction as the first spring prong **2410**. The second spring prong **2412** includes a curved tip **2422** that selectively engages the curved recess **2404** of the protrusion **2402** to inhibit rotation of the follower guide **2421** relative to the barrel cam cylinder **2420**. This aspect is described in further detail below. In some embodiments, the curved tip **2422** has a radius of about 0.040 inches and 0.015 inches of interference with the curved recess **2404** of the protrusion **2402**. Such dimensions facilitate both securement and slidable detachment of the second spring prong **2412** relative to the protrusion **2402**.

[0298] Interaction of the first and second spring prongs **2410**, **2412** with the protrusion **2402**, and the resulting motion of the follower guide **2421** relative to the barrel cam cylinder **2420**, are described with reference to the cam slot and aperture profiles illustrated in FIGS. 22A-22C. Referring first to FIG. 22A and upon a first actuation of the trigger assembly, the trigger pin initially moves from position BC1' to BC3'. As the trigger pin moves in this manner, the follower guide **2421** rotates together with the barrel cam cylinder **2420** because the curved tip **2418** of the first spring prong **2410** is engaged with the curved recess **2404** of the protrusion **2402**. Stated another way, the first spring prong **2410** engages the protrusion **2402** to initially hold the follower guide **2421** in the first locked position relative to the barrel cam cylinder **2420**.

[0299] Referring to FIG. 22B and by continuing the first actuation of the trigger assembly, the trigger pin moves from position BC3' to BC4'. In position BC3', the trigger pin engages the wall of the follower guide aperture. By moving toward position BC4', the trigger pin applies a force to the follower guide **2421** that causes the curved tip **2418** of the first spring prong **2410** to slip over and disengage the curved recess **2404** of the protrusion **2402**. As a result, the follower guide **2421** is "unlocked" and rotates relative to the barrel cam cylinder **2420** as the trigger pin moves from position BC3' to BC4'. As the trigger pin approaches position BC4', the curved tip **2422** of the second spring prong **2412** slips over and engages the curved recess **2404** of the protrusion **2402**. The follower guide **2421** thereby enters the second locked position relative to the barrel cam cylinder **2420**.

[0300] When the user releases the trigger assembly, the trigger pin moves from position BC4' to BC6'. As the trigger pin moves in this manner, the follower guide **2421** rotates together with the barrel cam cylinder **2420** because the curved tip **2422** of the second spring prong **2412** is engaged with the curved recess **2404** of the protrusion **2402**. Stated another way, the second spring prong **2412** engages the protrusion **2402** to hold the follower guide **2421** in the second locked position

relative to the barrel cam cylinder **2420** after the first actuation of the trigger assembly.
[0301] Still referring to FIG. **22B** and upon a second actuation of the trigger assembly, the trigger pin initially moves from position BC6' to BC8'. As the trigger pin moves in this manner, the follower guide **2421** rotates together with the barrel cam cylinder **2420** because the curved tip **2422** of the second spring prong **2412** is engaged with the curved recess **2404** of the protrusion **2402**. Stated another way, the second spring prong **2412** engages the protrusion **2402** to initially hold the follower guide **2421** in the second locked position relative to the barrel cam cylinder **2420**.

[0302] Referring to FIG. **22C** and by continuing the second actuation of the trigger assembly, the trigger pin moves from position BC8' to BC9'. In position BC8', the trigger pin engages the wall of the follower guide aperture. By moving toward position BC9', the trigger pin applies a force to the follower guide **2421** that causes the curved tip **2422** of the second spring prong **2412** to slip over and disengage the curved recess **2404** of the protrusion **2402**. As a result, the follower guide **2421** is "unlocked" and rotates relative to the barrel cam cylinder **2420** as the trigger pin moves from position BC8' to BC9'. As the trigger pin approaches position BC9', the curved tip **2418** of the first spring prong **2410** slips over and engages the curved recess **2404** of the protrusion **2402**. The follower guide **2421** thereby returns to the first locked position relative to the barrel cam cylinder **2420**.

[0303] When the user releases the trigger assembly, the trigger pin moves from position BC9' to BC1'. As the trigger pin moves in this manner, the follower guide **2421** rotates together with the barrel cam cylinder **2420** because the curved tip **2418** of the first spring prong **2410** is engaged with the curved recess **2404** of the protrusion **2402**. Stated another way, the first spring prong **2410** engages the protrusion **2402** to hold the follower guide **2421** in the first locked position relative to the barrel cam cylinder **2420** after the second actuation of the trigger assembly.

[0304] In addition and as shown in FIG. **24**, the first engagement surface **2413** of the follower guide **2421** engages the first engagement surface **2406** of the protrusion **2402** in the first locked position to inhibit the follower guide **2421** from rotating in a direction away from the second locked position. The second engagement surface **2414** of the follower guide **2421** engages the second engagement surface **2408** of the protrusion **2402** in the second locked position to inhibit the follower guide **2421** from rotating in a direction away from the first locked position.

[0305] Referring now to FIGS. **29-30**, an exemplary barrel cam assembly **2919** is depicted. The barrel cam assembly **2919** may be used with a surgical device, such as the surgical device **1206** described above, in place of the barrel cam assembly **1519**. The barrel cam assembly **2919** includes a barrel cam cylinder **2920** that rotatably carries a follower guide **2921**. The barrel cam cylinder **2920** and the follower guide **2921** may have the same features as any of the barrel cam cylinders and the follower guides, respectively, described herein (for example, the cam groove **1544** and the follower aperture **1565**, respectively), with the exception of the relative rotation-inhibiting mechanism.

[0306] The barrel cam assembly **2919** includes a relative rotation-inhibiting mechanism **2975** that inhibits some rotation of the follower guide **2921** relative to the barrel cam cylinder **2920**. Generally, the relative rotation-inhibiting mechanism **2975** permits the follower guide **2921** to move from a first relative rotation-inhibiting position (referred to as the "first locked position" for simplicity) to a second relative rotation-inhibiting position (referred to as the "second locked position" for simplicity) and vice versa. In the first locked position, the mechanism **2975** initially inhibits the follower guide **2921** from rotating in a first direction relative to the barrel cam cylinder **2920** (that is, toward the second locked position) and inhibits rotation of the follower guide **2921** in a second direction relative to the barrel cam cylinder **2920**. In the second locked position, the mechanism **2975** initially inhibits the follower guide **2921** from rotating in the second direction relative to the barrel cam cylinder **2920** (that is, toward the first locked position) and inhibits rotation of the follower guide **2921** in the first direction relative to the barrel cam cylinder **2920**.

[0307] The relative rotation-inhibiting mechanism **2975** includes a radially-outwardly projecting

pin **2902** carried near the proximal end of the barrel cam cylinder **2920**. The pin **2902** may be coupled to the barrel cam cylinder **2920** in various manners. For example, the pin **2902** may be press fitted or adhered in a hole formed on the barrel cam cylinder **2920**. In some embodiments, the relative rotation-inhibiting mechanism **2975** also includes a radially outwardly-facing curved recess **2904** formed near the proximal end of the barrel cam cylinder **2920**. The curved recess **2904** may be angularly offset from the pin **2902** about the longitudinal axis of the barrel cam cylinder **2920**. [0308] The relative rotation-inhibiting mechanism **2975** also includes a spring arm **2908**, a first transversely-facing engagement surface **2913**, and a second transversely-facing engagement surface **2914** formed near the proximal end of the follower guide **2921**.

[0309] The spring arm **2908** is cantilever from the remainder of the follower guide **2921** at a first end. At an opposite end, the spring arm **2908** includes a radially inwardly-facing curved finger **2910**. The finger **2910** selectively engages the curved recess **2904** of the barrel cam cylinder **2920** to inhibit rotation of the follower guide **2921** relative to the barrel cam cylinder **2920**. This aspect is described in further detail below.

[0310] Interaction of the spring arm **2908** with the barrel cam cylinder **2920**, and the resulting motion of the follower guide **2921** relative to the barrel cam cylinder **2920**, are described with reference to the cam slot and aperture profiles illustrated in FIGS. 22A-22C. Referring first to FIG. 22A and upon a first actuation of the trigger assembly, the trigger pin initially moves from position BC1' to BC3'. As the trigger pin moves in this manner, the follower guide **2921** rotates together with the barrel cam cylinder **2920** because the finger **2910** is engaged with the curved recess **2904** of the barrel cam cylinder **2920**. Stated another way, the finger **2910** engages the curved recess **2904** to initially hold the follower guide **2921** in the first locked position relative to the barrel cam cylinder **2920**.

[0311] Referring to FIG. 22B and by continuing the first actuation of the trigger assembly, the trigger pin moves from position BC3' to BC4'. In position BC3', the trigger pin engages the wall of the follower guide aperture. By moving toward position BC4', the trigger pin applies a force to the follower guide **2921** that causes the finger **2910** to slip over and disengage the curved recess **2904** of the barrel cam cylinder **2920**. As a result, the follower guide **2921** is "unlocked" and rotates relative to the barrel cam cylinder **2920** as the trigger pin moves from position BC3' to BC4'. The finger **2910** slips over and engages the outer surface of the barrel cam cylinder **2920** as the follower guide **2921** rotates relative to the barrel cam cylinder **2920**. As the trigger pin approaches position BC4', the finger **2910** remains engaged with the curved recess **2904** of the barrel cam cylinder **2920**. The follower guide **2921** thereby enters the second locked position relative to the barrel cam cylinder **2920**.

[0312] When the user releases the trigger assembly, the trigger pin moves from position BC4' to BC6'. As the trigger pin moves in this manner, the follower guide **2921** rotates together with the barrel cam cylinder **2920** because the finger **2910** is engaged with the outer surface of the barrel cam cylinder **2920**. Stated another way, the finger **2910** remains engaged with the outer surface of the barrel cam cylinder **2920** to hold the follower guide **2921** in the second locked position relative to the barrel cam cylinder **2920** after the first actuation of the trigger assembly.

[0313] Still referring to FIG. 22B and upon a second actuation of the trigger assembly, the trigger pin initially moves from position BC6' to BC8'. As the trigger pin moves in this manner, the follower guide **2921** rotates together with the barrel cam cylinder **2920** due to frictional forces between the finger **2910** and the outer surface of the barrel cam cylinder **2920**. Stated another way, the finger **2910** engages the outer surface of the barrel cam cylinder **2920** to initially hold the follower guide **2921** in the second locked position relative to the barrel cam cylinder **2920**.

[0314] Referring to FIG. 22C and by continuing the second actuation of the trigger assembly, the trigger pin moves from position BC8' to BC9'. In position BC8', the trigger pin engages the wall of the follower guide aperture. By moving toward position BC9', the trigger pin applies a force to the follower guide **2921** that overcomes the frictional forces between the finger **2910** and the outer

surface of the barrel cam cylinder **2920**. As a result, the finger **2910** slips over the outer surface of the barrel cam cylinder **2920**, and the follower guide **2921** is “unlocked” and rotates relative to the barrel cam cylinder **2920** as the trigger pin moves from position BC8’ to BC9’. As the trigger pin approaches position BC9’, the finger **2910** slips over and engages the curved recess **2904** of barrel cam cylinder **2920**. The follower guide **2921** thereby returns to the first locked position relative to the barrel cam cylinder **2920**.

[0315] When the user releases the trigger assembly, the trigger pin moves from position BC9’ to BC1’. As the trigger pin moves in this manner, the follower guide **2921** rotates together with the barrel cam cylinder **2920** because the finger **2910** is engaged with the curved recess **2904** of the barrel cam cylinder **2920**. Stated another way, the finger **2910** engages the protrusion **2902** to hold the follower guide **2921** in the first locked position relative to the barrel cam cylinder **2920** after the second actuation of the trigger assembly.

[0316] In addition and as shown in FIG. **29**, the first engagement surface **2913** of the follower guide **2921** engages the pin **2902** of the barrel cam cylinder **2920** in the first locked position to inhibit the follower guide **2921** from rotating in a direction away from the second locked position. The second engagement surface **2914** of the follower guide **2921** engages the pin **2902** in the second locked position to inhibit the follower guide **2921** from rotating in a direction away from the first locked position.

[0317] In some embodiments, the barrel cam cylinder **2920** includes a second radially outwardly-facing curved recess (not shown) that receives the finger **2910** in the second locked position of the follower guide **2921**. In some embodiments, the barrel cam cylinder **2920** lacks any radially outwardly-facing curved recesses. Instead, the finger **2910** remains in engagement with the outer surface of the barrel cam cylinder **2920** in the first locked position, the second locked position, and the unlocked position.

[0318] Referring now to FIG. **31**, an exemplary barrel cam assembly **3119** is depicted. The barrel cam assembly **3119** may be used with a surgical device, such as the surgical device **1206** described above, in place of the barrel cam assembly **1519**. The barrel cam assembly **3119** includes a barrel cam cylinder **3120** that rotatably carries a follower guide **3121**. The barrel cam cylinder **3120** and the follower guide **3121** may have the same features as any of the barrel cam cylinders and the follower guides, respectively, described herein (for example, the cam groove **1544** and the follower aperture **1565**, respectively), with the exception of the relative rotation-inhibiting mechanism.

[0319] The barrel cam assembly **3119** includes a relative rotation-inhibiting mechanism **3175** that inhibits some rotation of the follower guide **3121** relative to the barrel cam cylinder **3120**.

Generally, the relative rotation-inhibiting mechanism **3175** permits the follower guide **3121** to move from a first relative rotation-inhibiting position (referred to as the “first locked position” for simplicity) to a second relative rotation-inhibiting position (referred to as the “second locked position” for simplicity) and vice versa. In the first locked position, the mechanism **3175** initially inhibits the follower guide **3121** from rotating in a first direction relative to the barrel cam cylinder **3120** (that is, toward the second locked position) and inhibits rotation of the follower guide **3121** in a second direction relative to the barrel cam cylinder **3120**. In the second locked position, the mechanism **3175** initially inhibits the follower guide **3121** from rotating in the second direction relative to the barrel cam cylinder **3120** (that is, toward the first locked position) and inhibits rotation of the follower guide **3121** in the first direction relative to the barrel cam cylinder **3120**.

[0320] The relative rotation-inhibiting mechanism **3175** includes a radially-outwardly projecting pin **3102** carried near the proximal end of the barrel cam cylinder **3120**. The pin **3102** may be coupled to the barrel cam cylinder **3120** in various manners. For example, the pin **3102** may be press fitted or adhered in a hole formed on the barrel cam cylinder **3120**.

[0321] The relative rotation-inhibiting mechanism **3175** also includes first and second spherical elements **3108** and **3110**. The first and second spherical elements **3108** and **3110** are fixedly received in cylindrical blind holes formed in the outer surface of the barrel cam cylinder **3120**. The

first and second spherical elements **3108** and **3110** protrude from the holes to engage the inner surface of the follower guide **3121**. As such, the first and second spherical elements **3108** and **3110** facilitate frictional engagement between the barrel cam cylinder **3120** and the follower guide **3121**. [0322] In some embodiments, the first and second spherical elements **3108** and **3110** are press fitted into the holes on the barrel cam cylinder **3120**. In some embodiments, the first and second spherical elements **3108** and **3110** may be disposed near the proximal end and the distal end, respectively, of the barrel cam cylinder **3120**. In some embodiments, the first and second spherical elements **3108** and **3110** are formed of steel, polyethylene, or the like. In some embodiments, the first and second spherical elements **3108** and **3110** are similar to the rolling elements, or balls, of a ball bearing.

[0323] The relative rotation-inhibiting mechanism **3175** further includes a first transversely-facing engagement surface **3113** and a second transversely-facing engagement surface **3114** formed near the proximal end of the follower guide **3121**.

[0324] In some embodiments, the relative rotation-inhibiting mechanism **3175** further includes a through hole **3112** formed on the follower guide **3121**. The through hole **3112** has a diameter that is smaller than the diameter of the first spherical element **3108**. The through hole **3112** receives the first spherical element **3108** in the first locked position of the follower guide **3121**. Entry of the first spherical element **3108** in the through hole **3112** provides tactile feedback to the device user.

[0325] Interaction of the first and second spherical elements **3108** and **3110** with the follower guide **3121**, and the resulting motion of the follower guide **3121** relative to the barrel cam cylinder **3120**, are described with reference to the cam slot and aperture profiles illustrated in FIGS. 22A-22C. Referring first to FIG. 22A and upon a first actuation of the trigger assembly, the trigger pin initially moves from position BC1' to BC3'. As the trigger pin moves in this manner, the follower guide **3121** rotates together with the barrel cam cylinder **3120** due to the frictional forces between the first and second spherical elements **3108** and **3110** and the follower guide **3121**. Stated another way, the first and second spherical elements **3108** and **3110** initially hold the follower guide **3121** in the first locked position relative to the barrel cam cylinder **3120**.

[0326] Referring to FIG. 22B and by continuing the first actuation of the trigger assembly, the trigger pin moves from position BC3' to BC4'. In position BC3', the trigger pin engages the wall of the follower guide aperture. By moving toward position BC4', the trigger pin applies a force to the follower guide **3121** that overcomes the frictional forces between the first and second spherical elements **3108** and **3110** and the follower guide **3121**. As a result, the first and second spherical elements **3108** and **3110** slip against the follower guide **3121**, and the follower guide **3121** is "unlocked" and rotates relative to the barrel cam cylinder **3120** as the trigger pin moves from position BC3' to BC4'. When the trigger pin reaches position BC4', the trigger pin no longer applies the force to the follower guide **3121**.

[0327] When the user releases the trigger assembly, the trigger pin moves from position BC4' to BC6'. As the trigger pin moves in this manner, the follower guide **3121** rotates together with the barrel cam cylinder **3120** due to the frictional forces between the first and second spherical elements **3108** and **3110** and the follower guide **3121**. Stated another way, the first and second spherical elements **3108** and **3110** hold the follower guide **3121** in the second locked position relative to the barrel cam cylinder **3120** after the first actuation of the trigger assembly.

[0328] Still referring to FIG. 22B and upon a second actuation of the trigger assembly, the trigger pin initially moves from position BC6' to BC8'. As the trigger pin moves in this manner, the follower guide **3121** rotates together with the barrel cam cylinder **3120** due to the frictional forces between the first and second spherical elements **3108** and **3110** and the follower guide **3121**. Stated another way, the first and second spherical elements **3108** and **3110** initially hold the follower guide **3121** in the second locked position relative to the barrel cam cylinder **3120**.

[0329] Referring to FIG. 22C and by continuing the second actuation of the trigger assembly, the trigger pin moves from position BC8' to BC9'. In position BC8', the trigger pin engages the wall of

the follower guide aperture. By moving toward position BC9', the trigger pin applies a force to the follower guide **3121** that overcomes the frictional forces between the first and second spherical elements **3108** and **3110** and the follower guide **3121**. As a result, the first and second spherical elements **3108** and **3110** slip against the follower guide **3121**, and the follower guide **3121** is "unlocked" and rotates relative to the barrel cam cylinder **3120** as the trigger pin moves from position BC8' to BC9'. When the trigger pin reaches position BC9', the trigger pin no longer applies the force to the follower guide **3121**.

[0330] When the user releases the trigger assembly, the trigger pin moves from position BC9' to BC1'. As the trigger pin moves in this manner, the follower guide **3121** rotates together with the barrel cam cylinder **3120** due to the frictional forces between the first and second spherical elements **3108** and **3110** and the follower guide **3121**. Stated another way, the first and second spherical elements **3108** and **3110** hold the follower guide **3121** in the first locked position relative to the barrel cam cylinder **3120** after the second actuation of the trigger assembly.

[0331] In addition and as shown in FIG. **31**, the first engagement surface **3113** of the follower guide **3121** engages the pin **3102** of the barrel cam cylinder **3120** in the first locked position to inhibit the follower guide **3121** from rotating in a direction away from the second locked position. The second engagement surface **3114** of the follower guide **3121** engages the pin **3102** in the second locked position to inhibit the follower guide **3121** from rotating in a direction away from the first locked position.

[0332] In some embodiments, the barrel cam assembly **3119** includes one spherical element, or three or more spherical elements. In some embodiments, one or more of the spherical elements **3108** and/or **3110** are received in through holes formed on the follower guide **3121** in the first and/or second locked positions. Entry of the spherical elements in the through holes provides tactile feedback to the device user.

[0333] Referring now to FIG. **32**, an exemplary barrel cam assembly **3219** is depicted. The barrel cam assembly **3219** may be used with a surgical device, such as the surgical device **1206** described above, in place of the barrel cam assembly **1519**. The barrel cam assembly **3219** includes a barrel cam cylinder **3220** that rotatably carries a follower guide **3221**. The barrel cam cylinder **3220** and the follower guide **3221** may have the same features as any of the barrel cam cylinders and the follower guides, respectively, described herein (for example, the cam groove **1544** and the follower aperture **1565**, respectively), with the exception of the relative rotation-inhibiting mechanism.

[0334] The barrel cam assembly **3219** includes a relative rotation-inhibiting mechanism **3275** that inhibits some rotation of the follower guide **3221** relative to the barrel cam cylinder **3220**.

Generally, the relative rotation-inhibiting mechanism **3275** permits the follower guide **3221** to move from a first relative rotation-inhibiting position (referred to as the "first locked position" for simplicity) to a second relative rotation-inhibiting position (referred to as the "second locked position" for simplicity) and vice versa. In the first locked position, the mechanism **3275** initially inhibits the follower guide **3221** from rotating in a first direction relative to the barrel cam cylinder **3220** (that is, toward the second locked position) and inhibits rotation of the follower guide **3221** in a second direction relative to the barrel cam cylinder **3220**. In the second locked position, the mechanism **3275** initially inhibits the follower guide **3221** from rotating in the second direction relative to the barrel cam cylinder **3220** (that is, toward the first locked position) and inhibits rotation of the follower guide **3221** in the first direction relative to the barrel cam cylinder **3220**.

[0335] The relative rotation-inhibiting mechanism **3275** includes a longitudinally extending tab **3277** formed near the proximal end of the follower guide **3221**. The tab **3277** engages a semi-annular flange **3279** formed near the proximal end of the barrel cam cylinder **3220**. As shown in FIG. **31**, the tab **3277** engages one side of the flange **3279** in the first locked position to inhibit the follower guide **3221** from rotating in a direction away from the second locked position. The tab **3277** engages the other side of the flange **3279** in the second locked position to inhibit the follower guide **3221** from rotating in a direction away from the first locked position.

[0336] The relative rotation-inhibiting mechanism 3275 also includes a friction element 3208 that is fixedly carried by the barrel cam cylinder 3220. The friction element 3208 engages the follower guide 3221 to provide frictional engagement between the barrel cam cylinder 3220 and the follower guide 3221. In some embodiments, the friction element 3208 is formed by Teflon, polyethylene, nylon, or the like. In some embodiments, the friction element 3208 has a disk shape, a spring washer shape, or a wave washer shape. In some embodiments, the friction element 3208 is adhered to the barrel cam cylinder 3220.

[0337] Interaction of the friction element 3208 with the follower guide 3221, and the resulting motion of the follower guide 3221 relative to the barrel cam cylinder 3220, are described with reference to the cam slot and aperture profiles illustrated in FIGS. 22A-22C. Referring first to FIG. 22A and upon a first actuation of the trigger assembly, the trigger pin initially moves from position BC1' to BC3'. As the trigger pin moves in this manner, the follower guide 3221 rotates together with the barrel cam cylinder 3220 due to the frictional forces between the friction element 3208 and the follower guide 3221. Stated another way, the friction element 3208 initially holds the follower guide 3221 in the first locked position relative to the barrel cam cylinder 3220.

[0338] Referring to FIG. 22B and by continuing the first actuation of the trigger assembly, the trigger pin moves from position BC3' to BC4'. In position BC3', the trigger pin engages the wall of the follower guide aperture. By moving toward position BC4', the trigger pin applies a force to the follower guide 3221 that overcomes the frictional forces between the friction element 3208 and the follower guide 3221. As a result, the friction element 3208 slips against the follower guide 3221, and the follower guide 3221 is "unlocked" and rotates relative to the barrel cam cylinder 3220 as the trigger pin moves from position BC3' to BC4'. When the trigger pin reaches position BC4', the trigger pin no longer applies the force to the follower guide 3221.

[0339] When the user releases the trigger assembly, the trigger pin moves from position BC4' to BC6'. As the trigger pin moves in this manner, the follower guide 3221 rotates together with the barrel cam cylinder 3220 due to the frictional forces between the friction element 3208 and the follower guide 3221. Stated another way, the friction element 3208 holds the follower guide 3221 in the second locked position relative to the barrel cam cylinder 3220 after the first actuation of the trigger assembly.

[0340] Still referring to FIG. 22B and upon a second actuation of the trigger assembly, the trigger pin initially moves from position BC6' to BC8'. As the trigger pin moves in this manner, the follower guide 3221 rotates together with the barrel cam cylinder 3220 due to the frictional forces between the friction element 3208 and the follower guide 3221. Stated another way, the friction element 3208 initially holds the follower guide 3221 in the second locked position relative to the barrel cam cylinder 3220.

[0341] Referring to FIG. 22C and by continuing the second actuation of the trigger assembly, the trigger pin moves from position BC8' to BC9'. In position BC8', the trigger pin engages the wall of the follower guide aperture. By moving toward position BC9', the trigger pin applies a force to the follower guide 3221 that overcomes the frictional forces between the friction element 3208 and the follower guide 3221. As a result, the friction element 3208 slips against the follower guide 3221, and the follower guide 3221 is "unlocked" and rotates relative to the barrel cam cylinder 3220 as the trigger pin moves from position BC8' to BC9'. When the trigger pin reaches position BC9', the trigger pin no longer applies the force to the follower guide 3221.

[0342] When the user releases the trigger assembly, the trigger pin moves from position BC9' to BC1'. As the trigger pin moves in this manner, the follower guide 3221 rotates together with the barrel cam cylinder 3220 due to the frictional forces between the friction element 3208 and the follower guide 3221. Stated another way, the friction element 3208 holds the follower guide 3221 in the first locked position relative to the barrel cam cylinder 3220 after the second actuation of the trigger assembly.

[0343] In some embodiments, the barrel cam cylinder 3220 fixedly carries multiple friction

elements. In some embodiments, the follower guide **3321** fixedly carries one or more friction elements that engage the outer surface of the barrel cam cylinder **3320**.

[0344] Referring now to FIGS. **33** and **34**, an exemplary barrel cam assembly **3319** is depicted. The barrel cam assembly **3319** may be used with a surgical device, such as the surgical device **1206** described above, in place of the barrel cam assembly **1519**. The barrel cam assembly **3319** includes a barrel cam cylinder **3320** that rotatably carries a follower guide **3321**. The barrel cam cylinder **3320** and the follower guide **3321** may have the same features as any of the barrel cam cylinders and the follower guides, respectively, described herein (for example, the cam groove **1544** and the follower aperture **1565**, respectively), with the exception of the relative rotation-inhibiting mechanism.

[0345] The barrel cam assembly **3319** includes a relative rotation-inhibiting mechanism **3375** that inhibits some rotation of the follower guide **3321** relative to the barrel cam cylinder **3320**.

Generally, the relative rotation-inhibiting mechanism **3375** permits the follower guide **3321** to move from a first relative rotation-inhibiting position (referred to as the “first locked position” for simplicity) to a second relative rotation-inhibiting position (referred to as the “first locked position” for simplicity) and vice versa. In the first locked position, the mechanism **3375** initially inhibits the follower guide **3321** from rotating in a first direction relative to the barrel cam cylinder **3320** (that is, toward the second locked position) and inhibits rotation of the follower guide **3321** in a second direction relative to the barrel cam cylinder **3320**. In the second locked position, the mechanism **3375** initially inhibits the follower guide **3321** from rotating in the second direction relative to the barrel cam cylinder **3320** (that is, toward the first locked position) and inhibits rotation of the follower guide **3321** in the first direction relative to the barrel cam cylinder **3320**.

[0346] The relative rotation-inhibiting mechanism **3375** includes a longitudinally extending tab **3377** formed near the proximal end of the follower guide **3321**. The tab **3377** engages a semi-annular flange **3379** formed near the proximal end of the barrel cam cylinder **3320**. As shown in FIGS. **33** and **34**, the tab **3377** engages one side of the flange **3379** in the first locked position to inhibit the follower guide **3321** from rotating in a direction away from the second locked position. The tab **3377** engages the other side of the flange **3379** in the second locked position to inhibit the follower guide **3321** from rotating in a direction away from the first locked position.

[0347] The relative rotation-inhibiting mechanism **3375** also includes a first magnetic element **3308** that is fixedly carried by the barrel cam cylinder **3320**. The first magnetic element **3308** may be a magnet or may be formed from one or more ferromagnetic materials that are attracted to magnets (for example, steel). The relative rotation-inhibiting mechanism **3375** further includes second and third magnetic elements **3310** and **3312** that are fixedly carried by the follower guide **3321**. The second and third magnetic elements **3310** and **3312** may be magnets or may be formed from one or more ferromagnetic materials that are attracted to magnets (for example, steel) if the first magnetic element **3308** is a magnet.

[0348] Interaction of the first magnetic element **3308** with the second and third magnetic elements **3310** and **3312**, and the resulting motion of the follower guide **3321** relative to the barrel cam cylinder **3320**, are described with reference to the cam slot and aperture profiles illustrated in FIGS. **22A-22C**. Referring first to FIG. **22A** and upon a first actuation of the trigger assembly, the trigger pin initially moves from position **BC1'** to **BC3'**. As the trigger pin moves in this manner, the follower guide **3321** rotates together with the barrel cam cylinder **3320** due to the magnetic attraction forces between the first magnetic element **3308** and the second magnetic element **3310**. Stated another way, the first magnetic element **3308** and the second magnetic element **3310** initially hold the follower guide **3321** in the first locked position relative to the barrel cam cylinder **3320**.

[0349] Referring to FIG. **22B** and by continuing the first actuation of the trigger assembly, the trigger pin moves from position **BC3'** to **BC4'**. In position **BC3'**, the trigger pin engages the wall of the follower guide aperture. By moving toward position **BC4'**, the trigger pin applies a force to the follower guide **3321** that overcomes the magnetic attraction forces between the first magnetic

element **3308** and the second magnetic element **3310**. As a result, the first magnetic element **3308** and the second magnetic element **3310** move apart, and the follower guide **3321** is “unlocked” and rotates relative to the barrel cam cylinder **3320** as the trigger pin moves from position BC3’ to BC4’. As the trigger pin approaches position BC4’, the first magnetic element **3308** and the third magnetic element **3312** are magnetically attracted to each other. The follower guide **3321** thereby enters the second locked position relative to the barrel cam cylinder **3320**.

[0350] When the user releases the trigger assembly, the trigger pin moves from position BC4’ to BC6’. As the trigger pin moves in this manner, the follower guide **3321** rotates together with the barrel cam cylinder **3320** due to the magnetic attraction forces between the first magnetic element **3308** and the third magnetic element **3312**. Stated another way, the first magnetic element **3308** and the third magnetic element **3312** hold the follower guide **3321** in the second locked position relative to the barrel cam cylinder **3320** after the first actuation of the trigger assembly.

[0351] Still referring to FIG. 22B and upon a second actuation of the trigger assembly, the trigger pin initially moves from position BC6’ to BC8’. As the trigger pin moves in this manner, the follower guide **3321** rotates together with the barrel cam cylinder **3320** due to the magnetic attraction forces between the first magnetic element **3308** and the third magnetic element **3312**. Stated another way, the first magnetic element **3308** and the third magnetic element **3312** initially hold the follower guide **3321** in the second locked position relative to the barrel cam cylinder **3320**.

[0352] Referring to FIG. 22C and by continuing the second actuation of the trigger assembly, the trigger pin moves from position BC8’ to BC9’. In position BC8’, the trigger pin engages the wall of the follower guide aperture. By moving toward position BC9’, the trigger pin applies a force to the follower guide **3321** that overcomes the magnetic attraction forces between the first magnetic element **3308** and the third magnetic element **3312**. As a result, the first magnetic element **3308** and the third magnetic element **3312** move apart, and the follower guide **3321** is “unlocked” and rotates relative to the barrel cam cylinder **3320** as the trigger pin moves from position BC8’ to BC9’. As the trigger pin approaches position BC9’, the first magnetic element **3308** and the second magnetic element **3310** are magnetically attracted to each other. The follower guide **3321** thereby returns to the first locked position relative to the barrel cam cylinder **3320**.

[0353] When the user releases the trigger assembly, the trigger pin moves from position BC9’ to BC1’. As the trigger pin moves in this manner, the follower guide **3321** rotates together with the barrel cam cylinder **3320** due to the magnetic attraction forces between the first magnetic element **3308** and the second magnetic element **3310**. Stated another way, the first magnetic element **3308** and the second magnetic element **3310** hold the follower guide **3321** in the first locked position relative to the barrel cam cylinder **3320** after the second actuation of the trigger assembly.

[0354] In some embodiments, the follower guide **3321** carries one magnetic element, and the barrel cam cylinder **3320** carries two magnetic elements.

[0355] FIG. 35 depicts two-dimensional illustrations of the profile of a cutting tip slot **3516**, the profile of a barrel cam slot **3544**, and the profile of a follower guide aperture **3565** that may be formed on any of the cutting tips, barrel cam cylinders, and follower guides, respectively, described herein. FIGS. 36A-36C depict the how actuation of a trigger (which may be any of the triggers described herein), and the resulting movement of the trigger pin **3528** (which may be any of the trigger pins described herein), results in rotational movement of the barrel cam cylinder, the follower guide, and the cutting tip, and translation movement of the cutting tip. In these figures, a horizontal axis for the profiles of the slots **3516** and **3544** is the degree(s) of rotation of the cutting tip and the barrel cam cylinder. A vertical axis for the profile of the cam slot **3516** for the cutting tip is the amount of longitudinal displacement, if any, of the cutting tip. The vertical axis for the profile of the cam slot **3544** for the barrel cam cylinder is the amount of longitudinal displacement of the trigger assembly (and trigger pin).

[0356] In FIGS. 35A-35C, the aperture **3565** of the follower guide is shown as a dashed line and is overlaid on the profile of the cam slot **3544** for the barrel cam cylinder to illustrate the rotational

position of the aperture **3565** relative to the cam slot **3544**. As shown in FIGS. **35A-35C** and explained in further detail below, the rotational position of the aperture **3565** changes relative to the cam slot **3544** during actuation of the trigger.

[0357] Generally, an initial, or first, actuation of the trigger (that is, pulling the trigger as far as permitted by the handle assembly and then releasing the trigger so that it returns to its home position) results in a net rotational displacement of the cutting tip and the barrel cam cylinder about 254 degrees in one direction—clockwise when looking from the handle to the tip. The first actuation also extends the cutting tip from the outer band and then returns the cutting tip to the sheathed position as the cutting tip rotates. A subsequent, or second, actuation of the trigger results in a net rotational displacement of the cutting tip and the barrel cam cylinder about 254 degrees in the opposite direction—counter-clockwise when looking from the handle to the tip. The second actuation also extends the cutting tip from the outer band and then returns the cutting tip to the sheathed position as the cutting tip rotates. Additional “odd” actuations (that is, a third actuation, a fifth actuation, and so on) cause the same device motions as the first actuation of the trigger. Additional “even” actuations (that is, a fourth actuation, a sixth actuation, and so on) cause the same device motions as the second actuation of the trigger.

[0358] Referring specifically to FIG. **36A**, prior to the first actuation of the trigger, the trigger pin is at a first home position (BC1”) within the barrel cam cylinder, the guide pin is at its initial position (CT1”), and the cutting tip is at a recessed position within the outer sheath. In addition, the follower guide is at its first relative rotation-inhibiting position (GS1”) with respect to the barrel cam cylinder. Upon initiating the first actuation of the trigger, the trigger pin moves proximally, and the barrel cam cylinder, the cutting tip, and the follower guide rotate in a clockwise direction relative to the trigger pin (from a vantage point proximal of the barrel cam cylinder). When the trigger pin is at position BC2”, the trigger pin is in the intersection **3545** of the barrel cam slot **3544**. Additionally, when the trigger pin is at position BC2” (i) the barrel cam cylinder and cutting tip have rotated about 127 degrees in a clockwise direction since initiating the first actuation of the trigger, (ii) the guide pin is at position CT2”, and (iii) the cutting tip is at a partially extended position.

[0359] Continuing to refer to FIG. **36A**, when the trigger pin is at position BC2”, the trigger pin abuts a first curved wall **3567** of the follower guide aperture **3565**. Because the first curved wall **3567** of the follower guide aperture **3565** is aligned with one path of the intersection **3545** of the barrel cam slot **3544**, the follower guide prevents the trigger pin from traveling in the alternative path and guides the trigger pin straight through the intersection **3545** of the barrel cam slot **3544** as the trigger pin passes position BC2”. In some embodiments, the apex between the first curved wall **3567** and the second curved wall **3569** is offset from the wall of the barrel cam slot **3544** in a longitudinal direction (for example, by about 0.025 inches) to inhibit the trigger pin from engaging the apex. This facilitates guiding the trigger pin straight through the intersection **3545** of the barrel cam slot **3544**. In some embodiments, the first curved wall **3567** extends beyond the wall of the barrel cam slot **3544** (for example, the perpendicular distance between the barrel cam slot **3544** and the furthest point on the first curved wall **3567** may be about 0.010 inches) to facilitate smooth motion of the trigger pin as the trigger pin passes through the intersection **3545** and to permit a relatively small amount of misalignment between the barrel cam and the follower guide.

[0360] As the user continues to pull the trigger, and the trigger pin continues to move proximally, the barrel cam cylinder, the cutting tip, and the follower guide continue to rotate in a clockwise direction from position BC2” to position BC3”. When the trigger pin is at position BC3”, (i) the barrel cam cylinder and cutting tip have rotated about 140 degrees in a clockwise direction since initiating the first actuation of the trigger, (ii) the guide pin is at position CT3”, and (iii) the cutting tip is at a fully extended position.

[0361] As the user continues to pull the trigger, and the trigger pin continues to move proximally, and the barrel cam cylinder, the cutting tip, and the follower guide continue to rotate in a clockwise

direction. Specifically, the trigger pin moves from position BC3" to position BC4". When the trigger pin is at position BC4", (i) the barrel cam cylinder and cutting tip have rotated about 177 degrees in a clockwise direction since initiating the first actuation of the trigger, (ii) the guide pin is at position CT4", and (iii) the cutting tip is at a partially extended position.

[0362] When the trigger pin is at position BC4", the trigger pin abuts a wall **3571** of the follower guide aperture **3565**. As the trigger pin has moved from the first home position (BC1") to position BC4", the follower guide and barrel cam cylinder have rotated in unison with one another due to the presence of one of the relative rotation-inhibiting mechanisms described above. For example, the barrel cam cylinder and the follower guide include the relative rotation-inhibiting mechanism **2475** (see FIGS. **24-28**). In this case, the first spring prong **2410** on the follower guide has been engaged with the curved recess **2404** of the protrusion **2402** of the barrel cam cylinder, thereby preventing the follower guide and barrel cam cylinder from rotationally moving relative to one another. Referring to FIG. **36B**, as the trigger pin moves beyond position BC4" towards position BC5", the trigger pin engages the wall **3571** of the follower guide, thereby rotating the follower guide in a counter-clockwise direction relative to the barrel cam cylinder and forcing the follower guide from its first relative rotation-inhibiting position (GS1") to a movable position. That is, as the trigger pin moves from position BC4" to position BC5", the follower guide is in a movable position relative to the barrel cam cylinder. Once the trigger pin reaches position BC5", the follower guide and barrel cam cylinder are in a second relative rotation-inhibiting position (GS2"). In the second relative rotation-inhibiting position (GS2"), for example, the second spring prong **2412** on the follower guide engages the curved recess **2404** of the protrusion **2402** of the barrel cam cylinder, and a first engagement surface **2413** on the follower guide engages the protrusion **2402** to prevent the follower guide from further rotating in the counter-clockwise direction relative to the barrel cam cylinder.

[0363] Continuing to refer to FIG. **36B**, as the user continues to pull the trigger to move the trigger pin from position BC4" to position BC5", the barrel cam cylinder and the cutting tip continue to rotate in a clockwise direction relative to the trigger pin. The follower guide, however, does not continue to rotate relative to the trigger pin and thereby rotates relative to the barrel cam cylinder. Specifically, the follower guide rotates about 110 degrees in a counter-clockwise direction to a second relative rotation-inhibiting position (GS2") relative to the barrel cam cylinder. When the trigger pin is at position BC5" (i) the trigger has reached the end of its proximal travel, (ii) the barrel cam cylinder has rotated about 287 degrees in a clockwise direction since initiating the first actuation of the trigger, (iii) the guide pin is at position CT5", and (iv) the cutting tip is at its most recessed position. In addition, the cutting tip has rotated a total of about 284 degrees in a clockwise direction since initiating the first actuation of the trigger. The cutting tip rotates less than the barrel cam cylinder because the guide pin engages the wall of the cutting tip slot **3516** (that is, the guide pin reaches position CT5") before the trigger pin reaches position BC5". The barrel cam cylinder rotates about 3 degrees in the clockwise direction after the guide pin engages the wall of the cutting tip slot **3516**. As a result, the barrel cam cylinder rotates about 3 degrees relative to the cutting tip when the trigger pin approaches position BC5". The relative rotation between the barrel cam cylinder and the cutting tip is accounted for by rotational deflection of the flexible inner sheath.

[0364] After the trigger pin reaches position BC5", the user can release the trigger. Upon the user releasing the trigger, the trigger and trigger pin reverse direction and travel toward their distal position due to the constant force spring attached to the trigger. As the trigger and trigger pin begin to move toward their distal position, the barrel cam cylinder and cutting tip are rotationally stationary relative to the trigger pin. Accordingly, upon the user releasing the trigger, the trigger pin moves from position BC5" toward position BC6". When the trigger is at position BC6" (i) the barrel cam cylinder has rotated about 287 degrees in a clockwise direction since initiating the first actuation of the trigger (ii) the guide pin is still at position CT5", and (iii) the cutting tip is still at its most recessed position. At position BC6", however, the trigger pin is still not at its most distal

position. To reach the trigger pin's most distal position BC7", the barrel cam cylinder rotates about 33 degrees in a counter-clockwise direction. In some embodiments, the trigger pin engages a radiused corner of the wall of the barrel cam slot **3544** and/or the wall of the follower guide aperture **3565** (for example, a 0.050 inch radius) when moving from position BC6" to position BC7". Such a structure may facilitate reliably sliding the trigger pin to position BC7". When the trigger pin is at position BC7", the trigger pin is at a second home position within the barrel cam cylinder. When the trigger pin is at position BC7", (i) the barrel cam cylinder and the cutting tip have rotated a total of about 254 degrees in a clockwise direction since initiating the first actuation of the trigger, (ii) the guide pin is at position CT7", and (iii) the cutting tip is recessed within the outer sheath.

[0365] Still referring to FIG. **36B**, when the trigger pin is at its second home position (BC7"), the follower guide is at its second relative rotation-inhibiting position (GS2"). Upon initiating a second actuation of the trigger to move the trigger pin proximally, the follower guide and the barrel cam cylinder remain stationary with respect to one another, and the barrel cam cylinder, the cutting tip, and the follower guide rotate in a counter-clockwise direction. When the trigger pin is at position BC8", (i) the barrel cam cylinder and cutting tip have rotated about 114 degrees in a counter-clockwise direction since initiating the second actuation of the trigger, (ii) the guide pin is at position CT8", and (iii) the cutting tip is at its most extended position.

[0366] As the user continues the second actuation of the trigger, and the trigger pin continues to move proximally, the barrel cam cylinder, the cutting tip, and the follower guide continue to rotate in a counter-clockwise direction relative to the trigger pin. When the trigger pin is at position BC9", (i) the barrel cam cylinder and cutting tip have rotated about 127 degrees in a counter-clockwise direction since initiating the second actuation of the trigger, (ii) the guide pin is at position CT9", (iii) the cutting tip is at a partially extended position, and (iv) the trigger pin is at the intersection **3545** of the barrel cam slot **3544**. When the trigger pin is at position BC9", the trigger pin abuts a second curved wall **3569** of the follower guide aperture **3565**. Because the second curved wall **3569** of the follower guide aperture **3565** is aligned with one path of the intersection **3545** of the barrel cam slot **3544**, the follower guide prevents the trigger pin from traveling in the alternative path and guides the trigger pin straight through the intersection **3545** of the barrel cam slot **3544** as the trigger pin passes position BC9".

[0367] As the user continues to actuate the trigger, and the trigger pin continues to move proximally, the barrel cam cylinder, the cutting tip, and the follower guide continue to rotate in a counter-clockwise direction. When the trigger pin is at position BC10", (i) the barrel cam cylinder and cutting tip have rotated about 177 degrees in a clockwise direction since initiating the second actuation of the trigger, (ii) the guide pin is at position CT10", (iii) the cutting tip is at a partially extended position, and (iv) the trigger pin abuts a wall **3573** of the follower guide aperture **3565**.

[0368] As the trigger pin has moved from its second home position (BC7") to position BC10", the follower guide and barrel cam cylinder have rotated in unison with one another due to the presence of the relative rotation-inhibiting mechanism. For example, the second spring prong **2412** on the follower guide has been engaged with the curved recess **2404** of the protrusion **2402** of the barrel cam cylinder, thereby preventing the follower guide and barrel cam cylinder from rotationally moving relative to one another. Referring to FIG. **36C**, as the trigger pin moves beyond position BC10" towards position BC11", the trigger pin engages the wall **3573** of the follower guide, thereby rotating the follower guide in the clockwise direction relative to the barrel cam cylinder and forcing the follower guide from its second relative rotation-inhibiting position (GS2") to a movable position. That is, as the trigger pin moves from position BC10" to position BC11", the follower guide is in a movable position relative to the barrel cam cylinder. Once the trigger pin reaches position BC11", the follower guide and barrel cam cylinder have returned to the first relative rotation-inhibiting position (GS1"). In the first relative rotation-inhibiting position (GS1"), for example, the first spring prong **2410** on the follower guide engages the curved recess **2404** of

the protrusion **2402** of the barrel cam cylinder, and a second engagement surface **2414** on the follower guide (see FIG. 25) engages the protrusion **2402** to prevent the follower guide from further rotating in the clockwise direction relative to the barrel cam cylinder.

[0369] Continuing to refer to FIG. 36C, as the user continues the second actuation of the trigger, and the trigger pin continues to move proximally, the barrel cam cylinder and the cutting tip continue to rotate in a counter-clockwise direction. The follower guide, however, does not continue to rotate relative to the trigger pin, and the trigger pin rotates only relative to the barrel cam cylinder. Specifically, the follower guide rotates about 110 degrees in a clockwise direction to the first relative rotation-inhibiting position (GS1") relative to the barrel cam cylinder. When the trigger pin is at position BC11" (i) the trigger has reached the end of its proximal travel, (ii) the barrel cam cylinder and the cutting tip have rotated about 287 degrees in a counter-clockwise direction since initiating the second actuation of the trigger, (iii) the guide pin is at position CT11", and (iv) the cutting tip is at its most recessed position.

[0370] After the trigger pin reaches position BC11", the user can release the trigger. Upon the user releasing the trigger, the trigger and trigger pin reverse direction and travel toward their distal position due to the constant force spring attached to the trigger. As the trigger and trigger pin begin to move toward their distal position, the barrel cam cylinder and cutting tip are rotationally stationary relative to the trigger pin. Accordingly, upon the user releasing the trigger, the trigger pin moves from position BC11" toward position BC12". When the trigger is at position BC12" (i) the barrel cam cylinder and the cutting tip are still rotated about 287 degrees in a counter-clockwise direction since initiating the second actuation of the trigger (ii) the guide pin is still at position CT11", and (iii) the cutting tip is still at its most recessed position. As the trigger and trigger pin continue to move toward their distal position, the barrel cam cylinder and cutting tip rotate about 33 degrees in a clockwise direction relative to the trigger pin and the trigger pin returns to its first home position (BC1"). In some embodiments, the trigger pin engages a radiused corner of the wall of the barrel cam slot **3544** and/or the wall of the follower guide aperture **3565** (for example, a 0.050 inch radius) when moving from position BC12" to position BC1". Such a structure may facilitate reliably sliding the trigger pin to position BC1". When the trigger pin returns to its first home position (BC1"), the guide pin returns to its initial position (CT1"), and the cutting tip remains at a recessed position within the outer sheath. The user may then repeat the process, if so desired.

[0371] FIG. 37 depicts a two-dimensional illustration of the profile of a barrel cam slot **3744** that may be formed on any of the barrel cam cylinders described herein. The cam slot **3744** defines a generally "hourglass"-like or "figure eight"-like path for the follower (for example, the trigger pin **1528**). The trigger pin traverses about half of the cam slot **3744** when an initial, or first, actuation is applied to the trigger, and the trigger pin traverses the remainder of the cam slot **3744** (that is, about half of the cam slot **3744**) when a subsequent, or second, actuation is applied to the trigger. In each case, the follower guide (for example, the follower guide **1521**) causes the trigger pin to travel straight through the intersection (or crossing portion) **3745** of the cam slot **3744** during each actuation of the trigger. Stated another way, the follower guide causes the trigger pin to travel from a first leg **3747** of the cam slot **3744** to a second leg **3749** of the cam slot **3744**, and then from a third leg **3751** of the cam slot **3744** to a fourth leg **3753** of the cam slot **3744**.

[0372] Each of the legs **3747**, **3749**, **3751**, and **3753** is shaped to inhibit the trigger pin from engaging the walls of the legs **3747**, **3749**, **3751**, and **3753** when travelling through the intersection **3745** and potentially binding up with the barrel cam assembly. In some embodiments, each of the legs **3747**, **3749**, **3751**, and **3753** has a width that taperingly increases proceeding toward the intersection **3745**. Specifically, the first leg **3747** includes a first angled wall **3760** that is adjacent to the intersection **3745** and disposed on a distal side of the first leg **3747**. The first angled wall **3760** is angled away from the first trigger pin path **3762** (that is, the path traversed by the trigger pin during the first actuation of the trigger) proceeding toward the intersection **3745**. The first

angled wall **3760** may have a length in the range of about 0.25 to 0.40 inches, and more specifically about 0.33 inches, and may be angled away from the first trigger pin path **3762** by an angle in the range of about 1 to 7 degrees, and more specifically about 4 degrees. Similarly, the second leg **3749** includes a second angled wall **3764** that is adjacent to the intersection **3745** and disposed on a distal side of the second leg **3749**. The second angled wall **3764** is angled away from the first trigger pin path **3762** proceeding toward the intersection **3745**. The second angled wall **3764** may have a length in the range of about 0.18 to 0.32 inches, and more specifically about 0.25 inches, and may be angled away from the first trigger pin path **3762** by an angle in the range of about 4 to 10 degrees, and more specifically about 7 degrees. The third leg **3751** includes a third angled wall **3766** that is adjacent to the intersection **3745** and disposed on a distal side of the third leg **3751**. The third angled wall **3766** is angled away from the second trigger pin path **3768** (that is, the path traversed by the trigger pin during the second actuation of the trigger) proceeding toward the intersection **3745**. The third angled wall **3766** may have a length in the range of about 0.25 to 0.40 inches, and more specifically about 0.33 inches, and may be angled away from the second trigger pin path **3768** by an angle in the range of about 1 to 7 degrees, and more specifically about 4 degrees. The fourth leg **3753** includes a fourth angled wall **3770** that is adjacent to the intersection **3745** and disposed on a distal side of the fourth leg **3753**. The fourth angled wall **3770** is angled away from the second trigger pin path **3768** proceeding toward the intersection **3745**. The fourth angled wall **3770** may have a length in the range of about 0.18 to 0.32 inches, and more specifically about 0.25 inches, and may be angled away from the second trigger pin path **3768** by an angle in the range of about 4 to 10 degrees, and more specifically about 7 degrees.

[0373] The discussion above describes a barrel cam cylinder and a follower guide in the context of a medical device. However, barrel cam cylinders and follower guides according to the disclosure may be used with other types of devices (for example, non-medical devices) to convert a translational input to a rotational output. That is, barrel cam cylinders and follower guides according to the disclosure may be used such that a first translational input (for example, a first actuation of a translatable trigger) causes a first rotational output (for example, rotation of a shaft in a first direction) and a second translational input (for example, a second actuation of the translatable trigger) causes a second rotational output (for example, rotation of the shaft in a second direction).

[0374] The discussion above discusses that the inner sheath, including the cutting tip, travel at certain rates (e.g., constant and/or variable). However, the rates are also dependent upon the speed at which the inner sheath rotates and travels longitudinally (i.e., extends and/or retracts), and in turn, upon the speed of the actuation of the trigger assembly, including the longitudinal movement of the trigger and the rotational movement of the barrel cam cylinder. Accordingly, the discussion and/or comparison of the rates at which the blade travels assumes that the means for actuating extends the inner sheath at a relatively constant speed. Regardless of whether this assumption is correct, the greater the amount of blade extension per predetermined amount of rotation, the blade will extend at a greater rate and speed, thereby providing the surgical device with the ability to cut more tissue per rotation.

[0375] Referring to FIG. **38**, there is depicted an elevation view of an alternative embodiment of a distal portion of an assembled sheath assembly **3812** of the present disclosure. The sheath assembly **3812** includes an inner sheath assembly and an outer sheath assembly. Referring to FIG. **38A**, which illustrates a cross-sectional view of the distal end of the sheath assembly **3812**, and referring to FIG. **39**, which is an exploded illustration of the distal end of the sheath assembly **3812**, the sheath assembly **3812** may include some or all of the following components: an outer band **3836**; a guide pin **3840**; a cutting tip **3832**; a flexible inner sheath **3820**; a flexible outer sheath **3824**; an outer jacket **3828**; an inner key (not shown); an outer key (not shown); and a rigid inner tube (not shown). Although the inner key, the outer key, and the rigid inner tube are not shown, in FIG. **38**, those items are depicted in FIGS. **6-8** and **16-18**.

[0376] The sheath assembly **3812** depicted in FIGS. **38-39** is similar to the sheath assembly **112** depicted in FIGS. **6-8** and the sheath assembly **1212** depicted in FIGS. **16-18**, but the embodiment of the sheath assembly **112** depicted in FIGS. **38-39** creates a smaller profile. That is, the overall diameter of the sheath assembly **3812** depicted in FIGS. **38-39** is smaller than the outer diameter of the sheath assembly **112** depicted in FIGS. **6-8** and the sheath assembly **1212** depicted in FIGS. **16-18**. As illustrated in FIGS. **6 & 8**, the flexible inner sheath **620** is affixed to the cutting tip **632** in an overlapping manner. Specifically, the proximal end of the cutting tip **632** is inserted into the distal end of the flexible inner sheath **620** such that the flexible inner sheath **620** overlaps the proximal end of the cutting tip **632**. The sheath assembly **1212** depicted in FIGS. **16-18** has a similar overlapping design that depicted of the sheath assembly **112** of FIGS. **6-8**. This overlapping design may increase the overall thickness and diameter of the sheath assembly, thereby potentially increasing the difficulty level of the surgical device's ability, particularly the sheath assembly's ability, to navigate through small diameter vasculature.

[0377] The sheath assembly **3812** depicted in FIG. **39** provides a smaller profile in comparison to the sheath assemblies **112** and **1212** depicted in FIGS. **6-8** and FIGS. **16-18**, respectively, thereby increasing the surgical device's ease and ability, particularly the sheath assembly's ability, to navigate through small diameter vasculature. As illustrated in FIGS. **38-39**, the sheath assembly **3812** has a non-overlapping design. That is, flexible outer sheath **3824** abuts the outer band **3836** in a non-overlapping manner, and the inner sheath **3820** abuts the flexible inner sheath **3820** in a non-overlapping manner. Stated differently, the outer band **3836** is not inserted into the flexible outer sheath **3824**. Rather, the proximal end of the outer band **3836** abuts and is welded to the distal end of the flexible outer sheath **3824** such that the diameters of the lumens of the outer band **3836** and the flexible outer sheath **3824** diametrically align. Similarly, the cutting tip **3832** is not inserted into the flexible inner sheath **382**. Rather, the proximal end of the cutting tip **3832** abuts and is welded to the distal end of the flexible inner sheath **3820** such that the diameters of the lumens of the cutting tip **3832** and the flexible inner sheath **3820** diametrically align.

[0378] As a result of the non-overlapping design, the sheath assembly **3812**, particularly the outer jacket **3828**, outer sheath **3824**, and inner sheath **3820** may have the following dimensions:

TABLE-US-00001 TABLE 1 Outer Sheath Inner Sheath Outer Jacket Inner Outer Inner Outer									
Inner	Outer	Diameter	Diameter	Diameter	Diameter	Diameter	Diameter	Diameter	Size
(in.)	(in.)	+/-0.005	+/-0.005	+/-0.005	+/-0.005	9F	0.156-0.158	0.183-0.191	0.157 0.177 0.119
0.139	11F	0.182-0.184	0.209-0.217	0.183	0.203	0.145	0.165	13F	0.208-0.210 0.235-0.243 0.209
0.229		0.171	0.191						

[0379] Referring to FIGS. **40** and **40A**, there is depicted an enlarged view of the distal end of the outer sheath assembly **3802**. As will be discussed in more detail below, the outer sheath assembly may include a hypotube **3824**. As will also be discussed in more detail below, the hypotube **3824** includes a plurality of segments with different flexibilities resulting from the design of the kerfs in such segments. However, it may be preferable for the most distal end of the hypotube **3824** to include an uncut segment **3842**, which abuts the proximal end of the outer band **3836**. It will be this uncut segment **3842** of the hypotube **3824** that is welded to the outer band **3836** in a non-overlapping manner such that the diameters of the lumens of the outer band **3836** and the flexible outer sheath **3824** diametrically align. Referring to FIGS. **41** and **41A**, the inner sheath assembly may include a hypotube **3820**. And it may be preferable for the most distal end of the hypotube **3820** to include an uncut segment **3832**, which abuts the proximal end of the cutting tip **3832**. It will be this uncut segment **3842** of the hypotube **3824** that is welded to the cutting tip **3832** in a non-overlapping manner such that the diameters of the lumens of the cutting tip **3832** and the flexible inner sheath **3820** diametrically align.

[0380] Referring to FIGS. **42**, **42A** and **42B**, there is depicted the outer sheath **3824** constructed of a hypotube having multiple segments **3842**, **3844**, **3846**, **3848**. Segment **3842** is the most distal segment, and segment **3848** is the most proximal segment. Segments **3842**, **3848** are uncut

segments, and segments **3844**, **3846** are cut segments. That is, segments **3842**, **3848** do not include kerfs, and segments **3844**, **3846** include kerfs. Although segments **3844**, **3846** include kerfs, segment **3844** may have a consistent flexibility from its distal end to its proximal end, and segment **3846** may have a variable flexibility from its distal end to its proximal end. That is, the flexibility of the segment **3846** may vary in a manner such that the segment **3846** has a lower flexibility at its distal end and a greater flexibility at its proximal end. Stated differently, the flexibility of the segment **3846** may vary in a manner such that flexibility decreases from its distal end to its proximal end. Additionally, the distal cut segment **3844** may be more flexible in comparison to the proximal cut segment **3846**. Although only four segments of the hypotube are depicted in FIG. **42** and described herein, less than or more than four segments may be included within the hypotube. And the segments may include a variety of cut and uncut configurations to yield the flexibility characteristics of the hypotube.

[0381] Continuing to refer to FIGS. **42** and **42B**, uncut segment **3842** may have a length of about 0.025 inches, constant flexible (cut) segment **3844** may have a length of about 3.0 inches, variable flexible (cut) segment **3846** may have a length of about 3.732 inches, and uncut segment **3848** may have a length of about 2.0 inches. Segment **3844** may have a consistent flexibility because this segment has a consistent pattern of alternating kerfs **3849** and non-kerf **3851** portions over its length. For example, the pattern of alternating kerfs **3849** and non-kerf **3851** portions for segment **3844** may include a kerf **3849** having a width **3850** of about 0.001 inches and a length **3852** of about 1200 of the circumference followed by a non-kerf portion **3851** having a length **3854** of about 31.5° of the circumference over a longitudinal length of 3.0 inches. It may also be desirable for the kerfs **3849** (or portions thereof) adjacent one another along the axial length of the segment **3844** to be consistently spaced at a constant pitch (P) of about 0.025 inches. As depicted in FIG. **44**, the kerfs **3849** may also be offset from an axis perpendicular to the longitudinal axis of the outer sheath **3824** at a pitch angle (θ) of about 2.0° to about 3.0°, particularly about 2.0°, 2.10, 2.20, 2.30, 2.40, 2.50, 2.60, 2.70, 2.80, 2.90 or 3.0°. The pitch angle may be the same over the entire length of segment **3844**, the pitch angle may increase from the distal end of segment **3844** to its proximal end, or the pitch angle may decrease from the distal end of the segment **3844** to its proximal end.

[0382] Continuing to refer to FIG. **42** variable flexible (cut) segment **3846** has a decreasing flexibility from its distal end to its proximal end because this pitch of the kerfs **3849** increases from its distal end to its proximal end. That is, although segment **3846** has a consistent pattern of alternating kerfs **3849** and non-kerf portions **3851** over its length, for example, alternating kerfs **3849** for about 1200 of the circumference followed by a non-kerf **3851** portion for about 31.5° of the circumference over a longitudinal length of 3.732 inches, the distance between the kerfs **3849** (or portions thereof) adjacent one another along the axial length of the segment **3844** is spaced at an increasing pitch of about 0.025 inches to 0.080 inches from its distal end to its proximal end.

[0383] The kerfs **3849** may also be offset from an axis perpendicular to the longitudinal axis of the outer sheath **3824** at a pitch angle (θ) of about 2.0° to about 10.0°, particularly about 2.0°, 3.0°, 4.0°, 5.0°, 6.0°, 7.0°, 8.0°, 9.0° or 10.0°, including any increment of 0.1° thereof. The pitch angle (θ) may be constant or vary along segment **3846**. That is, the pitch angle may be the same over the entire length of segment **3846**, the pitch angle may increase from the distal end of segment **3846** to its proximal end, or the pitch angle may decrease from the distal end of the segment **3846** to its proximal end. For example, the pitch (e.g., pitch 0.025) and pitch angle (e.g., pitch angle (θ) of 2.0°, 2.2°, 2.6°) may be the same for entirety of segment **3844**, and the pitch and/or pitch angle may increase from distal end (e.g., having a pitch of 0.025 inches and a pitch angle (θ) of 2.0°, 2.2°, 2.6°) to the proximal end of segment **3846** (e.g., having a pitch of 0.0080 inches and pitch angle (θ) of 6.3°, 7.1°, 8.2° respectively) while maintaining the same pattern of alternating kerfs portions **3849** (e.g., 1200 cut) and non-kerf portions **3851** (e.g., 31.5° uncut) for both segments **3844**, **3846**. The varying pattern in segment **3846** provides the inner sheath **3824** with a relatively gradual transition in flexibility from distal, flexible cut segment **3844** to proximal, rigid uncut

segment **3848**.

[0384] Referring to FIGS. **43**, **43A** and **43B**, there is depicted the inner sheath **3820** constructed of a hypotube having multiple segments **3862**, **3864**, **3866**, **3868**. Segment **3862** is the most distal segment and segment **3868** is the most proximal segment. Segments **3862**, **3868** are uncut segments, and segments **3864**, **3866** are cut segments. That is, segments **3862**, **3868** do not include kerfs, and segments **3864**, **3866** include kerfs. Although segments **3864**, **3866** include kerfs, segment **3864** has a consistent flexibility from its distal end to its proximal end, and segment **3866** has a variable flexibility from its distal end to its proximal end. That is, the flexibility of the segment **3866** may vary in a manner such that the segment **3866** has a lower flexibility at its distal end and a greater flexibility at its proximal end. Stated differently, the flexibility of the segment **3866** may vary in a manner such that flexibility decreases from its distal end to its proximal end. Additionally, the distal cut segment **3864** may be more flexible in comparison to the proximal cut segment **3866**. Although only four segments of the hypotube are depicted in FIG. **43** and described herein, less than or more than four segments may be included within the hypotube. And the segments may include a variety of cut and uncut configurations to yield the flexibility characteristics of the hypotube.

[0385] Continuing to refer to FIGS. **43** and **43B**, uncut segment **3862** may have a length of about 0.025 inches, constant flexible (cut) segment **3864** may have a length of about 3.0 inches to about 3.50 inches, variable flexible (cut) segment **3866** may have a length of about 3.725 inches, and uncut segment **3868** may have a length of about 7.0 inches. Segment **3864** may have a consistent flexibility because this segment may have a consistent pattern of alternating kerfs **3869** and non-kerf **3871** portions over its length. For example, the pattern of alternating kerfs **3869** and non-kerf **3871** portions for segment **3864** may include a kerf **3869** having a width **3870** of about 0.001 inches and a length **3872** for about 1200 of the circumference followed by a non-kerf portion **3871** having a length **3872** of about 31.5° of the circumference over a longitudinal length of 3.0 inches. It may also be desirable for the kerfs **3869** (or portions thereof) adjacent one another along the axial length of the segment **3864** to be spaced at a variable pitch (P) of about 0.025 inches to about 0.080 inches from the distal end of segment **3864** to the proximal end of segment **3864**. The kerfs **3869** may also be offset from an axis perpendicular to the longitudinal axis of the inner sheath **3820** at a pitch angle (θ) of about 2.0° to about 4.0°, particularly about 2.0°, 2.10, 2.20, 2.30, 2.40, 2.50, 2.60, 2.70, 2.80, 2.90, 3.0°, 3.1°, 3.2°, 3.3°, 3.4°, 3.5°, 3.6°, 3.7°, 3.8°, 3.9° or 4.0°. The pitch angle may be the same over the entire length of segment **3864**, the pitch angle may increase from the distal end of segment **3864** to its proximal end, or the pitch angle may decrease from the distal end of the segment **3864** to its proximal end.

[0386] Continuing to refer to FIG. **43** variable flexible (cut) segment **3866** may have a decreasing flexibility from its distal end to its proximal end because this pitch of the kerfs increases from its distal end to its proximal end. That is, although segment **3866** has a consistent pattern of alternating kerfs and non-kerf portions over its length, for example, alternating kerfs for about 120° of the circumference followed by a non-kerf portion for about 31.5° of the circumference over a longitudinal length of 3.846 inches, the distance between the kerfs **3870** (or portions thereof) adjacent one another along the axial length of the segment **3864** is spaced at an increasing pitch of about 0.025 inches to 0.080 inches from its distal end to its proximal end.

[0387] The kerfs may also be offset from an axis perpendicular to the longitudinal axis of the inner sheath **3820** at a pitch angle (θ) of about 2.0° to about 15.0°, particularly about 5.0°, 6.0°, 7.0°, 8.0°, 9.0°, 10.0° or 15.0°, including any increment of 0.1° thereof. The pitch angle (θ) may be constant or vary along segment **3866**. That is, the pitch angle may be the same over the entire length of segment **3866**, the pitch angle may increase from the distal end of segment **3866** to its proximal end, or the pitch angle may decrease from the distal end of the segment **3866** to its proximal end. For example, the pitch (e.g., pitch 0.025 inches) and pitch angle (e.g., pitch angle (θ) of 2.4°, 2.8°, 3.3°) may be the same for entirety of segment **3864**, and the pitch and/or pitch angle

may increase from distal end (e.g., having a pitch of 0.025 inches and a pitch angle (θ) of 2.4°, 2.8°, 3.3°) to the proximal end of segment **3866** (e.g., having a pitch of 0.0080 inches and a pitch angle (θ) of 7.6°, 8.8°, 10.4° respectively) while maintaining the same pattern of alternating kerfs portions (e.g., **1200** cut) and non-kerf portions (e.g., 31.5° uncut) for both segments **3864**, **3866**. The varying pattern in segment **3866** provides the inner sheath **3820** with a relatively gradual transition in flexibility from distal, flexible cut segment **3864** to proximal, rigid uncut segment **3868**.

[0388] When comparing the outer sheath (outer hypotube) **3824** with and the inner sheath (inner hypotube) **3820**, the inner sheath **3820** may be more flexible than the outer sheath **3824**. Specifically, the flexible segment **3864** of the inner sheath **3820** may be more flexible in comparison to flexible segment **3844** of the outer sheath **3824**, and flexible segment **3866** of the inner sheath **3820** may be more flexible in comparison to flexible segment **3846** of the outer sheath **3824**.

[0389] Referring to FIG. 45A, there is depicted a block diagram of the outer sheath (outer hypotube) **3824** with respect to the inner sheath (inner hypotube) **3820** illustrating the axial alignment of corresponding segments thereof. The entirety of uncut distal segment **3842** (including its distal and proximal ends) of the outer hypotube **3824** substantially aligns axially with entirety of uncut distal segment **3862** (including its distal and proximal ends) of the inner hypotube **3820**. Although the distal ends of segments of flexible segments **3844**, **3864** having constant flexibility are substantially axially aligned along the longitudinal axes of the outer hypotube **3824** and the inner hypotube **3820**, the length of flexible segment **3864** of the inner hypotube **3820** is longer than the length of flexible segment **3844** of the outer hypotube **3824**, thereby preventing the proximal ends of the flexible segments **3844**, **3864** from axially aligning and creating overlap between the proximal end of flexible segment **3864** (having a constant flexibility) of the inner hypotube **3820** with both the proximal end of flexible segment **3844** (having a constant flexibility) and the distal end of flexible segment **3846** (having a variable flexibility) of the outer hypotube **3824**.

Accordingly, the distal ends of flexible segments **3846**, **3866** do not axially align, thereby creating overlap between the distal end of flexible segment **3846** (having a variable flexibility) of the outer hypotube **3824** with both the proximal end flexible segment **3864** (having a constant flexibility) and the distal end of flexible segment **3866** (having a variable flexibility) of the inner hypotube **3820**. And the proximal ends of flexible segments **3846**, **3866** also do not axially align. The proximal end of flexible segment **3866** (having a variable flexibility) of the inner hypotube **3820** overlaps with both the proximal end of flexible segment **3846** (having a variable flexibility) and the distal end of the proximal uncut segment **3848** of the outer hypotube **3824**. And portions (e.g., distal portion, central portion, and proximal portion) of uncut segments **3848**, **3868** of the outer hypotube **3824** and the inner hypotube **3820** overlap. This arrangement of overlapping segments between the inner and outer hypotubes creates differing flexibilities between the inner and outer hypotubes, thus preventing the likelihood of potential kinking of the hypotubes.

[0390] Although FIG. 45A depicts a block diagram of the outer sheath (outer hypotube) **3824** with respect to the inner sheath (inner hypotube) **3820** having a particular axial alignment of corresponding segments, the segments of the outer sheath (outer hypotube) **3824** and the inner sheath (inner hypotube) **3820** may have alternative arrangements. For example, referring to FIG. 45B, the entirety of uncut distal segment **3842'** (including its distal and proximal ends) of the outer hypotube **3824'** substantially aligns axially with entirety of uncut distal segment **3862'** (including its distal and proximal ends) of the inner hypotube **3820'**. The distal ends of segments of flexible segments **3844'**, **3864'** having constant flexibility are substantially axially aligned along the longitudinal axes of the outer hypotube **3824'** and the inner hypotube **3820'**, and the length of flexible segment **3864'** of the inner hypotube **3820'** is shorter than the length of flexible segment **3844'** of the outer hypotube **3824'**, thereby preventing the proximal ends of the flexible segments **3844'**, **3864'** from axially aligning and creating overlap between the proximal end of flexible

segment **3844'** (having a constant flexibility) of the outer hypotube **3824'** with both the proximal end of flexible segment **3864'** (having a constant flexibility) and the distal end of flexible segment **3866'** (having a variable flexibility) of the inner hypotube **3820'**. Accordingly, the distal ends of flexible segments **3846'**, **3866'** do not axially align, thereby creating overlap between the distal end of flexible segment **3866'** (having a variable flexibility) of the inner hypotube **3820'** with both the proximal end flexible segment **3844'** (having a constant flexibility) and the distal end of flexible segment **3846'** (having a variable flexibility) of the outer hypotube **3824'**. And the proximal ends of flexible segments **3846'**, **3866'** also do not axially align. The proximal end of flexible segment **3866'** (having a variable flexibility) of the inner hypotube **3820'** overlaps with both the proximal end of flexible segment **3846'** (having a variable flexibility) and the distal end of the proximal uncut segment **3848'** of the inner hypotube **3820'**. And portions (e.g., distal portion, central portion, and proximal portion) of uncut segments **3848'**, **3868'** of the outer hypotube **3824'** and the inner hypotube **3820'** overlap.

[0391] Referring to FIG. 45C, the entirety of uncut distal segment **3842"** (including its distal and proximal ends) of the outer hypotube **3824"** substantially aligns axially with entirety of uncut distal segment **3862"** (including its distal and proximal ends) of the inner hypotube **3820"**. The distal ends of segments of flexible segments **3844"**, **3864"** having constant flexibility are substantially axially aligned along the longitudinal axes of the outer hypotube **3824"** and the inner hypotube **3820"**, and the length of flexible segment **3864"** of the inner hypotube **3820"** is shorter than the length of flexible segment **3844"** of the outer hypotube **3824"**, thereby preventing the proximal ends of the flexible segments **3844"**, **3864"** from axially aligning and creating overlap between the proximal end of flexible segment **3844"** (having a constant flexibility) of the outer hypotube **3824"** with both the proximal end of flexible segment **3864"** (having a constant flexibility) and the distal end of flexible segment **3866"** (having a variable flexibility) of the inner hypotube **3820"**.

Accordingly, the distal ends of flexible segments **3846"**, **3866"** do not axially align, thereby creating overlap between the distal end of flexible segment **3866"** (having a variable flexibility) of the inner hypotube **3820"** with both the proximal end flexible segment **3844"** (having a constant flexibility) and the distal end of flexible segment **3846"** (having a variable flexibility) of the outer hypotube **3824"**. And the proximal ends of flexible segments **3846"**, **3866"** also do not axially align. The proximal end of flexible segment **3846"** (having a variable flexibility) of the outer hypotube **3824"** overlaps with both the proximal end of flexible segment **3866"** (having a variable flexibility) and the distal end of the proximal uncut segment **3868"** of the inner hypotube **3820"**. And portions (e.g., distal portion, central portion, and proximal portion) of uncut segments **3848"**, **3868"** of the outer hypotube **3824"** and the inner hypotube **3820"** overlap.

[0392] Referring to FIG. 45D, the entirety of uncut distal segment **3842'''** (including its distal and proximal ends) of the outer hypotube **3824'''** substantially aligns axially with entirety of uncut distal segment **3862'''** (including its distal and proximal ends) of the inner hypotube **3820'''**. The distal ends of segments of flexible segments **3844'''**, **3864'''** having constant flexibility are substantially axially aligned along the longitudinal axes of the outer hypotube **3824'''** and the inner hypotube **3820'''**, and the length of flexible segment **3864'''** of the inner hypotube **3820'''** is longer than the length of flexible segment **3844'''** of the outer hypotube **3824'''**, thereby preventing the proximal ends of the flexible segments **3844'''**, **3864'''** from axially aligning and creating overlap between the proximal end of flexible segment **3864'''** (having a constant flexibility) of the inner hypotube **3820'''** with both the proximal end of flexible segment **3844'''** (having a constant flexibility) and the distal end of flexible segment **3846'''** (having a variable flexibility) of the outer hypotube **3824'''**. Accordingly, the distal ends of flexible segments **3846'''**, **3866'''** do not axially align, thereby creating overlap between the distal end of flexible segment **3846'''** (having a variable flexibility) of the outer hypotube **3824'''** with both the proximal end flexible segment **3864'''** (having a constant flexibility) and the distal end of flexible segment **3866'''** (having a variable flexibility) of the inner hypotube **3820'''**. And the proximal ends of flexible segments **3846'''**, **3866'''** also do not

axially align. The proximal end of flexible segment **3846''** (having a variable flexibility) of the outer hypotube **3824'''** overlaps with both the proximal end of flexible segment **3866'''** (having a variable flexibility) and the distal end of the proximal uncut segment **3868''** of the inner hypotube **3820'''**. And portions (e.g., distal portion, central portion, and proximal portion) of uncut segments **3848'''**, **3868'''** of the outer hypotube **3824'''** and the inner hypotube **3820'''** overlap.

[0393] Although not shown in any particular figure, the scope of this disclosure shall include any and all combinations of the inner hypotube and the outer hypotube, as well as the cut and the uncut segments of such hypotubes, as depicted in FIGS. **45A**, **45B**, **45C** and **45D**.

[0394] In the appended figures, similar components and/or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a letter that distinguishes among the similar components. If only the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

[0395] A number of variations and modifications of the disclosure may be used. It would be possible to provide for some features of the disclosure without providing others.

[0396] In some embodiments, the systems and methods of this disclosure may be implemented in conjunction with a special purpose computer, a programmed microprocessor or microcontroller and peripheral integrated circuit element(s), an ASIC or other integrated circuit, a digital signal processor, a hard-wired electronic or logic circuit such as discrete element circuit, a programmable logic device or gate array such as PLD, PLA, FPGA, PAL, special purpose computer, any comparable means, or the like. In general, any device(s) or means capable of implementing the methodology illustrated herein may be used to implement the various aspects of this disclosure. Exemplary hardware that may be used for the disclosed embodiments, configurations and aspects includes computers, handheld devices, telephones (e.g., cellular, Internet enabled, digital, analog, hybrids, and others), and other hardware known in the art. Some of these devices include processors (e.g., a single or multiple microprocessors), memory, nonvolatile storage, input devices, and output devices. Furthermore, alternative software implementations including, but not limited to, distributed processing or component/object distributed processing, parallel processing, or virtual machine processing may also be constructed to implement the methods described herein.

[0397] The present disclosure, in various aspects, embodiments, and/or configurations, includes components, methods, processes, systems and/or apparatus substantially as depicted and described herein, including various aspects, embodiments, configurations embodiments, sub combinations, and/or subsets thereof. Those of skill in the art will understand how to make and use the disclosed aspects, embodiments, and/or configurations after understanding the present disclosure. The present disclosure, in various aspects, embodiments, and/or configurations, includes providing devices and processes in the absence of items not depicted and/or described herein or in various aspects, embodiments, and/or configurations hereof, including in the absence of such items as may have been used in previous devices or processes, e.g., for improving performance, achieving ease and/or reducing cost of implementation.

[0398] For example, although a pin and slot cam configuration is discussed within this disclosure, other possible cam configurations may be used. For example, a captured ring cam configuration may be used. A captured ring cam configuration may include a ring that is attached to at least one of the inner sheath (or inner member attached to the inner sheath) or the outer sheath (or outer member attached to the outer sheath) and that is captured by two angled lobes on the other sheath (or member). Although the ring may be captured by one lobe, it may be preferred for the ring to be captured by two lobes-one on each side of the ring-such that cutting surface may be forced in both a proximal direction (toward a retraction position) and distal direction (toward an extended direction). The benefit of being able to force the cutting surface in both directions with the aid of the captured cam configuration potentially negates the need for a spring or other retraction mechanism to force the inner sheath (or inner member) and cutting surface back within the outer

sheath (or outer member.

[0399] Another example of an alternate embodiment may include replacing the cutting tip with a dilator top or separator tip. A further example of an alternate embodiment may include varying the degrees of rotation of the inner sheath assembly or the barrel cam cylinder in the clockwise and/or counter-clockwise direction. An even further example of an alternate embodiment may include the barrel cam cylinder and the inner sheath assembly first rotating in a counter-clockwise direction followed by rotating in a clockwise direction.

[0400] The foregoing discussion has been presented for purposes of illustration and description. The foregoing is not intended to limit the disclosure to the form or forms disclosed herein. In the foregoing Summary for example, various features of the disclosure are grouped together in one or more aspects, embodiments, and/or configurations for the purpose of streamlining the disclosure. The features of the aspects, embodiments, and/or configurations of the disclosure may be combined in alternate aspects, embodiments, and/or configurations other than those discussed above. This method of disclosure is not to be interpreted as reflecting an intention that the claims require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive aspects lie in less than all features of a single foregoing disclosed aspect, embodiment, and/or configuration. Thus, the following claims are hereby incorporated into this Summary, with each claim standing on its own as a separate preferred embodiment of the disclosure.

[0401] Moreover, though the description has included description of one or more aspects, embodiments, and/or configurations and certain variations and modifications, other variations, combinations, and modifications are within the scope of the disclosure, e.g., as may be within the skill and knowledge of those in the art, after understanding the present disclosure. It is intended to obtain rights which include alternative aspects, embodiments, and/or configurations to the extent permitted, including alternate, interchangeable and/or equivalent structures, functions, ranges or steps to those claimed, whether or not such alternate, interchangeable and/or equivalent structures, functions, ranges or steps are disclosed herein, and without intending to publicly dedicate any patentable subject matter.

Claims

1. An apparatus, comprising: an elongate sheath assembly configured to be advanced through a blood vessel of a patient body, wherein the elongate sheath assembly comprises an outer sheath and an inner sheath, wherein the inner sheath is received inside a lumen of the outer sheath, wherein a lumen of the inner sheath is configured to receive an object separate from the elongate sheath assembly, wherein the elongate sheath assembly comprises a plurality of flexible segments comprising a plurality of cuts, wherein the outer sheath comprises a first flexible segment of the plurality of flexible segments and the inner sheath comprises a second flexible segment of the plurality of flexible segments, wherein the first flexible segment and the second flexible segment are unaligned along a length of the elongate sheath assembly to prevent kinking.
2. The apparatus of claim 1, wherein at least one of: a distal end of the first flexible segment is unaligned with a distal end of the second flexible segment; or a proximal end of the first flexible segment is unaligned with a proximal end of the second flexible segment.
3. The apparatus of claim 1, wherein the outer sheath further comprises a third flexible segment of the plurality of flexible segments and the inner sheath comprises a fourth flexible segment of the plurality of flexible segments.
4. The apparatus of claim 3, wherein the third flexible segment is proximal of the first flexible segment, wherein the fourth flexible segment is proximal of the second flexible segment.
5. The apparatus of claim 3, wherein at least one of: the plurality of cuts in the first flexible segment is configured to provide a constant flexibility and the plurality of cuts in the third flexible segment is configured to provide a variable flexibility; or the plurality of cuts in the second flexible

segment is configured to provide a constant flexibility and the plurality of cuts in the fourth flexible segment is configured to provide a variable flexibility.

6. The apparatus of claim 3, wherein the third flexible segment and the fourth flexible segment are unaligned along the length of the elongate sheath assembly.

7. The apparatus of claim 6, wherein a proximal end of the first flexible segment is unaligned with a proximal end of the second flexible segment, and wherein a distal end of the third flexible segment is unaligned with a distal end of the fourth flexible segment.

8. The apparatus of claim 7, wherein a proximal end of the third flexible segment is unaligned with a proximal end of the fourth flexible segment.

9. The apparatus of claim 1, wherein the inner sheath is longer than the outer sheath.

10. The apparatus of claim 1, wherein the object comprises a medical lead implanted in the patient body, wherein the lumen of the inner sheath is configured to receive the medical lead for removal of the medical lead from the patient body.
